



Transformer

show sharing

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Pad-mounted transformer

Pad-mounted transformer

Single-phase or three-phase



Loop or radial units



Live-front or dead-front units

Live-front units

Live-front units have the primary cables terminated in a stress cone supported by a bushing. Thus the primary has exposed energized metal, or “live,” parts.

Dead-front units

Dead-front units use primary cables that are terminated with high-voltage separable insulated connectors. Thus the primary has all “dead” parts — no exposed energized metal



Three Phase Pad-mounted transformer

Three Phase Pad-mounted transformer ---Standard

pad-mounted transformer: An outdoor transformer utilized as part of an underground distribution system, with enclosed compartment(s) for high-voltage and low-voltage cables entering from below, and mounted on a foundation pad.

IEEE C57.12.34: IEEE Standard Requirements for Pad-Mounted, Compartmental-Type, Self-Cooled, Three-Phase Distribution Transformers, 10 MVA and Smaller; High-Voltage, 34.5 kV Nominal System Voltage and Below; Low-Voltage, 15 kV Nominal System Voltage and Below

CSA 227.4: Scope

CSA C227.4:21

Three-phase, pad-mounted distribution transformers with separable insulated high-voltage connectors

CSA C227.5:08

Three-phase live-front pad-mounted distribution transformers

- IEEE C57.12.00
- IEEE C57.12.28
- IEEE C57.12.34
- IEEE C57.12.90
- CSA C227.4
- CSA C227.5
- CCN:XPLH



Three Phase Pad-mounted transformer -----Scope

IEEE C57.12.34: This standard covers certain electrical, dimensional, and mechanical characteristics and takes into consideration certain safety features of three-phase, 60 Hz, liquid-filled, self-cooled, pad-mounted, compartmental-type distribution transformers. These transformers are rated 10 MVA and smaller, with the high-voltage limit of 34.5 kV system nominal voltage and below, and with low-voltage limit of 15 Kv system nominal voltage and below. These transformers are generally used for step-down purposes from an underground primary cable supply. This standard covers the connector, bushing, and terminal arrangements for radial or loop-feed systems. Either certain minimum dimensions or certain specific dimensions shall be specified. This standard does not cover the electrical and mechanical requirements of any accessory devices that may be supplied with the transformer

CSA 227.4: Scope

This Standard specifies the requirements for three-phase, pad-mounted distribution transformers, consisting of a transformer and a cable entrance compartment with provision for separable insulated high-voltage connectors, intended primarily for operation by electric utilities on three-phase underground distribution systems having primary voltages as shown in Clause 4.6.

C227.5: 1 Scope

1.1

This Standard specifies requirements for three-phase live-front pad-mounted distribution transformer units that

- (a) consist of a transformer and a separable or integral cable entrance compartment;
- (b) are rated 75 to 3000 kV•A and intended for operation on three-phase underground distribution systems having the preferred rated voltages specified in Tables 1 and 2; and
- (c) are intended for mounting outdoors on pads without additional protective enclosures

Rating	US IEEE C57.12.34	CSA C227.4	CSA C227.5
Power limited	<=10MVA	<=3MA	75~3000KVA
Voltage	<= 34.5KV	<= 34.5KV	<= 25KV
Frequency	60 Hz	60 Hz	Not define
Cooling method	self-cooled,	-	Not define

Three Phase Pad-mounted transformer ----Power Rating

IEEE C57.12.34

Kilovolt-ampere ratings are for continuous operation and based on **not exceeding either a 65 °C average winding temperature rise or an 80 °C hot-spot conductor temperature rise**. The temperature rise of the insulating oil shall not exceed 65 °C when measured near the top of the tank. These kilovolt-ampere ratings are based on the usual temperature and altitude service conditions specified in IEEE Std C57.12.00™. The kilovolt-ampere **ratings shall be as in Table 1:**

IEEE C57.12.34	KVA rating	
	C227.4	C227.5
45	-	-
75	75	75
112.5	-	-
150	150	150
225	225	225
300	300	300
500	500	500
750	750	750
1000	1000	1000
1500	1500	1500
2000	2000	2000
2500	2500	2500
-	3000	3000
3750	-	-
5000	-	-
7500	-	-
10000	-	-

IEEE C57.12.34

Table 1—Kilovolt-ampere ratings

45	1000
75	1500
112.5	2000
150	2500
225	3750
300	5000
500	7500
750	10 000

C227.4

4.1 kVA ratings

The kVA ratings at rated voltages **shall be** 75, 150, 225, 300, 500, 750, 1000, 1500, 2000, 2500, and 3000.

C227.5

4.1 kV•A ratings

The rated kV•A capacities at rated voltage shall be 75, 150, 225, 300, 500, 750, 1000, 1500, 2000, 2500, and 3000.

Three Phase Pad-mounted transformer ----Voltage Rating

IEEE C57.12.34

Voltage ratings shall be in accordance with Table 3 of this standard

Table 3—Range of kVA and voltage ratings

High-voltage rating (V)	Minimum BIL (kV)	Rating ^a (kVA)			
		For low-voltage rating 208Y/120 ^c or 240 V	For low-voltage rating 480Y/277 ^c , 480, 600Y/347 ^c , or 600 V	For standard low-voltage rating 4800 Δ through 2400 Δ ^a as specified in the high-voltage rating column of this table	For standard low-voltage rating 13 800 Grd Y/7970 or 13 800 Δ through 6900 Δ ^a as specified in the high-voltage rating column of this table
2400 Δ	45	45 through 750	45 through 750		
4160 Grd Y/2400 or 4160 Δ ^c	60	45 through 1000	45 through 1500		
4800 Δ	60	45 through 1000	45 through 2000		
6900 Δ	75	45 through 1000	45 through 2000	750 through 10 000	
8300 Grd Y/4800 ^c	75	45 through 1000	45 through 3750	750 through 10 000	
12 000 Grd Y/6930 ^c or 12 000 Δ	95	45 through 1000	45 through 3750	750 through 10 000	
12 470 Grd Y/7200 ^c	95	45 through 1000	45 through 3750	750 through 10 000	
13 200 Grd Y/7620 ^c	95	45 through 1000	45 through 3750	750 through 10 000	
13 800 Grd Y/7970 ^c or 13 800 Δ	95	45 through 1000	45 through 3750	750 through 10 000	
16 340 Δ	95	45 through 1000	45 through 3750	750 through 10 000	
20 780 Grd Y/12 000 ^c	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
22 860 Grd Y/13 200 ^c	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
23 000 Δ	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
23 900 Grd Y/13 800 ^c	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
24 940 Grd Y/14 400 ^c	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
34 500 Grd Y/19 920 ^{c,d}	150	75 through 1000	75 through 3750	750 through 10 000	750 through 10 000
34 500 Δ ^b	150	75 through 1000	75 through 3750	750 through 10 000	750 through 10 000

^a Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.

^b The highest BIL level for separable insulated connectors is 150 kV BIL. If 200 kV BIL level is required, bushings shall be used.

^c Unsymmetrical excitation or loading of Y-Y connected units may cause heating of their tanks in excess of that which would be produced by balanced conditions. To reduce the probability of tank heating, such units shall be provided with a core construction that will not saturate when 33% zero-sequence voltage is applied.

^d When specifying 125 kV BIL, adequate grounding and surge protection studies should be made.

^e Care should be taken in that some of these kVA ranges result in providing over 200 A at the rated voltage, thus requiring a 600 A connector. Consult IEEE Std 386™ for complete connector ratings.

Three Phase Pad-mounted transformer ----Voltage Rating

CSA 227.4

Voltage designation

The designation of the voltage rating is represented in Table 1.

Table 2
Low- and high-voltage electrical characteristics
of transformers and assembled terminals
(See Clauses [4.6.1](#), [4.6.2](#), and [6.1.2.2](#).)

High voltage		Low voltage		Assembled HV terminals, 200 A	
Rated voltage (V)	BIL (kV)	Rated voltage (V)	BIL (kV)	Voltage class (kV)	BIL (kV)
4160GrdY/2400	60	208Y/120† 480Y/277 600Y/347	30 30 30	15*	95
4160	60			15*	95
8320GrdY/4800	75			15*	95
8320	75			15*	95
12470GrdY/7200	95			15	95
12470	95			15	95
13860GrdY/8000	95			15	95
13860	95			15	95
24940GrdY/14400	125			25	125
27600GrdY/15935	125			28	125
24940	150			35	150
27600	150			35	150
34500GrdY/19920	150			35	150
34500‡	150			35	150

Three Phase Pad-mounted transformer ----Voltage Rating

CSA 227.5

Voltage designation

The designation of the voltage rating is represented in Table 1.

Table 2
Preferred high-voltage ratings and required BIL ratings
(See Clauses 1.1 and 4.6.1.)

Insulation class, kV	Preferred high voltage	BIL, kV
5.0	2 400	60
	4 160 GrdY/2 400	60
	4 160	60
8.7	4 800	75
	7 200	75
	8 320 GrdY/4 800	75
	8 320	75
15.0	12 470 GrdY/7 200	95
	12 470	95
	13 860 GrdY/8 000	95
	13 860	95
18.0	24 940 GrdY/14 400	125
	27 600 GrdY/15 935	125
25.0*	24 940	125

*The dielectric test levels are the same as those for the 18.0 kV insulation class (see Table 10).

Note: If single-pole switching is planned for units of an insulation class of 15 kV or higher, the probability of ferroresonance should be considered.

Table 3
Preferred low-voltage ratings and required BIL ratings
(See Clause 4.6.2.)

Insulation class, kV	Preferred low voltage	BIL, kV
1.2	208Y/120 or 208 GrdY/120*	30
	480Y/277 or 480 GrdY/277†	30
	600‡	30
	600Y/347 or 600 GrdY/347	30

*The maximum unit size shall be 1000 kV•A.

†The maximum unit size shall be 2000 kV•A.

‡With GrdY primary voltages, a Δ secondary voltage should not be used.

Three Phase Pad-mounted transformer ---- Voltage Rating

CSA 227.4 & C227.3

Table 1
Designation of voltage ratings (schematic representation)
(See Clauses 1.1 and 4.2.1.)

Nomenclature	Example of nameplate marking	Typical winding diagram	Usage
E	4 160		E shall indicate a winding that is permanently Δ -connected for operation on an E volt system.
$E_1 \text{GrdY}/E$	12 470GrdY/7 200		$E_1 \text{GrdY}/E$ shall indicate a winding that has reduced insulation and is permanently Y-connected. The neutral end may be connected in the tank or brought out and effectively grounded for operation on an E_1 volt system with E volts available from line to neutral.
$V \times V_1$	4 160 $\times$ 12 470		$V \times V_1$ shall indicate a winding, the sections of which may be connected in multiple to obtain one of the voltage ratings of V, or may be connected in series to obtain one of the voltage ratings of V_1 . Windings are permanently Δ - or Y-connected.
	4 160GrdY/2 400 $\times$ 470GrdY/7 200		
$E_1 Y/E$	208Y/120		$E_1 Y/E$ shall indicate a winding that is permanently Y-connected with a fully insulated neutral brought out for operation on an E_1 volt system, with E volts available from line to neutral.

Note: $E_1 = \sqrt{3}E$.

Table 1
Designation of voltage ratings (schematic representation)
(See Clauses 1.1 and 4.2.1.)

Nomenclature	Example of nameplate marking	Typical winding diagram	Usage
E	4 160		E shall indicate a winding that is permanently Δ -connected for operation on an E volt system.
$E_1 \text{GrdY}/E$	12 470GrdY/7 200		$E_1 \text{GrdY}/E$ shall indicate a winding that has reduced insulation and is permanently Y-connected. The neutral end may be connected in the tank or brought out and effectively grounded for operation on an E_1 volt system with E volts available from line to neutral.
$V \times V_1$	4 160 $\times$ 12 470		$V \times V_1$ shall indicate a winding, the sections of which may be connected in multiple to obtain one of the voltage ratings of V, or may be connected in series to obtain one of the voltage ratings of V_1 . Windings are permanently Δ - or Y-connected.
	4 160GrdY/2 400 $\times$ 470GrdY/7 200		
$E_1 Y/E$	208Y/120		$E_1 Y/E$ shall indicate a winding that is permanently Y-connected with a fully insulated neutral brought out for operation on an E_1 volt system, with E volts available from line to neutral.

Note: $E_1 = \sqrt{3}E$.

Three Phase Pad-mounted transformer ----Tap Rating

4.3 Tap ratings

When specified, high-voltage taps shall be composed of two 2.5% above the high-voltage rating and two 2.5% below the high-voltage rating. Transformers with a 208Y/120 low-voltage rating may be specified with four 2.5% taps below the high-voltage rating. The tap changer handle or the area next to the handle in the terminating compartment shall be marked indicating suitability for de-energized operation only.

4.5 Off-circuit voltage taps

4.5.1

Transformers without taps are the standard.

4.5.2

When required by the purchaser, transformers shall be equipped with four high voltage taps that are $\pm 2.5\%$ and $\pm 5\%$ of the rated voltage. All taps shall be full-capacity.

Note: Taps in dual voltage transformers can negatively impact transformer reliability, and some dual voltage coil configurations can present operational problems. If taps are required on the lower voltage or on both voltages, the purchaser should discuss the requirement with the manufacturer to evaluate the associated risks given the coil configuration. If the dual voltage ratio is higher than 2:1, taps should only be on the higher voltage connection.

4.5 Off-circuit voltage taps

Note: Transformers without taps are the standard. Taps are included as an option in [Clause 10\(k\)](#).

4.5.1

Unless otherwise specified by the purchaser, high-voltage taps shall be $\pm 2.5\%$ and $\pm 5\%$ of the rated voltage.

4.5.2

When a transformer is connected on a tap below rated voltage, the kV•A capacity shall be reduced in direct proportion to the voltage of the tap. When a transformer is connected on a tap above rated voltage, the capacity shall be the rated kV•A of the transformer.

Three Phase Pad-mounted transformer ----Impedance voltage

The percent impedance voltage, as measured on the **rated voltage connection**, shall be as shown in Table 2.

IEEE C57.12.34

Table 2—Nominal percent impedance voltage

Rating (kVA)	For low-voltage rating 600 V and below	For low-voltage rating 2400 Δ through 4800 Δ	For low-voltage rating 6900 Δ through 13 800 Grd Y/7970 or 13 800 Δ
45	2.70 through 5.75 *	2.70 through 5.75 *	2.70 through 5.75 *
75	2.70 through 5.75 *	2.70 through 5.75 *	2.70 through 5.75 *
112.5	3.10 through 5.75 *	3.10 through 5.75 *	3.10 through 5.75 *
150	3.10 through 5.75 *	3.10 through 5.75 *	3.10 through 5.75 *
225	3.10 through 5.75 *	3.10 through 5.75 *	3.10 through 5.75 *
300	3.10 through 5.75 *	3.10 through 5.75 *	3.10 through 5.75 *
500	4.35 through 5.75 *	4.35 through 5.75 *	4.35 through 5.75 *
750	5.75	5.75	5.75
1000	5.75	5.75	5.75
1500	5.75	5.75	5.75
2000	5.75	5.75	5.75
2500	5.75	5.75	5.75
3750	5.75	5.75	6.00
5000		6.00	6.50
7500		6.00	6.50
10 000		6.00	6.50

a These impedance values were designed to meet Table 15 of IEEE Std C57.12.00-2010 which defines the maximum per-unit short-circuit withstand capability. It is recommended that the end user evaluate other requirements such as voltage regulation, system impedance, or available fault current when using these values.

Table 6
Minimum transformer impedance
(See Clause 4.11.)

Transformer size, kVA	Minimum transformer impedance, %
0–150	1.8
225–300	2.0
500	3.0
750–1000	4.0
> 1000	5.0

Table 7
Minimum transformer impedance
(See Clause 4.11.)

Transformer size, kV•A	Minimum transformer impedance, %
0–150	1.8
225–300	2.0
500	3.0
750–1000	4.0
> 1000	5.0

Note: The impedance value can result in per-unit short circuit levels exceeding the distribution transformer withstand capability required by Table 5; the end user should evaluate other requirements such as system impedance or available fault current when using these values

CSA 227.4

4.11 Impedance

The absolute minimum impedance shall be in accordance with Table 6 unless otherwise specified [see Clause 10 m)].

CSA 227.5

4.11 Impedance

The minimum transformer impedance shall be as specified in Table 7 unless otherwise specified by the purchaser (see Clause 10).

Three Phase Pad-mounted transformer ----Impedance voltage

Rating (kVA)	Minimum transformer impedance			CSA 227.4&C227.5	IEEE C57.12.34 &CSA227.4& C227.5
	IEEE C57.12.34 For low-voltage rating 600 V and below	For low-voltage rating 2400 Δ through 4800 Δ	For low-voltage rating 6900 Δ through 13 800 Grd Y/7970 or 13 800 Δ		
45	2.70 ~5.75 a	2.70 ~5.75 a	2.70 ~5.75 a	1.8	2.70 ~5.75 a
75	2.70 ~5.75 a	2.70 ~5.75 a	2.70 ~5.75 a	1.8	2.70 ~5.75 a
112.5	3.1 ~5.75 a	3.1 ~5.75 a	3.1 ~5.75 a	1.8	3.1 ~5.75 a
150	3.1 ~5.75 a	3.1 ~5.75 a	3.1 ~5.75 a	1.8	3.1 ~5.75 a
225	3.1 ~5.75 a	3.1 ~5.75 a	3.1 ~5.75 a	2	3.1 ~5.75 a
300	3.1 ~5.75 a	3.1 ~5.75 a	3.1 ~5.75 a	2	3.1 ~5.75 a
500	4.35 ~5.75 a	4.35 ~5.75 a	4.35 ~5.75 a	3	4.35 ~5.75 a
750	5.75	5.75	5.75	4	5.75
1000	5.75	5.75	5.75	4	5.75
1500	5.75	5.75	5.75	5	5.75
2000	5.75	5.75	5.75	5	5.75
2500	5.75	5.75	5.75	5	5.75
3750	5.75	5.75	6	5	6
5000	--	6	6.5	5	6.5
7500	--	6	6.5	5	6.5
10000	--	6	6.5		6.5

Three Phase Pad-mounted transformer ----Tolerance on impedance voltage

IEEE C57.12.00

IEEE C57.12.34 refer IEEE C57.12.00	CSA C227.4	C227.5
The impedance of a transformer shall not vary from the guaranteed value by more than $\pm 7.5\%$ for transformers with a guaranteed value of 2.5% or higher	The impedance of a two-winding transformer with an impedance voltage greater than 2.5% shall have a tolerance of $\pm 7.5\%$ of the specified value	Not defined
those with an impedance voltage of 2.5% or less shall have a tolerance of $\pm 10\%$ of the specified value.	and $\pm 15.0\%$ for transformers with a guaranteed value below 2.5%.	
Differences of impedance between duplicate two-winding transformers, when two or more units of a given rating are produced by one manufacturer at the same time, shall not exceed 7.5% of the specified value.	-	
The impedance of a transformer having three or more windings, or having zigzag windings, shall have a tolerance of $\pm 10\%$ of the specified value.	-	
The impedance of an autotransformer shall have a tolerance of $\pm 10\%$ of the specified value.	-	

9.2 Tolerances for impedance

The tolerances for impedance shall be as follows:

- The impedance of a two-winding transformer with an impedance voltage greater than 2.5% shall have a tolerance of $\pm 7.5\%$ of the specified value, and those with an impedance voltage of 2.5% or less shall have a tolerance of $\pm 10\%$ of the specified value. Differences of impedance between duplicate two-winding transformers, when two or more units of a given rating are produced by one manufacturer at the same time, shall not exceed 7.5% of the specified value.
- The impedance of a transformer having three or more windings, or having zigzag windings, shall have a tolerance of $\pm 10\%$ of the specified value. Differences of impedance between duplicate three-winding or zigzag transformers, when two or more units of a given rating are produced by one manufacturer at the same time, shall not exceed 10% of the specified value.
- The impedance of an autotransformer shall have a tolerance of $\pm 10\%$ of the specified value.

CSA 227.4

7.3.3 Tolerance for impedance

7.3.3.1 Guarantee

The impedance of a transformer shall not vary from the guaranteed value by more than $\pm 7.5\%$ for transformers with a guaranteed value of 2.5% or higher, and $\pm 15.0\%$ for transformers with a guaranteed value below 2.5%.

Three Phase Pad-mounted transformer ----Tolerance on impedance voltage

IEEE C57.12.00

IEEE C57.12.34 refer IEEE C57.12.00	CSA C227.4	C227.5
The impedance of a transformer shall not vary from the guaranteed value by more than $\pm 7.5\%$ for transformers with a guaranteed value of 2.5% or higher	The impedance of a two-winding transformer with an impedance voltage greater than 2.5% shall have a tolerance of $\pm 7.5\%$ of the specified value	Not defined
those with an impedance voltage of 2.5% or less shall have a tolerance of $\pm 10\%$ of the specified value.	and $\pm 15.0\%$ for transformers with a guaranteed value below 2.5%.	
Differences of impedance between duplicate two-winding transformers, when two or more units of a given rating are produced by one manufacturer at the same time, shall not exceed 7.5% of the specified value.	-	
The impedance of a transformer having three or more windings, or having zigzag windings, shall have a tolerance of $\pm 10\%$ of the specified value.	-	
The impedance of an autotransformer shall have a tolerance of $\pm 10\%$ of the specified value.	-	

9.2 Tolerances for impedance

The tolerances for impedance shall be as follows:

- The impedance of a two-winding transformer with an impedance voltage greater than 2.5% shall have a tolerance of $\pm 7.5\%$ of the specified value, and those with an impedance voltage of 2.5% or less shall have a tolerance of $\pm 10\%$ of the specified value. Differences of impedance between duplicate two-winding transformers, when two or more units of a given rating are produced by one manufacturer at the same time, shall not exceed 7.5% of the specified value.
- The impedance of a transformer having three or more windings, or having zigzag windings, shall have a tolerance of $\pm 10\%$ of the specified value. Differences of impedance between duplicate three-winding or zigzag transformers, when two or more units of a given rating are produced by one manufacturer at the same time, shall not exceed 10% of the specified value.
- The impedance of an autotransformer shall have a tolerance of $\pm 10\%$ of the specified value.

CSA 227.4

7.3.3 Tolerance for impedance

7.3.3.1 Guarantee

The impedance of a transformer shall not vary from the guaranteed value by more than $\pm 7.5\%$ for transformers with a guaranteed value of 2.5% or higher, and $\pm 15.0\%$ for transformers with a guaranteed value below 2.5%.

Three Phase Pad-mounted transformer ---- Basic lightning impulse

Table 3—Range of kVA and voltage ratings

High-voltage rating (V)	Minimum BIL (kV)	Rating ^a (kVA)			
		For low-voltage rating 208Y/120 ^c or 240 V	For low-voltage rating 480Y/277 ^c , 480, 600Y/347 ^c , or 600 V	For standard low-voltage rating 4800 Δ through 2400 Δ ^a as specified in the high-voltage rating column of this table	For standard low-voltage rating 13 800 Grd Y/7970 or 13 800 Δ through 6900 Δ ^a as specified in the high-voltage rating column of this table
2400 Δ	45	45 through 750	45 through 750		
4160 Grd Y/2400 or 4160 Δ ^c	60	45 through 1000	45 through 1500		
4800 Δ	60	45 through 1000	45 through 2000		
6900 Δ	75	45 through 1000	45 through 2000	750 through 10 000	
8300 Grd Y/4800 ^c	75	45 through 1000	45 through 3750	750 through 10 000	
12 000 Grd Y/6930 ^c or 12 000 Δ	95	45 through 1000	45 through 3750	750 through 10 000	
12 470 Grd Y/7200 ^c	95	45 through 1000	45 through 3750	750 through 10 000	
13 200 Grd Y/7620 ^c	95	45 through 1000	45 through 3750	750 through 10 000	
13 800 Grd Y/7970 ^c or 13 800 Δ	95	45 through 1000	45 through 3750	750 through 10 000	
16 340 Δ	95	45 through 1000	45 through 3750	750 through 10 000	
20 780 Grd Y/12 000 ^c	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
22 860 Grd Y/13 200 ^c	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
23 000 Δ	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
23 900 Grd Y/13 800 ^c	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
24 940 Grd Y/14 400 ^c	125	45 through 1000	45 through 3750	750 through 10 000	750 through 10 000
34 500 Grd Y/19 920 ^{c, d}	150	75 through 1000	75 through 3750	750 through 10 000	750 through 10 000
34 500 Δ ^b	150	75 through 1000	75 through 3750	750 through 10 000	750 through 10 000

a Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.

b The highest BIL level for separable insulated connectors is 150 kV BIL. If 200 kV BIL level is required, bushings shall be used.

c Unsymmetrical excitation or loading of Y-Y connected units may cause heating of their tanks in excess of that which would be produced by balanced conditions. To reduce the probability of tank heating, such units shall be provided with a core construction that will not saturate when 33% zero-sequence voltage is applied.

d When specifying 125 kV BIL, adequate grounding and surge protection studies should be made.

e Care should be taken in that some of these kVA ranges result in providing over 200 A at the rated voltage, thus requiring a 600 A connector. Consult IEEE Std 386™ for complete connector ratings.

Table 9
Dielectric test values
(See Clauses 7.5.2, 7.5.3.2, and 7.5.4.)

Rated voltage	Applied voltage test 60 Hz, kV	Impulse test BIL-full wave, kV
600 or less	10	30
4160GrdY/2400 or 4160	19	60
8320GrdY/4800 or 4800	26	75
12470GrdY/7200 or 12470	34	95
13860GrdY/8000 or 13860	34	95
24940GrdY/14400	40	125
27600GrdY/15935	40	125
27600	50	150
24940	50	150
34500GrdY/19920	50	150
34500	50	150

Table 2
Preferred high-voltage ratings and required BIL ratings
(See Clauses 1.1 and 4.6.1.)

Insulation class, kV	Preferred high voltage	BIL, kV
5.0	2 400	60
	4 160 GrdY/2 400	60
	4 160	60
8.7	4 800	75
	7 200	75
	8 320 GrdY/4 800	75
	8 320	75
15.0	12 470 GrdY/7 200	95
	12 470	95
	13 860 GrdY/8 000	95
	13 860	95
18.0	24 940 GrdY/14 400	125
	27 600 GrdY/15 935	125
25.0*	24 940	125

*The dielectric test levels are the same as those for the 18.0 kV insulation class (see Table 10).

Note: If single-pole switching is planned for units of an insulation class of 15 kV or higher, the probability of ferroresonance should be considered.

Three Phase Pad-mounted transformer ---- Basic lightning impulse

IEEE C57.12.34		CSA 227.4		CSA 227.5	
High-voltage rating (V)	Minimum BIL (kV)	High-voltage rating (V)	Minimum BIL (kV)	High-voltage rating (V)	Minimum BIL (kV)
--		600 or less	30	--	--
2400 Δ	45	--	--	2 400	60
4160 Grd Y/2400 or 4160 Δc	60	4160GrdY/2400 or 4160	60	4 160 GrdY/2 400 or 4160	60
4800 Δ	60	--	--	4800	75
6900 Δ	75	--	--	7200	75
8300 Grd Y/4800c	75	8320GrdY/4800 or 4800	75	8320Grd Y/4800c or 8320	75
12000 Grd Y/6930c or 12 000 Δ	95	--	--	--	----
12470 Grd Y/7200c	95	12470GrdY/7200 or 12470	95	12 470 GrdY/7 200 or 12470	95
13200 Grd Y/7620c	95	--	--	--	--
13800 Grd Y/7970c or 13 800 Δ	95	13860GrdY/8000 or 13860	95	13 860 GrdY/8 000 or 13800	95
16 340 Δ	95	--	--	--	--
20780 Grd Y/12 000c	125	--	--	--	--
22860 Grd Y/13 200c	125	--	--	--	--
23 000 Δ	125	--	--	--	--
23900 Grd Y/13 800c	125	--	--	--	--
24940 Grd Y/14 400c	125	24940GrdY/14400	125	24 940 GrdY/14 400	125
--	--	27600GrdY/15935	125	27 600 GrdY/15 935	125
--	--	27600	150	--	--
--	--	24940	150	24 940	125
34500 Grd Y/19 920c, d	125	34500GrdY/19920	150	--	--
34 500 Δb	150	34500	150	--	--

Three Phase Pad-mounted transformer ---- Basic lightning impulse

IEEE C57.12.34

7. Tests

7.1 General

Except as specified in 7.2 and 7.3 of this standard, tests shall be performed as specified in IEEE Std C57.12.00 and IEEE Std C57.12.90™

7.1 General **CSA C227.4**

Where details of the procedure, methods, precautions, and calculations used in conducting the tests in Clauses 7.2 to 7.8 are not specified, the requirements of IEEE C57.12.90 shall apply.

7 Tests

7.1

CSA C227.5

Transformers shall be tested in accordance with CAN/CSA-C2.1. The test values for dielectric tests shall be as specified in Table 10.

7.2

The transformer and cable entrance compartment assembly shall be of a design that has been type tested to demonstrate that it will successfully withstand overvoltage testing equivalent to testing specified in Clause 7.1 when equipped with the high-voltage accessories (if any) provided by the manufacturer.

7.3

The transformer and cable entrance compartment assembly shall be of a design that has been type tested in accordance with the enclosure security requirements specified in IEEE C57.12.28 to demonstrate that it will meet the tamper resistance requirements specified in Clause 5.2.4.

Three Phase Pad-mounted transformer ---- Basic lightning impulse

IEEE C57.12.34

7. Tests

7.1 General

Except as specified in 7.2 and 7.3 of this standard, tests shall be performed as specified in IEEE Std C57.12.00 and IEEE Std C57.12.90™

7.1 General

CSA C227.4

Where details of the procedure, methods, precautions, and calculations used in conducting the tests in Clauses 7.2 to 7.8 are not specified, the requirements of IEEE C57.12.90 shall apply.

7 Tests

7.1

CSA C227.5

Transformers shall be tested in accordance with CAN/CSA-C2.1. The test values for dielectric tests shall be as specified in Table 10.

7.2

The transformer and cable entrance compartment assembly shall be of a design that has been type tested to demonstrate that it will successfully withstand overvoltage testing equivalent to testing specified in Clause 7.1 when equipped with the high-voltage accessories (if any) provided by the manufacturer.

7.3

The transformer and cable entrance compartment assembly shall be of a design that has been type tested in accordance with the enclosure security requirements specified in IEEE C57.12.28 to demonstrate that it will meet the tamper resistance requirements specified in Clause 5.2.4.

Three Phase Pad-mounted transformer ---- Basic lightning impulse

CSA C227.4 routing test

- g) induced voltage on all transformers; for loop feed transformers, induced voltage shall be done with the **SWA and SWB switches in the closed position**;
- h) transformer tank leak-detection test. This test shall be performed on each transformer tank to verify the integrity of the welds and on the final assembled unit **without the pressure relief device**; and
- i) **a continuity test**. This shall be performed between H1A and H1B, H2A and H2B, and H3A and H3B for loop feed transformers.

CSA C227.4 type test

- f) transformer integrity;
- h) negative pressure withstand. The tank design shall be tested per the requirements of Clause **5.1.2** using the most negative vacuum pressure (–35 kPa)

Three Phase Pad-mounted transformer ----TEST

Ratio, Polarity and phase relation tests	IEEE C57.12.90(R)	6, 7
	CSA C2.1(R)	9.2, 9.3.1
	CSA 227.4(R)	7.2a), b) 7.3.1
Resistance Measurements	IEEE C57.12.90(R)	5
	CSA C2.1(T)	9.4.2
	CSA 227.4(T)	7.4.2a)
Insulation resistance	IEEE C57.12.90(T)	10.11
No-load losses and excitation current	IEEE C57.12.90(R)	8
	CSA C2.1(R)	9.2, 9.3.2
	CSA 227.4(R)	7.2c), d), 7.3.2
Measurement of short-circuit impedance and load loss	IEEE C57.12.90(R)	9
	CSA C2.1(R)	9.2, 9.3.2
	CSA 227.4(R)	7.2e), d), 7.3.3
Audible sound levels	IEEE C57.12.90(0)	13
	CSA C2.1(0)	16.2g), 16.8.1
	CSA 227.4(T)	7.4.2g), 4.9

Three Phase Pad-mounted transformer ---- spacing

- IEEE C57.12.34
- ←→
- IEEE C57.12.00

Table 4—Electrical characteristics and minimum electrical clearances of high-voltage bushings and low-voltage terminals

Transformer voltage	BIL ^{a,c} (kV)	60-Hz dry-1-min withstand ^{b,c} (kV)	Minimum clearance—live parts ^a		
			Phase to ground ^c	Phase to phase ^c	Phase to non- hygroscopic insulating barrier
208Y/120, 240	30	10	25 (1.0)	25 (1.0)	
480Y/277, 480	30	10	25 (1.0)	25 (1.0)	
600Y/347, 600	30	10	25 (1.0)	25 (1.0)	
2400	45	15	51 (2.0)	51 (2.0)	
4160 to 4800	60	21	64 (2.5)	64 (2.5)	
7200	75	27	89 (3.5)	102 (4.0)	76 (3.0)
12 000 to 13 800	95	35	127 (5.0)	140 (5.5)	76 (3.0)
22 860 Grd Y to 24 940 Grd Y	125	42	146 (5.75)	159 (6.25)	76 (3.0)
34 500 Grd Y ^b	150	70	203 (8.0)	229 (9.0)	102 (4.0)

NOTE—These dimensions should be increased wherever possible to allow for ease in making connections by the user.

^aAll dimensions are in millimeters (inches).
^bThe use of barriers shall not reduce these electrical characteristics.
^cWhere clearances are less than those shown, an adequate non-hygroscopic insulating barrier shall be provided.

Table 10—Minimum external clearances between transformer live parts of different phases of the same voltage

Nominal system voltage, rms	Maximum system voltage, rms (from ANSI C84.1)	Minimum clearance between live parts of different phases ^a				Minimum clearance between top shed of insulator of bushings of different phases ^a			
		Distribution transformers		Power transformers		Distribution transformers		Power transformers	
(kV)	(kV)	mm	(in)	mm	(in)	mm	(in)	mm	(in)
1.2	—	25	(1)	51	(2)	25	(1)	25	(1)
2.5	—	51	(2)	76	(3)	25	(1)	38	(1.5)
5.0	—	64	(2.5)	102	(4)	38	(1.5)	51	(2)
8.7	—	102	(4)	127	(5)	51	(2)	64	(2.5)
15	—	140	(5.5)	165	(6.5)	76	(3)	89	(3.5)
25	—	178 ^b	(7)	229	(9)	114	(4.5)	152	(6)
34.5	—	330 ^b	(13)	330	(13)	203	(8)	203	(8)

Three Phase Pad-mounted transformer ---- TEST

Temperature Rise	IEEE C57.12.90(T)	11
	CSA C2.1(T)	16.2f
	CSA 227.4(T)	7.4.2b), 4.3, 7.6
Lightning impulse	IEEE C57.12.90(T)	10.3
	CSA C2.1(T)	9.5.5
	CSA 227.4(T)	7.4.2c)
Applied-voltage tests	IEEE C57.12.90(R)	10.6
	CSA C2.1(R)	9.5.1
	CSA 227.4(R)	7.2f)
Induced-voltage test	IEEE C57.12.90(R)	10.7
	CSA C2.1(R)	9.5.2
	CSA 227.4(R)	7.2g)
Dissolved gas in oil analysis	IEEE C57.12.00(0)	Table 17
	---	--
Pressure and Leak	IEEE C57.12.00(R)	Table 17
	CSA C2.1(R)	8.2
	CSA 227.4(R)	7.2h), 5.1.2
radio-influence voltage	IEEE C57.12.90	Annex A
	CSA 227.4(R)	7.4.2d), 4.8

Three Phase Pad-mounted transformer ---- TEST

Test items	Standard	Clause
Partial discharge test	IEEE C57.12.90(0)	10.9
	CSA 227.4(O)	10w)
Short-circuit capability	IEEE C57.12.90(T)	12
	CSA C2.1(0)	9.7
	CSA 227.4(T)	7.4.2e), 4.10, 7.7

Three Phase Pad-mounted transformer ---- **Compartment**

8.1 General

A pad-mounted, compartmental-type transformer shall consist of a tank with high-voltage and low-voltage cable terminating compartments, as shown in this standard. The compartment shall be **separated by a barrier** of metal or other rigid material.

8.2 Compartment configuration

The high-voltage and low-voltage compartments shall be located side-by-side **on one side of the transformer tank**. When viewed from the front, **the lower-voltage compartment shall be on the right**.

5.2.1 Compartment arrangement

The arrangement of the transformer and cable entrance compartment shall be such that the **high- and low-voltage facilities are mounted side by side on the transformer tank, with the high-voltage facility on the left when viewed from the front**, as shown in Figures 3 to 8. Access to the cable entrance compartment shall be by means of doors.

10 Optional items

r) **a barrier between the high-voltage and low-voltage compartments**. A barrier can **impact dimensions** of the transformer;

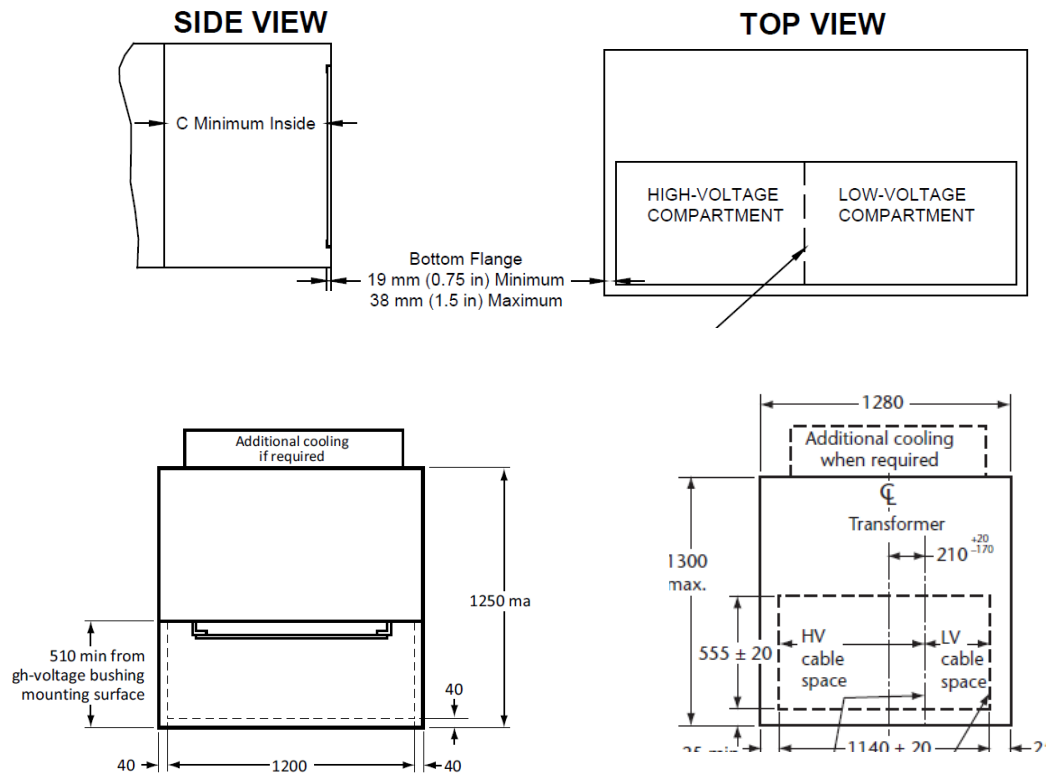
5.2.1 Arrangement

The transformer and cable entrance compartment shall be arranged in such a way that the **high-voltage and low-voltage facilities are mounted side by side** on the transformer tank as shown in **Figure 1, 2, 3, 4, or 5**, as applicable. The cable entrance compartment may be an integral part of the transformer tank or bolted to the front of the transformer tank from the inside of the enclosure. The cable entrance compartment shall be **divided by a suitable metal barrier** into a high-voltage compartment and a low-voltage compartment, each accessible by **a separate door**.

5.2.2 Metal barrier

The metal barrier separating the high-voltage and low-voltage compartments shall have **two holes for 1/2 inch bolts in the lower section, not more than 100 mm from its bottom edge**, to allow the user to attach a bolted **extension in order to separate the cable entry hole** in the foundation pad into high-voltage and low-voltage sections.

Three Phase Pad-mounted transformer ---- Construction



Three Phase Pad-mounted transformer ---- Construction



Canada pad-mounted



Three Phase Pad-mounted transformer ---- Lifting and Jacking

8.6 Lifting----IEEE C57.12.34

The construction of the unit shall be such that it can be lifted, skidded, or slid, or any combination of these, into place on the mounting surface without disturbing the high-voltage or low-voltage cables.

- a) Jack bosses or jacking facilities shall be provided on the tank for transformers rated 500 kVA or larger. The vertical clearance for a jack shall be 38 mm (1.5 in) minimum, 165 mm (6.5 in) maximum.
- b) The transformer shall be arranged for rolling in two directions; parallel to and at right angles to one side of the transformer.
- c) The transformer shall be provided with lifting provisions permanently attached and arranged on the tank to provide a distributed balanced lift in a vertical direction for the completely assembled transformer and shall be designed to provide a safety factor of five when the horizontal angle of loading on the lifting sling does not exceed 30° as described in ASME B30.9.

5.1.3 Lifting provisions----CSA C227.4

Construction of the unit shall be such that it can be lifted, tied down during shipment, skidded, or slid into place on the mounting pad without disturbing high- or low-voltage cables. Jacking and lifting provisions shall be arranged on the tank so as to prevent overturning when lifted.

5.1.4 Jacking provision

Jack steps or equivalent jacking facilities shall be provided on the tank. Vertical clearance for a jack shall be 50 mm minimum and 165 mm maximum.

5.1.5 Transformer base

The transformer base of units rated 750 kVA and larger shall be arranged for rolling in four directions, parallel to and at right angles to one side of the transformer.

5.1.3 Lifting and jacking---CSA C227.5

The unit shall be constructed in such a way that it can be lifted, tied down during shipment, skidded, or slid into place on the mounting pad without disturbing high-voltage or low-voltage cables. Jacking and lifting provisions shall be arranged on the tank in a way that prevents overturning. Jack steps or equivalent jacking facilities shall be provided on the tank. Jacks shall have a vertical clearance of at least 150 mm and not more than 165 mm.

5.1.4 Transformer base

The transformer base of units rated 750 kV•A or more shall be arranged for rolling in four directions, i.e., parallel and at right angles to one side of the transformer

Three Phase Pad-mounted transformer ---- electrical clearances

Table 4—Electrical characteristics and minimum electrical clearances of high-voltage bushings and low-voltage terminals

Transformer voltage	BIL ^{b, c} (kV)	60-Hz dry-one- minute withstand ^{b, c} (kV)	Minimum clearance—live parts ^a		
			Phase to ground ^c	Phase to phase ^c	Phase to non-hygroscopic insulating barrier
208Y/120, 240	30	10	25 (1.0) ^a	25 (1.0)	
480Y/277, 480	30	10	25 (1.0)	25 (1.0)	
600Y/347, 600	30	10	25 (1.0)	25 (1.0)	
2400	45	15	51 (2.0)	51 (2.0)	
4160 through 4800	60	21	64 (2.5)	64 (2.5)	
7200	75	27	89 (3.5)	102 (4.0)	76 (3.0)
12 000 through 13 800	95	35	127 (5.0)	140 (5.5)	76 (3.0)
22 860 Grd Y through 24 940 Grd Y	125	42	146 (5.75)	159 (6.25)	76 (3.0)
34 500 Grd Y ^b	150	70	203 (8.0)	229 (9.0)	102 (4.0)

NOTE—These dimensions should be increased wherever possible to allow for ease in making connections by the user.^d

^a All dimensions are in millimeters (inches).

^b The use of barriers shall not reduce these electrical characteristics.

^c Where clearances are less than those shown, an adequate non-hygroscopic insulating barrier shall be provided.

^d Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

Table 5—Electrical characteristics and ratings of high-voltage connectors

Transformer		Electrical characteristics of completely assembled high-voltage connectors ^c			
High-voltage ratings (V)	BIL (kV)	High-voltage ratings		BIL (kV)	60-Hz Dry-one-minute withstand (kV)
		Phase to ground (kV)	Phase to ground/ Phase to phase (kV)		
2400	45	8.3	8.3/14.4	95	34
4160 through 4800	60	8.3	8.3/14.4	95	34
7200	75	8.3	8.3/14.4	95	34
12 000 through 13 800 ^a	95	8.3 or 15.2	8.3/14.4 or 15.2/26.3	95 or 125	34 or 40
22 860 Grd Y through 24 940 Grd Y	125	15.2	15.2/26.3	125	40
34 500 Grd Y ^b	150	21.1	21.1/36.6	150	50

^a The required connector rating should be specified.

^b When specifying 125 kV BIL, adequate grounding and surge protection studies should be made.

^c IEEE Std 386 should be consulted for complete connector ratings.

Physical sizes of transformers----CSA C227.4

5.1.1 Dimensions

Transformers with 125 kV BIL and below shall have the dimensions shown in Table 7. Transformers with 150 kV BIL shall meet the IEEE C57.12.34 minimum dimensions and bushing layouts with the exception that a barrier between the HV and LV compartments is not required unless otherwise specified by the purchaser

Rated kVA capacity	Type of feed	Transformer maximum BIL, kV	Width, mm	Figure
75, 150	Radial	95	1280*	3
75, 150, 225, 300	Loop	95	1480	4
75, 150, 225, 300, 500	Radial	125	1480	5
75, 150, 225, 300, 500	Loop	125	1480	6
750, 1000, 1500, 2000, 2500, 3000	Radial	125	1730	7
750, 1000, 1500, 2000, 2500, 3000	Loop	125	1730	8

* For radial feed transformers rated at 75, 150 kVA, a width of 1480 mm should be allowed if agreed by the purchaser.

75kVA 150KVA Radial CSA C227.4

Rated kVA capacity	Type of feed	Transformer maximum BIL, kV	Width, mm	Figure
75, 150	Radial	95	1280*	3
75, 150, 225, 300	Loop	95	1480	4
75, 150, 225, 300, 500	Radial	125	1480	5
75, 150, 225, 300, 500	Loop	125	1480	6
750, 1000, 1500, 2000, 2500, 3000	Radial	125	1730	7
750, 1000, 1500, 2000, 2500, 3000	Loop	125	1730	8

* For radial feed transformers rated at 75, 150 kVA, a width of 1480 mm should be allowed if agreed by the purchaser.

Figure 3 a)

Size 1280 × 1250 radial feed — Plan view

(See Clauses [5.2.1](#), [6.1.1.1](#), and [9.3](#) and Table [7](#).)

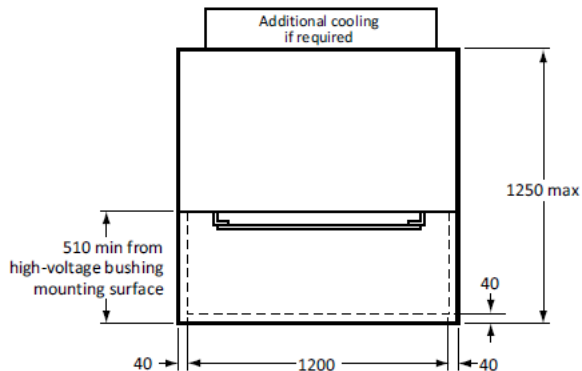


Figure 3 b)

Size 1280 × 1250 radial feed — Side view

(See Clauses [5.2.1](#), [6.1.1.1](#), and [9.3](#) and Table [7](#).)

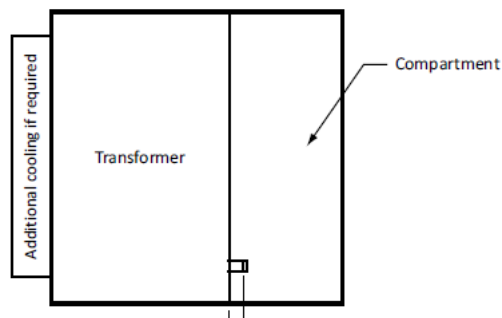
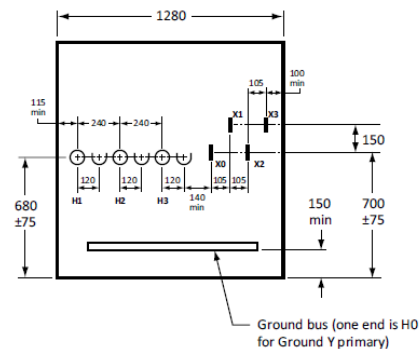


Figure 3 c)

Size 1280 × 1250 radial feed — Front view

(See Clauses [5.2.1](#), [6.1.1.1](#), and [9.3](#) and Table [7](#).)



Notes for Figure 3 a), b), and c):

- 1) All dimensions are in millimetres.
(Tolerances are not cumulative.)
- 2) For drawing clarity, bushing and spade terminal markings have been shown at locations other than above their respective bushings and terminals where the markings should actually appear.

Three Phase Pad-mounted transformer ---- Low voltage greater than 600 V

Table 4—Electrical characteristics and minimum electrical clearances of high-voltage bushings and low-voltage terminals

Transformer voltage	BIL ^{b, c} (kV)	60-Hz dry-one- minute withstand ^{b, c} (kV)	Minimum clearance—live parts ^a		
			Phase to ground ^c	Phase to phase ^c	Phase to non-hygroscopic insulating barrier
208Y/120, 240	30	10	25 (1.0) ^a	25 (1.0)	
480Y/277, 480	30	10	25 (1.0)	25 (1.0)	
600Y/347, 600	30	10	25 (1.0)	25 (1.0)	
2400	45	15	51 (2.0)	51 (2.0)	
4160 through 4800	60	21	64 (2.5)	64 (2.5)	
7200	75	27	89 (3.5)	102 (4.0)	76 (3.0)
12 000 through 13 800	95	35	127 (5.0)	140 (5.5)	76 (3.0)
22 860 Grd Y through 24 940 Grd Y	125	42	146 (5.75)	159 (6.25)	76 (3.0)
34 500 Grd Y ^b	150	70	203 (8.0)	229 (9.0)	102 (4.0)

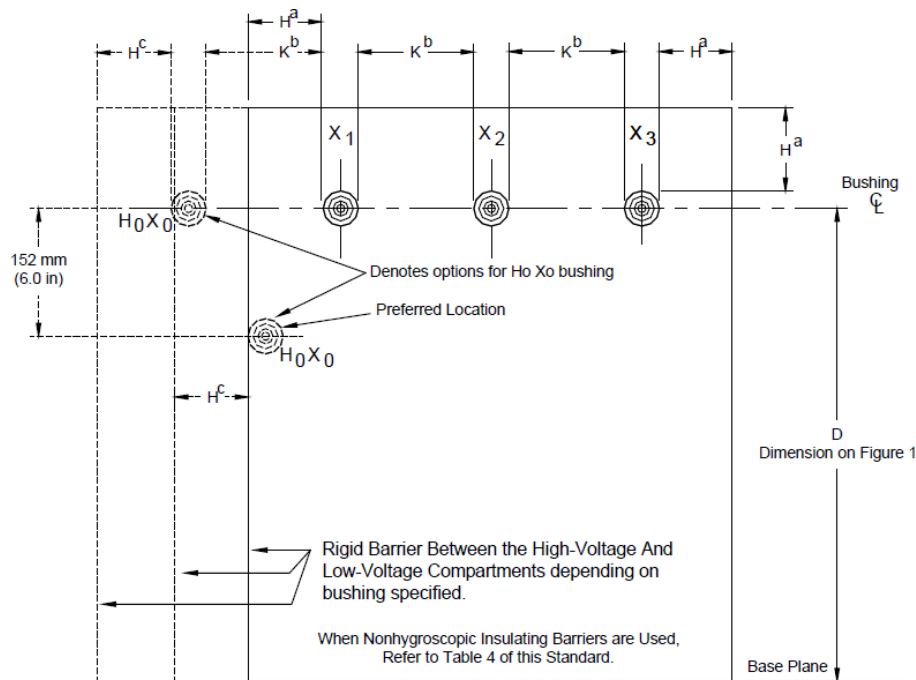
NOTE—These dimensions should be increased wherever possible to allow for ease in making connections by the user.^d

^a All dimensions are in millimeters (inches).

^b The use of barriers shall not reduce these electrical characteristics.

^c Where clearances are less than those shown, an adequate non-hygroscopic insulating barrier shall be provided.

^d Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.



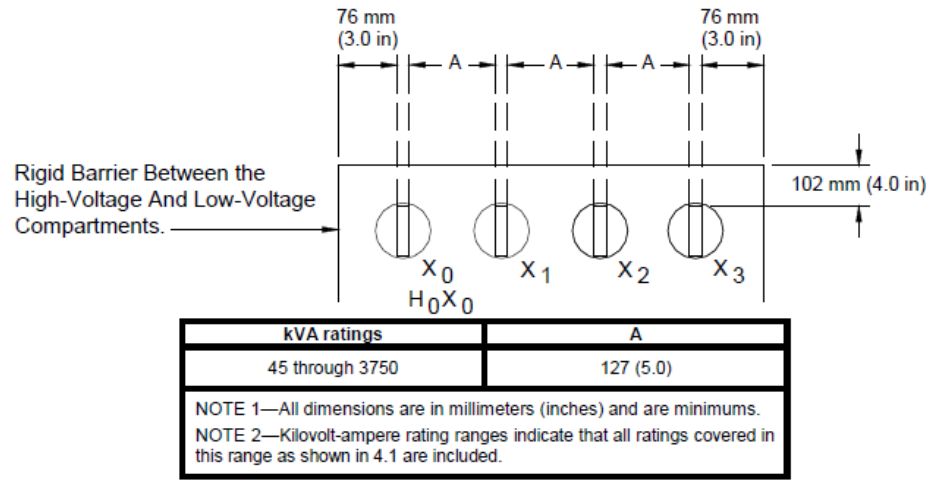
^a H is the minimum phase to ground clearance as specified in Table 4.

^b K is the minimum phase to phase clearance as specified in Table 4.

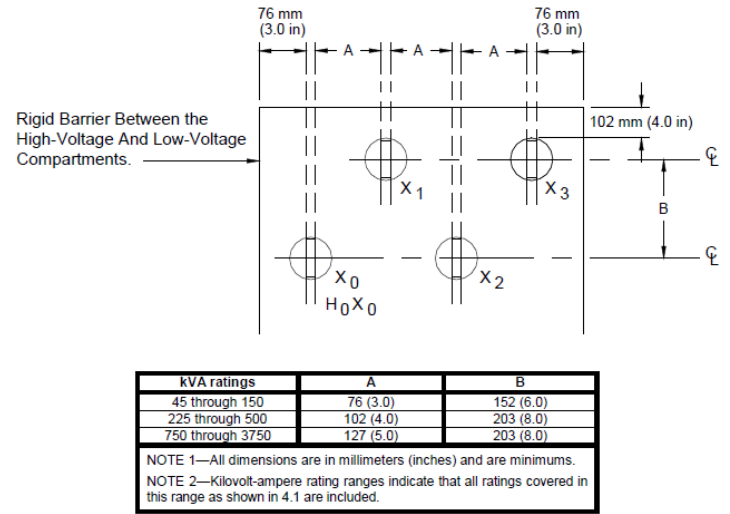
^c If effectively grounded, this dimension will be 76 mm (3.0 in).

Figure 3—Low-voltage terminal arrangements greater than 600 V with minimum dimensions

Three Phase Pad-mounted transformer ----- Low voltage 600 V and below



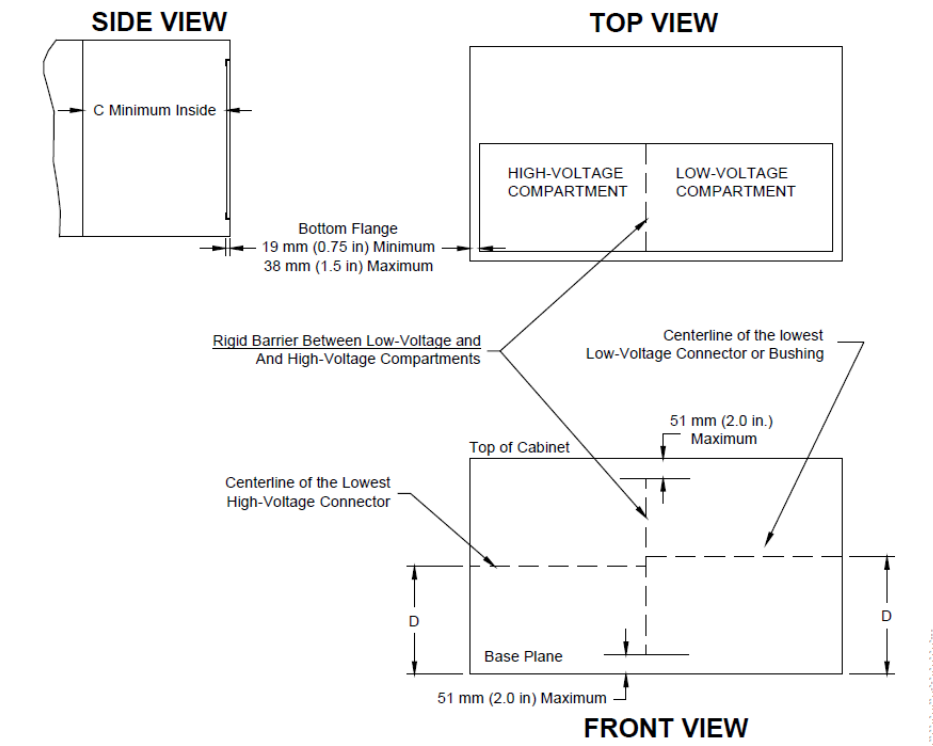
(B) In-line terminal arrangement



(A) Staggered terminal arrangement

Figure 4—600 V and below terminal arrangements with minimum dimensions

Three Phase Pad-mounted transformer ---- radial feed-with high-voltage connectors---C57.12.34



Separable insulated connectors with high-voltage ratings as follows:					
kVA ratings	8.3 kV or 15.2 kV 200 A interface		8.3 kV or 15.2 kV 600 A or 900 A interface		21.1 kV and 21.1 kV/36.6 kV 200 A interface
	D	C	D	C	D C
45 through 500	533 (21.0)	457 (18.0)			533 (21.0) 533 (21.0)
750 through 3750	686 (27.0)	457 (18.0)			686 (27.0) 533 (21.0)
5000 through 10 000			686 (27.0)	610 (24.0)	686 (27.0) 533 (21.0)

NOTE 1—All dimensions are in millimeters (inches) and are minimums.
NOTE 2—Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.

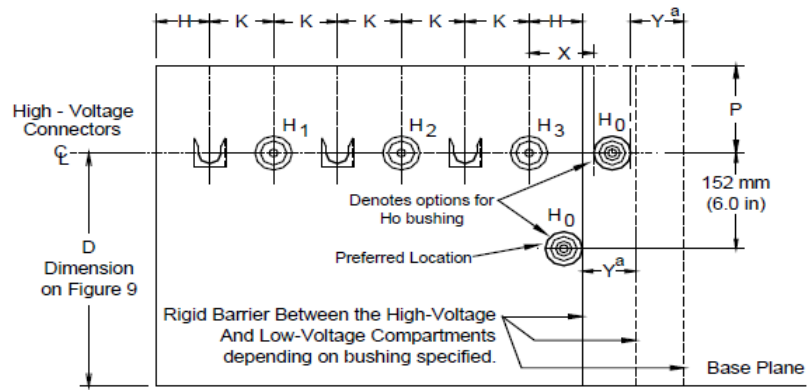
Figure 9—Compartment designations and minimum dimensions for loop-feed or radial-feed transformers with high-voltage connectors

Table 7
Physical sizes of transformers
(See Clause 5.1.1.)

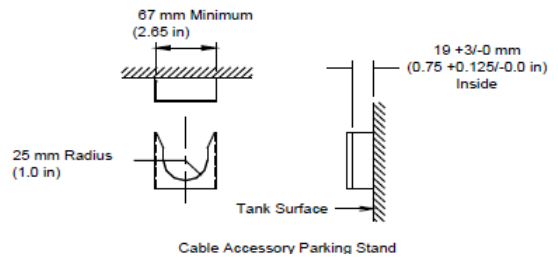
Rated kVA capacity	Type of feed	Transformer maximum BIL, kV	Width, mm	Figure
75, 150	Radial	95	1280*	3
75, 150, 225, 300	Loop	95	1480	4
75, 150, 225, 300, 500	Radial	125	1480	5
75, 150, 225, 300, 500	Loop	125	1480	6
750, 1000, 1500, 2000, 2500, 3000	Radial	125	1730	7
750, 1000, 1500, 2000, 2500, 3000	Loop	125	1730	8

* For radial feed transformers rated at 75, 150 kVA, a width of 1480 mm should be allowed if agreed by the purchaser.

Three Phase Pad-mounted transformer ---- radial feed-with high-voltage connectors---C57.12.34



^a If effectively grounded, this dimension will be 76 mm (3.0 in).



Cable Accessory Parking Stand

Cable Accessory Parking Stand									
Separable insulated connectors with high-voltage ratings as follows:									
kVA ratings	8.3 kV and 8.3 kV/14.4 kV			15.2 kV and 15.2 kV/26.3 kV			21.1 kV and 21.1 kV/36.6 kV ^a		
	H	K	P	H	K	P	H	K	P
45 through 150 without balls	58 (2.3)	102 (4.0)	114 (4.5)	74 (2.9)	102 (4.0)	114 (4.5)	102 (4.0)	152 (6.0)	191 (7.5)
45 through 150 with balls	58 (2.3)	102 (4.0)	279 (11.0)	74 (2.9)	102 (4.0)	356 (14.0)	102 (4.0)	152 (6.0)	432 (17.0)
225 through 500 without balls	76 (3.0)	102 (4.0)	114 (4.5)	76 (3.0)	102 (4.0)	114 (4.5)	102 (4.0)	152 (6.0)	191 (7.5)
225 through 500 with balls	76 (3.0)	102 (4.0)	279 (11.0)	76 (3.0)	102 (4.0)	356 (14.0)	102 (4.0)	152 (6.0)	432 (17.0)
750 through 10 000 without balls	81 (3.2)	102 (4.0)	114 (4.5)	81 (3.2)	102 (4.0)	114 (4.5)	102 (4.0)	152 (6.0)	191 (7.5)
750 through 10 000 with balls	81 (3.2)	102 (4.0)	279 (11.0)	81 (3.2)	102 (4.0)	356 (14.0)	102 (4.0)	152 (6.0)	432 (17.0)
X is the minimum phase to phase clearance as specified in Table 4.									
Y is the minimum phase to phase clearance as specified in Table 4.									
NOTE 1—All dimensions are in millimeters (inches) and are minimums.									
NOTE 2—Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.									
NOTE 3—The above minimum dimensions do not provide for all types of parking-stand mounted devices.									

^a The configuration of the 21.1 kV/36.6 kV connectors is based on IEEE Std 386.

Figure 10—Minimum dimensions for radial-feed transformers with high-voltage connectors

Separable insulated connectors with high-voltage ratings as follows:					
kVA ratings	8.3 kV or 15.2 kV 200 A interface		8.3 kV or 15.2 kV 600 A or 900 A interface		21.1 kV and 21.1 kV/36.6 kV 200 A interface
	D	C	D	C	D
45 through 500	533 (21.0)	457 (18.0)			533 (21.0)
750 through 3750	686 (27.0)	457 (18.0)			533 (21.0)
5000 through 10 000			686 (27.0)	610 (24.0)	533 (21.0)
NOTE 1—All dimensions are in millimeters (inches) and are minimums.					
NOTE 2—Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.					

Figure 9—Compartment designations and minimum dimensions for loop-feed or radial-feed transformers with high-voltage connectors

Three Phase Pad-mounted transformer ---- radial feed-with high-voltage connectors---C227.4

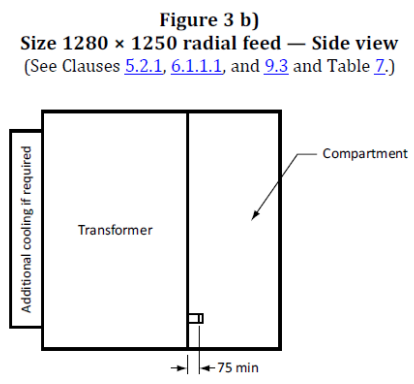
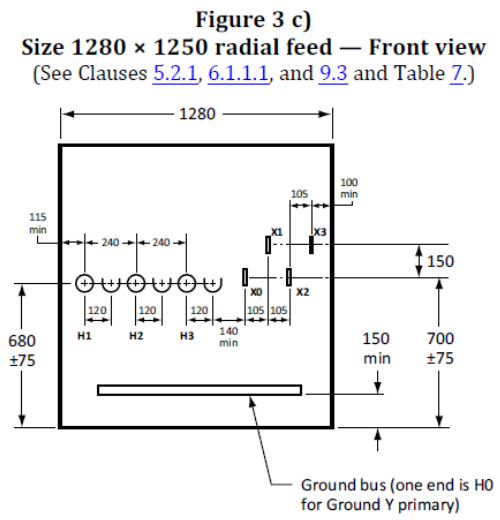
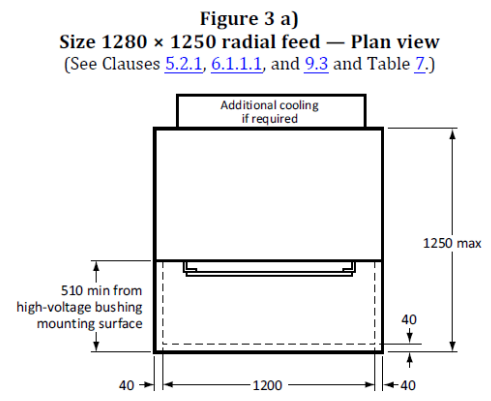


Table 7
Physical sizes of transformers
(See Clause 5.1.1.)

Rated kVA capacity	Type of feed	Transformer maximum BIL, kV	Width, mm	Figure
75, 150	Radial	95	1280*	3
75, 150, 225, 300	Loop	95	1480	4
75, 150, 225, 300, 500	Radial	125	1480	5
75, 150, 225, 300, 500	Loop	125	1480	6
750, 1000, 1500, 2000, 2500, 3000	Radial	125	1730	7
750, 1000, 1500, 2000, 2500, 3000	Loop	125	1730	8

* For radial feed transformers rated at 75, 150 kVA, a width of 1480 mm should be allowed if agreed by the purchaser.

Three Phase Pad-mounted transformer ---- radial feed-with high-voltage connectors ---C227.4

Figure 3 a)

Size 1280 × 1250 radial feed — Plan view
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)

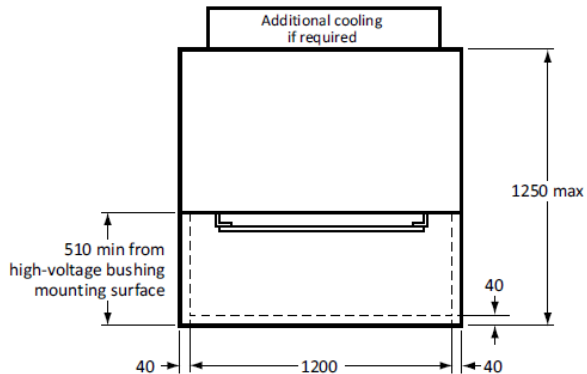


Figure 3 c)

Size 1280 × 1250 radial feed — Front view
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)

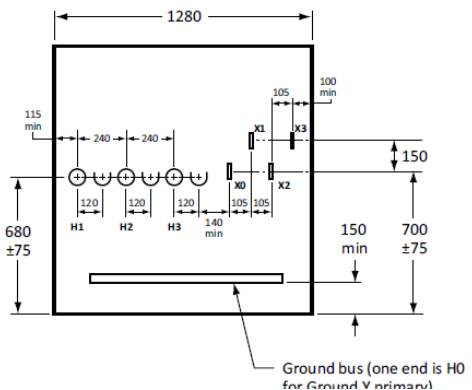


Figure 3 b)

Size 1280 × 1250 radial feed — Side view
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)

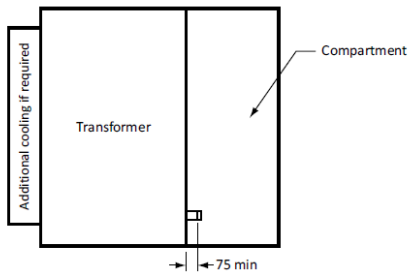


Table 7
Physical sizes of transformers
(See Clause 5.1.1.)

Rated kVA capacity	Type of feed	Transformer maximum BIL, kV	Width, mm	Figure
75, 150	Radial	95	1280*	3
75, 150, 225, 300	Loop	95	1480	4
75, 150, 225, 300, 500	Radial	125	1480	5
75, 150, 225, 300, 500	Loop	125	1480	6
750, 1000, 1500, 2000, 2500, 3000	Radial	125	1730	7
750, 1000, 1500, 2000, 2500, 3000	Loop	125	1730	8

* For radial feed transformers rated at 75, 150 kVA, a width of 1480 mm should be allowed if agreed by the purchaser.

Three Phase Pad-mounted transformer ---- radial feed-with high-voltage connectors ---C227.4

Figure 5 a)

Size 1480 × 1370 radial feed — Plan view
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)

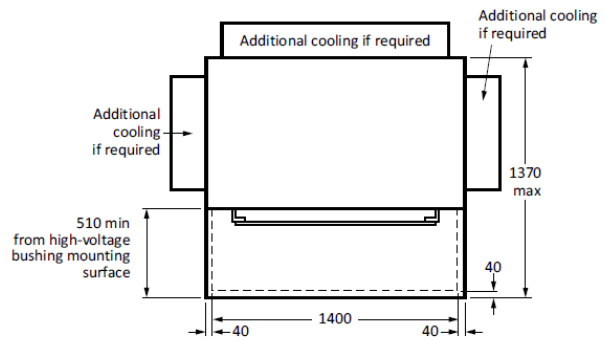


Figure 5 b)

Size 1480 × 1370 radial feed — Side view
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)

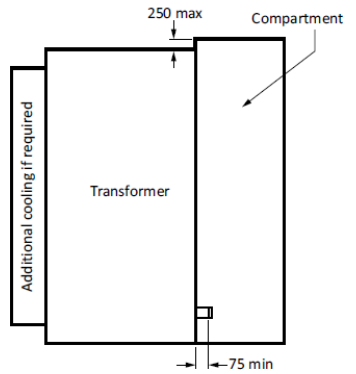


Figure 5 c)

Size 1480 × 1370 radial feed — Front view
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)

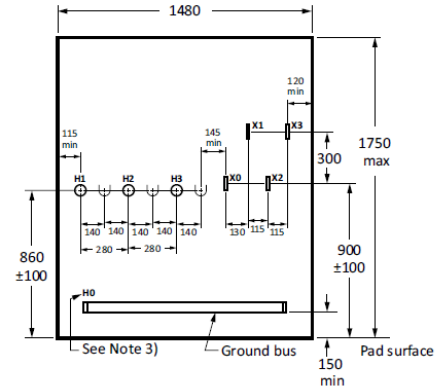


Table 7
Physical sizes of transformers
(See Clause 5.1.1.)

Rated kVA capacity	Type of feed	Transformer maximum BIL, kV	Width, mm	Figure
75, 150	Radial	95	1280*	3
75, 150, 225, 300	Loop	95	1480	4
75, 150, 225, 300, 500	Radial	125	1480	5
75, 150, 225, 300, 500	Loop	125	1480	6
750, 1000, 1500, 2000, 2500, 3000	Radial	125	1730	7
750, 1000, 1500, 2000, 2500, 3000	Loop	125	1730	8

* For radial feed transformers rated at 75, 150 kVA, a width of 1480 mm should be allowed if agreed by the purchaser.

Three Phase Pad-mounted transformer ---- radial feed-with high-voltage connectors-- C227.4

Figure 7 a)

Size 1730 × 1575 radial feed — Plan view
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)

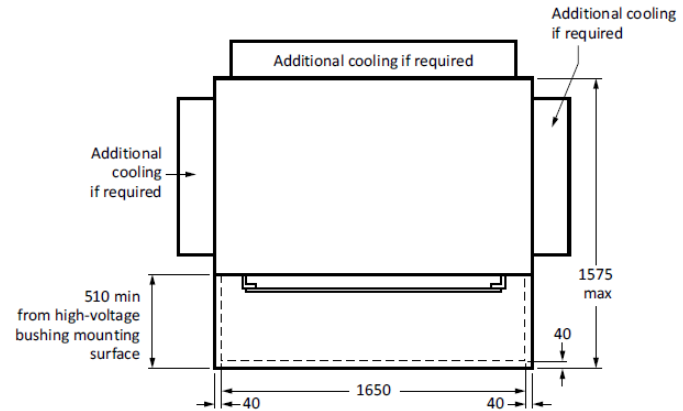


Figure 7 b)

Size 1730 × 1575 radial feed — Side view
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)

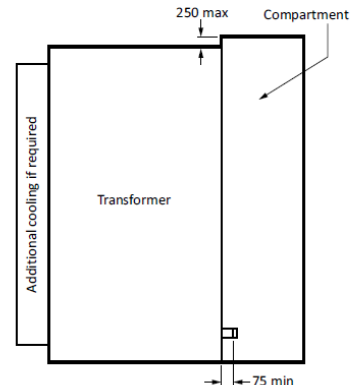


Figure 7 c)

Size 1730 × 1575 radial feed — Front view
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)

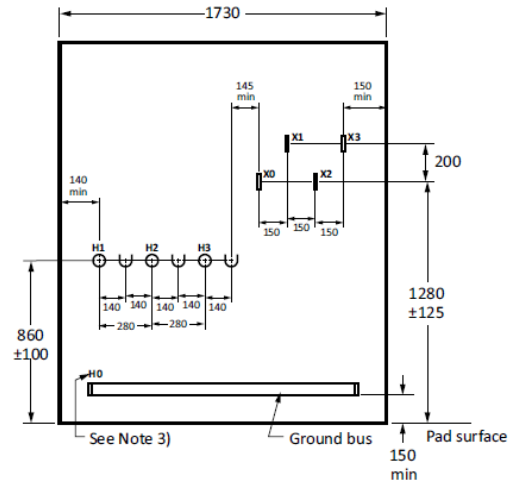
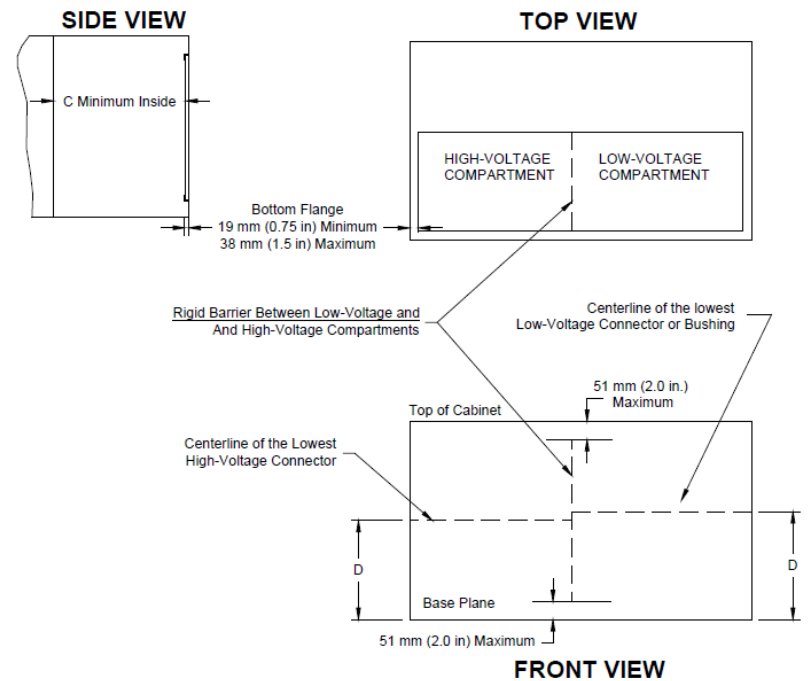


Table 7
Physical sizes of transformers
(See Clause 5.1.1.)

Rated kVA capacity	Type of feed	Transformer maximum BIL, kV	Width, mm	Figure
75, 150	Radial	95	1280*	3
75, 150, 225, 300	Loop	95	1480	4
75, 150, 225, 300, 500	Radial	125	1480	5
75, 150, 225, 300, 500	Loop	125	1480	6
750, 1000, 1500, 2000, 2500, 3000	Radial	125	1730	7
750, 1000, 1500, 2000, 2500, 3000	Loop	125	1730	8

* For radial feed transformers rated at 75, 150 kVA, a width of 1480 mm should be allowed if agreed by the purchaser.

Three Phase Pad-mounted transformer ---- loop-feed with high-voltage connectors---C57.12.34



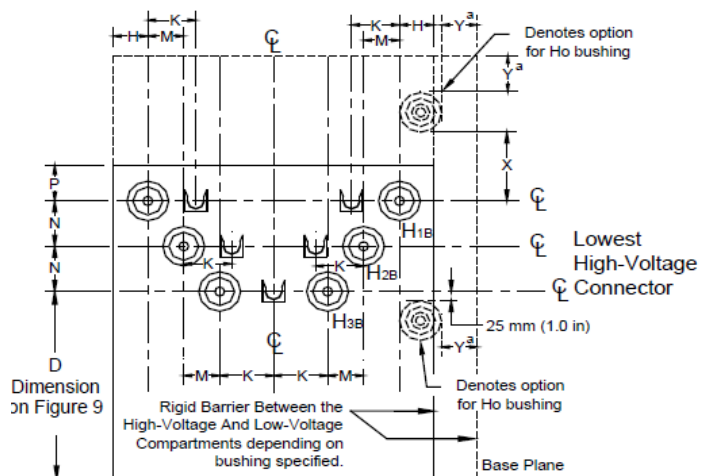
Separable insulated connectors with high-voltage ratings as follows:						
kVA ratings	8.3 kV or 15.2 kV 200 A interface		8.3 kV or 15.2 kV 600 A or 900 A interface		21.1 kV and 21.1 kV/36.6 kV 200 A interface	
	D	C	D	C	D	C
45 through 500	533 (21.0)	457 (18.0)			533 (21.0)	533 (21.0)
750 through 3750	686 (27.0)	457 (18.0)			686 (27.0)	533 (21.0)
5000 through 10 000			686 (27.0)	610 (24.0)	686 (27.0)	533 (21.0)

NOTE 1—All dimensions are in millimeters (inches) and are minimums.

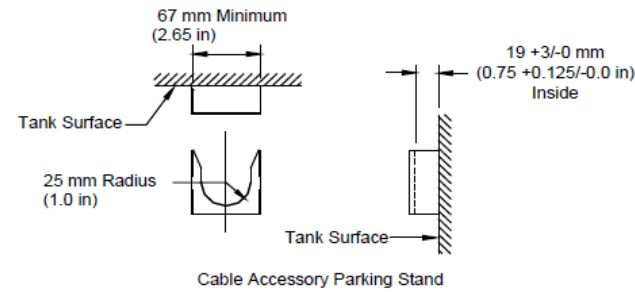
NOTE 2—Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.

Figure 9—Compartment designations and minimum dimensions for loop-feed or radial-feed transformers with high-voltage connectors

Three Phase Pad-mounted transformer ---- loop-feed with high-voltage connectors---IEE C57.12.34



a If effectively grounded, this dimension will be 76 mm (3.0 in).



Separable insulated connectors with high-voltage ratings as follows:															
kVA ratings	8.3 kV and 8.3 kV/14.4 kV					15.2 kV and 15.2kV/26.3 kV					21.1 kV and 21.1 kV/36.6 kV ^a				
	H	N	M	K	P	H	N	M	K	P	H	N	M	K	P
45 through 10 000 without bails	76 (3.0)	102 (4.0)	76 (3.0)	102 (4.0)	114 (4.5)	76 (3.0)	102 (4.0)	89 (3.5)	114 (4.5)	114 (4.5)	102 (4.0)	102 (4.0)	95 (3.75)	152 (6.0)	191 (7.5)
45 through 10 000 with bails	76 (3.0)	102 (4.0)	76 (3.0)	102 (4.0)	279 (11.0)	76 (3.0)	102 (4.0)	89 (3.5)	114 (4.5)	356 (14.0)	102 (4.0)	102 (4.0)	95 (3.75)	152 (6.0)	432 (17.0)
X is the minimum phase to phase clearance as specified in Table 4.															
Y is the minimum phase to phase clearance as specified in Table 4.															
NOTE 1—All dimensions are in millimeters (inches) and are minimums.															
NOTE 2—Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.															
NOTE 3—The above minimum dimensions do not provide for all types of parking-stand-mounted devices.															

^a The configuration of the 21.1 kV/36.6 kV connectors is based on IEEE Std 386.

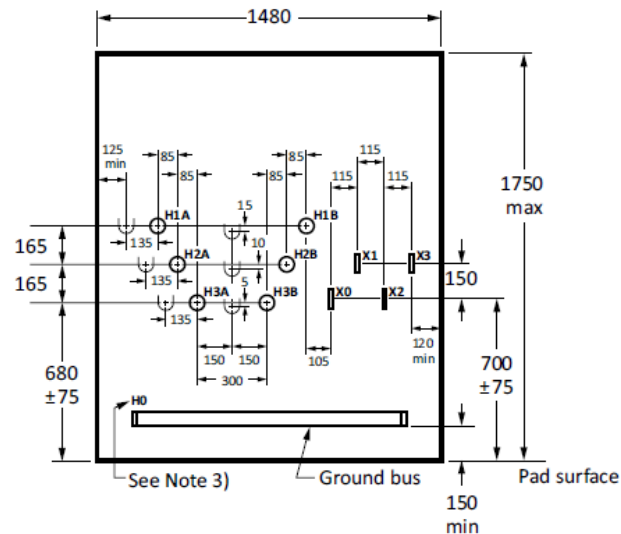
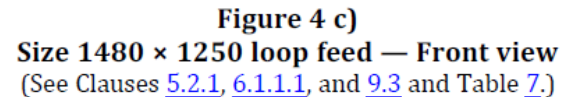
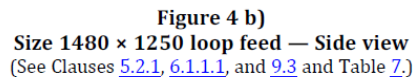
Figure 11—Minimum dimensions for loop-feed transformers with high-voltage connectors

Separable insulated connectors with high-voltage ratings as follows:						
kVA ratings	8.3 kV or 15.2 kV 200 A interface		8.3 kV or 15.2 kV 600 A or 900 A interface		21.1 kV and 21.1 kV/36.6 kV 200 A interface	
	D	C	D	C	D	C
45 through 500	533 (21.0)	457 (18.0)			533 (21.0)	533 (21.0)
750 through 3750	686 (27.0)	457 (18.0)			686 (27.0)	533 (21.0)
5000 through 10 000			686 (27.0)	610 (24.0)	686 (27.0)	533 (21.0)

NOTE 1—All dimensions are in millimeters (inches) and are minimums.
NOTE 2—Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.

Figure 9—Compartment designations and minimum dimensions for loop-feed or radial-feed transformers with high-voltage connectors

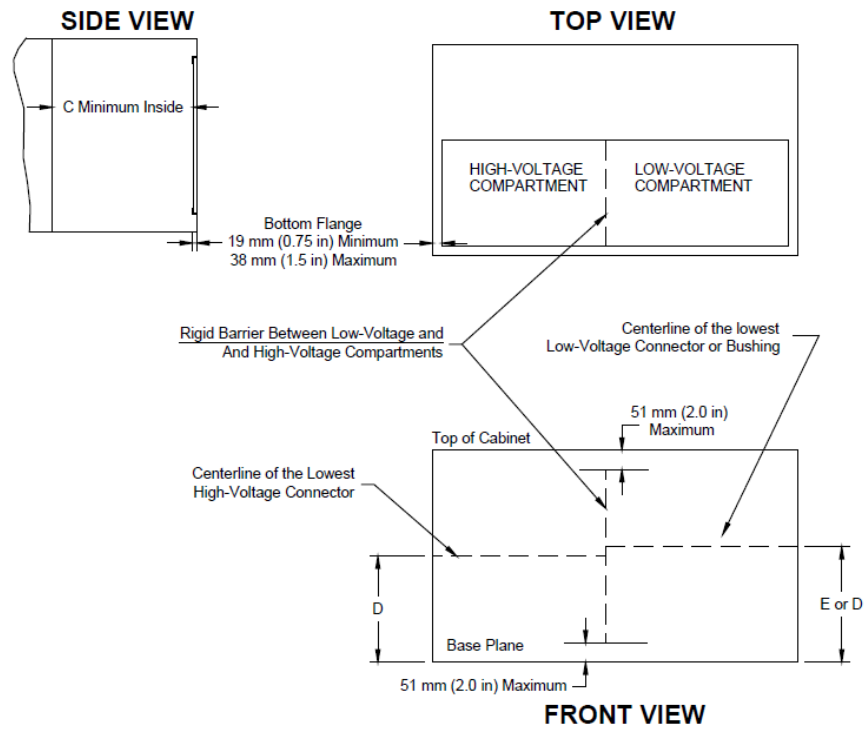
(See Clauses 5.2.1, 6.1.1.1, and 9.3 and Table 7.)



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Figure 1: Typical layout of a 1000A switchgear. The diagram shows a top-down view of a switchgear enclosure with dimensions in millimeters. The overall width is 1730 mm and the height is 1280 mm ± 125 mm. Internal components include circuit breakers (H0A, H0B, H0C, H0D, H0E, H0F, H0G, H0H, H0I, H0J, H0K, H0L, H0M, H0N, H0O, H0P, H0Q, H0R, H0S, H0T, H0U, H0V, H0W, H0X, H0Y, H0Z) and busbars (X0, X1, X2, X3). Dimensions for clearances and spacings are provided for various components.

Three Phase Pad-mounted transformer ---- radial-feed with high-voltage connectors

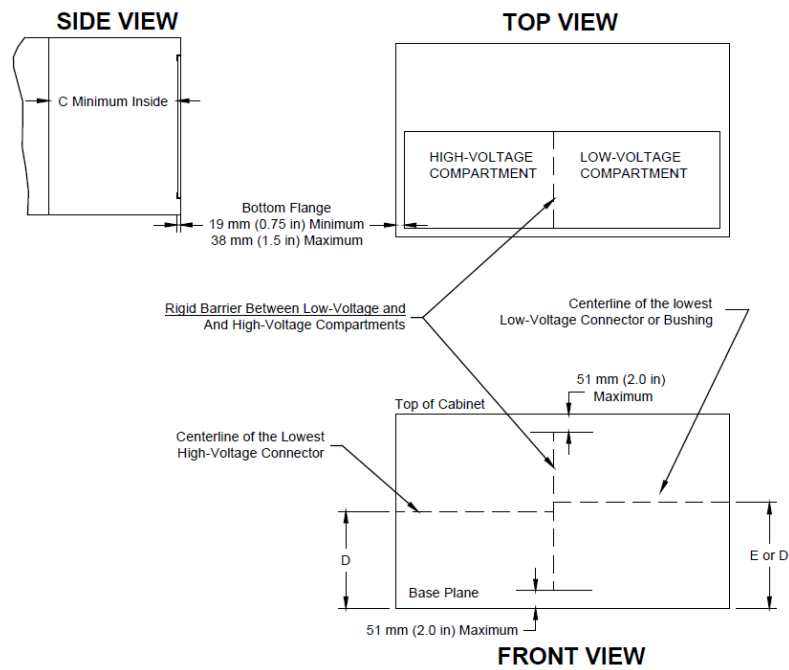


kVA ratings	C			D			E
	95 kV BIL and less	125 kV BIL	150 kV BIL	95 kV BIL and less	125 kV BIL	150 kV BIL	30 kV BIL
45 through 500	406 (16.0)	457 (18.0)	483 (19.0)	610 (24.0)	610 (24.0)	610 (24.0)	508 (20.0)
750 through 2500	406 (16.0)	457 (18.0)	533 (21.0)	914 (36.0)	914 (36.0)	914 (36.0)	686 (27.0)
3750 through 10 000	406 (16.0)	457 (18.0)	533 (21.0)	1219 (48.0)	1219 (48.0)	1219 (48.0)	864 (34.0)

NOTE 1—All dimensions are in millimeters (inches) and are minimums unless otherwise indicated.
NOTE 2—The BIL is shown for the transformer's high-voltage rating, refer to Table 3.
NOTE 3—Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.

Figure 1—Compartment designations and minimum dimensions for radial-feed transformers with high-voltage bushings

Three Phase Pad-mounted transformer ---- radial-feed with high-voltage bushing---IEEE C57.12.34

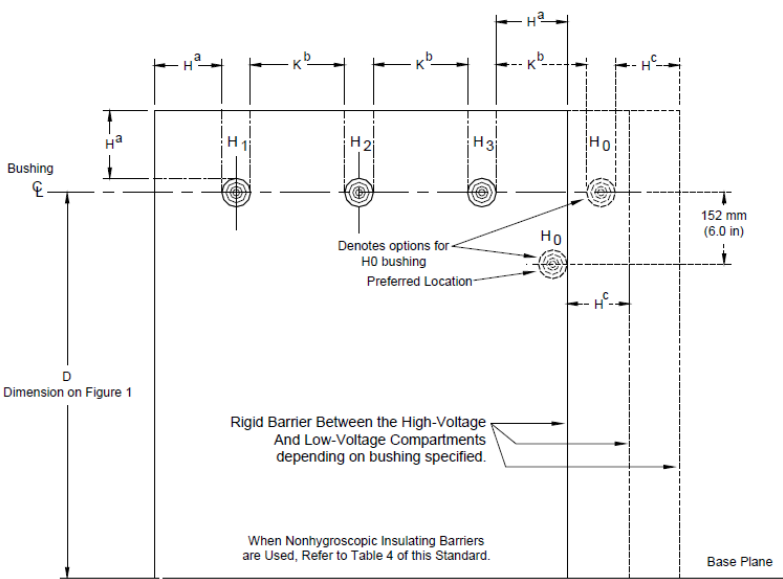


kVA ratings	C			D			E
	95 kV BIL and less	125 kV BIL	150 kV BIL	95 kV BIL and less	125 kV BIL	150 kV BIL	30 kV BIL
45 through 500	406 (16.0)	457 (18.0)	483 (19.0)	610 (24.0)	610 (24.0)	610 (24.0)	508 (20.0)
750 through 2500	406 (16.0)	457 (18.0)	533 (21.0)	914 (36.0)	914 (36.0)	914 (36.0)	686 (27.0)
3750 through 10 000	406 (16.0)	457 (18.0)	533 (21.0)	1219 (48.0)	1219 (48.0)	1219 (48.0)	864 (34.0)

NOTE 1—All dimensions are in millimeters (inches) and are minimums unless otherwise indicated.
NOTE 2—The BIL is shown for the transformer's high-voltage rating, refer to Table 3.
NOTE 3—Kilovolt-ampere rating ranges indicate that all ratings covered in this range as shown in 4.1 are included.

Figure 1—Compartment designations and minimum dimensions for radial-feed transformers with high-voltage bushings

Three Phase Pad-mounted transformer ---- radial-feed with high-voltage bushing--- IEEE C57.12.34



^a H is the minimum phase to ground clearance as specified in Table 4.
^b K is the minimum phase to phase clearance as specified in Table 4.
^c If effectively grounded, this dimension will be 76 mm (3.0 in).

Figure 2—Minimum dimensions for radial-feed transformers with high-voltage bushings

Table 4—Electrical characteristics and minimum electrical clearances of high-voltage bushings and low-voltage terminals

Transformer voltage	BIL ^{b, c} (kV)	60-Hz dry-one- minute withstand ^{b, c} (kV)	Minimum clearance—live parts ^a		
			Phase to ground ^c	Phase to phase ^c	Phase to non-hygroscopic insulating barrier
208Y/120, 240	30	10	25 (1.0) ^a	25 (1.0)	
480Y/277, 480	30	10	25 (1.0)	25 (1.0)	
600Y/347, 600	30	10	25 (1.0)	25 (1.0)	
2400	45	15	51 (2.0)	51 (2.0)	
4160 through 4800	60	21	64 (2.5)	64 (2.5)	
7200	75	27	89 (3.5)	102 (4.0)	76 (3.0)
12 000 through 13 800	95	35	127 (5.0)	140 (5.5)	76 (3.0)
22 860 Grd Y through 24 940 Grd Y	125	42	146 (5.75)	159 (6.25)	76 (3.0)
34 500 Grd Y ^b	150	70	203 (8.0)	229 (9.0)	102 (4.0)

NOTE—These dimensions should be increased wherever possible to allow for ease in making connections by the user.^d

^a All dimensions are in millimeters (inches).
^b The use of barriers shall not reduce these electrical characteristics.
^c Where clearances are less than those shown, an adequate non-hygroscopic insulating barrier shall be provided.
^d Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

Three Phase Pad-mounted transformer ---- Hinges

Locking bolts and associated threaded receptacles, hinges, and hinge pins shall be AISI6 type 304 stainless steel or material of equivalent corrosion resistance.---IEEE C57.12.34 refer to IEEE C57.12.28

5.2.3 Hinges ----CSA C227.4

Hinge assemblies and pins shall be 304 or 304 L stainless steel or material of equivalent corrosion resistance. They shall be of sufficient strength to withstand adverse operating conditions.

5.2.5 Hinges---CSA C227.5

Hinge assemblies and pins shall meet the requirements of IEEE C57.12.28, be made of corrosion-resistant material, and be of sufficient strength to withstand adverse operating conditions.

Three Phase Pad-mounted transformer ---- Grounding

8.11 Grounding provisions---IEEE C57.12.34

8.11.1 500 kVA and below

Two steel pads, each with a ½-inch 13 UNC tapped hole and a minimum thread depth of 11 mm (0.44 in) shall be provided.

8.11.2 Above 500 kVA

Two unpainted, copper-faced steel or stainless steel pads, 51 × 89 mm (2.0 × 3.5 in) each with two holes spaced on 44 mm (1.75 in) centers and tapped for ½-inch 13 UNC thread shall be provided. The minimum thickness of the copper facing shall be 0.5 mm (0.02 in). Minimum thread depth of the holes shall be 13 mm (0.5 in).

8.11.3 Location

The ground pads described in 8.11.1 and 8.11.2 shall be welded on or near the transformer base, one in the high voltage compartment and one in the low-voltage compartment. In cases where the transformer tank and compartments are separate, these pads shall be electrically bonded.

6.2.1 Grounding spade terminals---CSA 227.4

Two vertically mounted grounding spade terminals shall be welded to the tank wall at a minimum height of 150 mm, and at or above the level of the sill referred to in Clause 5.2.4; one shall be on the high-voltage side and one on the low-voltage side. For GrdY transformers, the terminal on the high voltage side shall be designated and marked H0. Both terminals shall be unpainted, fabricated from stainless steel, with a minimum width of 40 mm and a minimum thickness of 6 mm, and provided with two 14 mm (9/16 in) holes, spaced 44.5 mm apart.

The grounding assembly shall consist of a rigid copper bus bolted to and joining the two grounding spade terminals specified in Clause 6.2.1, as shown in Figure 12. There shall be at least one hole per high-voltage bushing, located below each bushing well and 45 mm from the grounding bracket mounting hardware, and four holes on the low-voltage side. All holes shall be 14 mm (9/16 in) in diameter. A grounding bracket as shown in Figure 12 shall be bolted to the ground bus

to facilitate attachment of working ground clamps. The neutral terminal X0 shall be connected to the ground bus by a conductor of suitable ampacity.

6.4.1 Grounding spade terminals---CSA 227.5

Two grounding spade terminals shall be welded to the tank wall above the level of the sill. One shall be located on the high-voltage side and the other on the low-voltage side. For GrdY transformers, one terminal shall be designated and marked H0. Terminals shall

- (a) be unpainted;
- (b) be fabricated from stainless steel;
- (c) be at least 40 mm wide and 6 mm thick; and
- (d) have two 14.3 mm holes spaced on 44.5 mm centres.

Three Phase Pad-mounted transformer ---- Grounding 227.5

6.4.2 Grounding assembly ---227.5

The grounding assembly shall consist of a rigid copper bus that is at least 40 mm wide and 6 mm thick and joins the high-voltage and low-voltage compartments. It shall be mounted at the front of the compartments, 25 mm above the sill, and connected to the two grounding spade terminals specified in [Clause 6.4.1](#). The bus shall have at least six holes on the high-voltage side aligned approximately vertically below the high-voltage bushings, and four holes on the low-voltage side. Holes shall be 14.3 mm in diameter.

6.4.3 Neutral terminal

If the low-voltage winding is connected Y or GrdY, the neutral terminal X0 shall be connected to the grounding bus with a removable copper conductor of suitable ampacity.

6.4.4 Bonding of bolted tank cover

The transformer cover, when bolted, shall

- (a) be electrically bonded to the tank with a copper strap using copper or bronze hardware of a minimum size of 1/4 in; and
- (b) not be accessible to the public.

Three Phase Pad-mounted transformer ---- **Grounding**

Ground bus installation — Loop feed
(See Clause 6.2.2.)

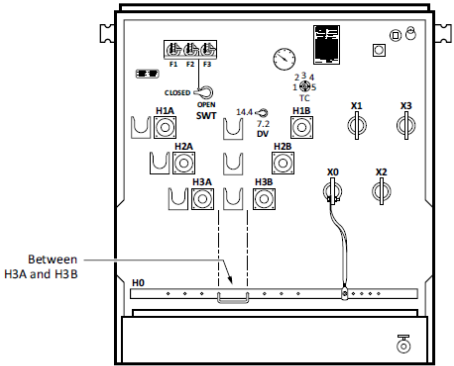
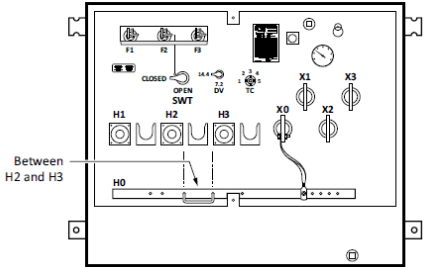


Figure 12 b)
Ground bus installation — Radial feed
(See Clause 6.2.2.)

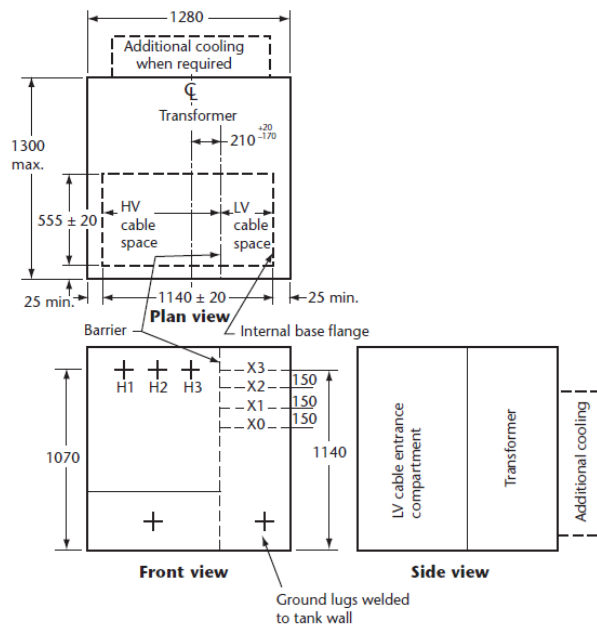


Three Phase Pad-mounted transformer ---- radial-feed with high-voltage bushing---C227.5

Table 8
Transformer dimensions
(See [Clause 5.1.1.](#))

Rated kV•A capacity	Type of feed	Transformer maximum BIL, kV	Width, mm	Figure
75, 150	Radial	95	1030	1
75, 150, 225, 300	Loop	95	1280	2
75, 150, 225, 300, 500	Radial	125	1480	3
75, 150, 225, 300, 500	Loop	125	1480	3
750, 1000, 1500, 2000, 2500, 3000	Radial	125	1730	4
750, 1000, 1500, 2000, 2500, 3000	Loop	125	1730, 1830	5

Three Phase Pad-mounted transformer ---- radial-feed with high-voltage bushing---C227.5

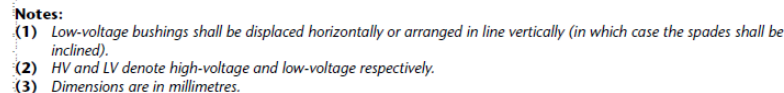


Notes:

- (1) Low-voltage bushings shall be displaced horizontally or arranged in line vertically (in which case the spades shall be inclined).
- (2) Smaller sizes may be used where equipment permits.
- (3) HV and LV denote high-voltage and low-voltage respectively.
- (4) Dimensions are in millimetres.

Figure 2
Transformer dimensions (II)

(See Clauses 5.1.1, 5.2.1, 5.2.8, 6.1.1, and 9.3, Table 8, and Figures 4 and 5.)



(See [Clauses 5.1.1, 5.2.1, 5.2.8, 6.1.1, and 9.3](#), [Table 8](#), and [Figures 4 and 5](#).)

Type of feed	Terminal connection height H , mm
Radial feed, direct connected	1420
Radial feed with disconnects	1420
Loop feed with disconnects	
5.0 and 8.7 kV	800
15 kV	550
Radial feed, direct connected with termination(s)	—

Notes:

(1) Low-voltage bushings shall be displaced horizontally or arranged in line vertically (in which case the spades shall be inclined).

(2) HV and LV denote high-voltage and low-voltage respectively.

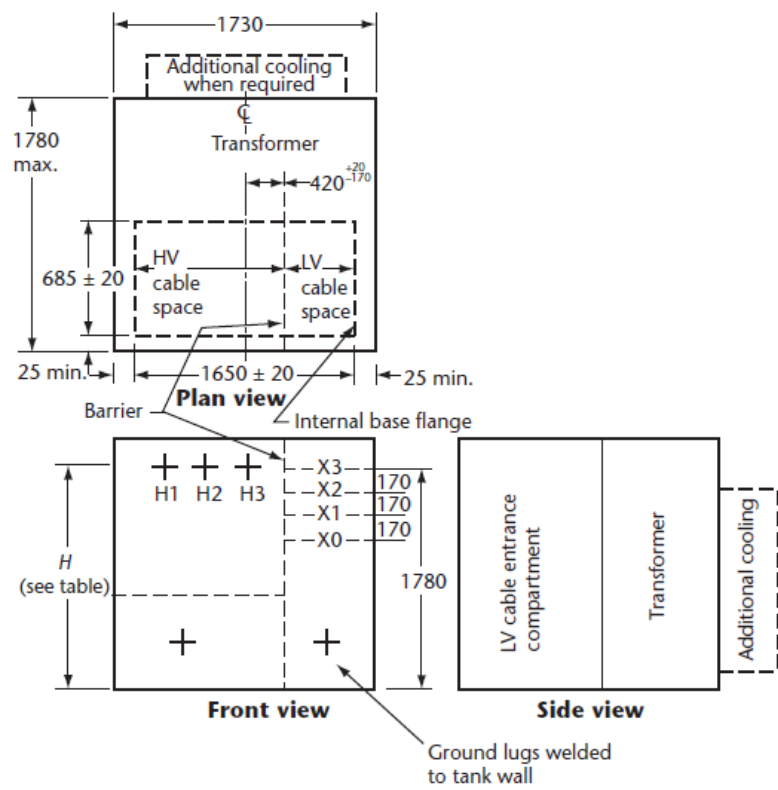
(3) Dimensions are in millimetres.

Figure 3

Transformer dimensions (III)

(See [Clauses 5.1.1, 5.2.1, 5.2.8, 6.1.1, and 9.3](#), [Table 8](#), and [Figures 4 and 5](#).)

Three Phase Pad-mounted transformer ---- radial-feed with high-voltage bushing---C227.5

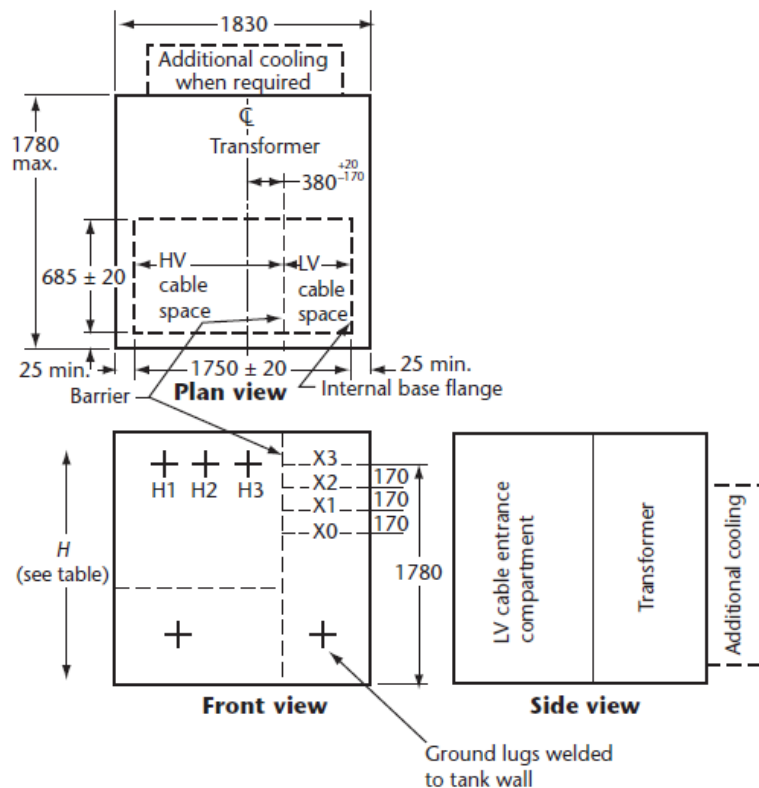


Type of feed	Terminal connector height H, mm
Radial feed, direct connected	1780
Radial feed with disconnects	1780
Loop feed with disconnects	
5.0 or 8.7 kV	800
15 kV	550
18 or 25 kV	990
Radial feed, direct connected with termination(s)	—

- Notes:**
- (1) Low-voltage bushings shall be displaced horizontally or arranged in line vertically (in which case the spades shall be inclined).
 - (2) The values in Figures 2 and 3 may be used to determine high-voltage and low-voltage cable connector heights where equipment permits.
 - (3) HV and LV denote high-voltage and low-voltage respectively.
 - (4) Dimensions are in millimetres.

Figure 4
Transformer dimensions (IV)
(See Clauses 5.1.1, 5.2.1, 5.2.8, 6.1.1, and 9.3 and Table 8.)

Three Phase Pad-mounted transformer ---- radial-feed with high-voltage bushing---C227.5



Type of feed	Terminal connector height H , mm
Radial feed, direct connected	1780
Radial feed with disconnects	1780
Loop feed with disconnects	
5.0 or 8.7 kV	800
15 kV	550
18 or 25 kV	990
Radial feed, direct connected with termination(s)	—

Three Phase Pad-mounted transformer ---- Low-voltage bushings

8.7.3 600 V and below terminals

8.7.3.1 General

The phase and neutral terminals shall be in accordance with Figure 4 or Figure 8 of this standard and arranged for vertical take-off. Terminal dimensions shall be as shown in Figure 19 of this standard or, if specified, Figure 20 of this standard.

8.7.3.2 Support

Terminals equipped with ten or more hole spades shall be furnished with additional support. A manufacturer-designed support shall be attached at the end that is the farthest from the tank wall and mounted in such a way that it will not interfere with the use of any of the holes of the spade or any spade extenders/connectors that the user may want to install.

6.1.3.1----C227.4 ---6.1.3 Low-voltage bushings

Low-voltage terminals for all sizes of transformers shall be the spade type in accordance with Table 8 and Figure 11 and shall have a thickness that will provide adequate ampacity. Terminals shall be of plated corrosion-resistant metal, suitable for use with copper or aluminum connectors. Low-voltage terminals for stud-type bushings shall be of plated copper suitable for use with copper or aluminum connectors.

6.1.3.2

The low-voltage neutral shall be connected to an insulated terminal, designated X0, with a current-carrying capacity equal to the other low-voltage terminals.

6.1.3.3

The low-voltage terminals shall be capable of withstanding a minimum vertical cantilever force of 750 N continuously applied at the outermost hole of the terminal without causing a leak, damage, or permanent distortion.

6.3 Low-voltage bushings ----C227.5 ---6.1.3 Low-voltage bushings

6.3.1

Low-voltage terminals shall be the spade type as specified in Table 9 and shown in Figure 7, with a thickness that will provide adequate ampacity. Terminals shall be made of plated corrosion-resistant metal suitable for use with copper or aluminum connectors. Low-voltage terminals for stud-type bushings shall be made of plated copper suitable for use with copper or aluminum connectors.

6.3.2

The low-voltage neutral, if supplied, shall be connected to an insulated neutral terminal, designated X0, having a current-carrying capacity equal to the other low-voltage terminals.

6.3.3

The low-voltage terminals shall be capable of withstanding a minimum vertical cantilever force of 750 N continuously applied at the outermost hole of the terminal without leakage, damage, or permanent distortion.

Three Phase Pad-mounted transformer ---- Nameplate(1)

CSA: The lettering or numbering required in Items c), d), e), i), k), and l) shall be 4 mm high and shall be legible from outside the cable entrance compartment.

IEEE C57.12.00	CSA C227.4<=3000KVA	C88.16 >3000KVA
Serial number	(d) identification number (e.g., serial number);	manufacturer's serial number
Month/year of manufacture	(m) year and month of manufacture;	year of manufacture;
Class (ONAN, ONAF, etc.) b	(a) type (ONAN, LNaN, or KNAN) 65 °C;	type designation of cooling
Number of phases	(b) number of phases;	number of phases;
Frequency	60 Hz,	frequency;
kVA (or MVA) rating a, b	(c) rating in kilovolt amperes;	rating in MVA at usual service conditions
Voltage ratings a, c	(i) rated high and low voltages;	
Tap voltages d	(k) tap voltages in per cent of rated voltage and corresponding position of tap switch;	rated voltages and tap designations
Temperature rise, °C	----	temperature rise in degrees
Polarity (single-phase transformers)	(l) terminal markings, including physical identification and diagram of connections, and vector diagram for three-phase transformers;	terminal arrangement (plan view) with a mechanical
Percent impedance e	(e) per cent impedance at 85 °C (one decimal minimum);	impedance voltages expressed as percentages (measured values)
Basic lightning impulse insulation levels (BIL)	(g) BIL (basic impulse level);	lightning—impulse levels (full wave in kV)
Approximate total mass in kg or weight in lb h	total mass (including liquid);	mass in kilograms

Three Phase Pad-mounted transformer ---- Nameplate(2)

IEEE C57.12.00	CSA C227.4<=3000KVA	C88.16 >3000KVA
Connection diagram i	--	connection diagram
Name and location (country) of manufacture	(p) name of manufacturer and address;	1. manufacturer's name; and
Installation and operating instructions reference	Same as l	reference to instruction manual;
The word transformer or autotransformer	a) PADMOUNT TYPE	kind of transformer
Type of insulating liquid (generic name preferred) j	----	identification of insulating liquid;
Conductor material (of each winding)	---	identification of winding material;
Step-up operation suitability k	---	--
Maximum value of primary voltage l	---	--
Tank, pressure, and liquid data m	----	operating pressure, positive and negative, for sealed units.tank pressure/vacuum strength
DOE compliant n	(q) energy efficiency in accordance with CAN/CSA-C802.1.	---
-----	(f) oil volume in litres;	quantity of oil or liquid (litres);
-----	(j) < 2 ppm PCB;	--
-----	CSA Standard number;	--
	k) bayonet fuse and CLF identification;	
--	--	duty if other than continuously rated
--	--	load capability in MVA at 0 °C ambient temperature
--	--	switching—impulse levels (full wave in kV);
--	--	ratings of instrument and auxiliary transformers
--	--	cooling water requirements where applicable (litres per second);

space for purchaser's serial number

Substation Transformers--- Information on the inside

C227.4

9.4.2 Information on the inside of the transformer compartment

The following information shall be permanently marked on the exterior of the tank inside the compartment with **15 to 25 mm** lettering required unless otherwise indicated below (see Figure 14):

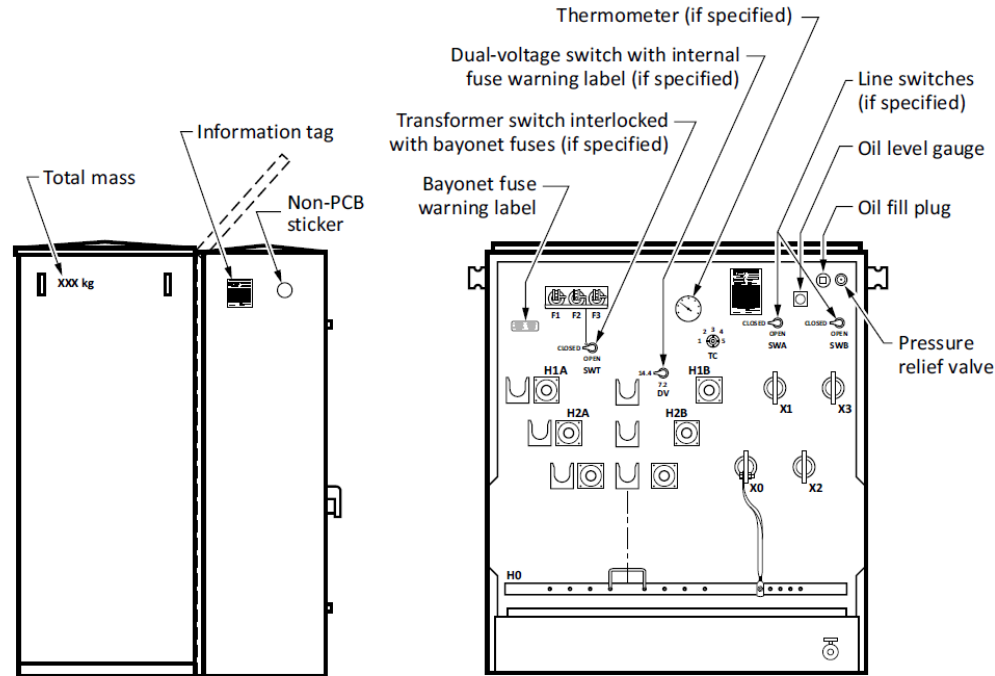
- a) high-voltage terminal designations;
- b) low-voltage terminal designations;
- c) DV (dual-voltage switch), if present;
- d) TC (tap changer switch), if present;
- e) voltage markings of voltage selector switch positions;
- f) tap position of the tap changer switch in numerals or letters (if indicator plate is provided). This marking shall be **at least 8 mm high**; and
- g) internal load-break switch positions and designation, if present

Substation Transformers---Warning Marking: Transformer markings — Loop feed

Figure 14 a)

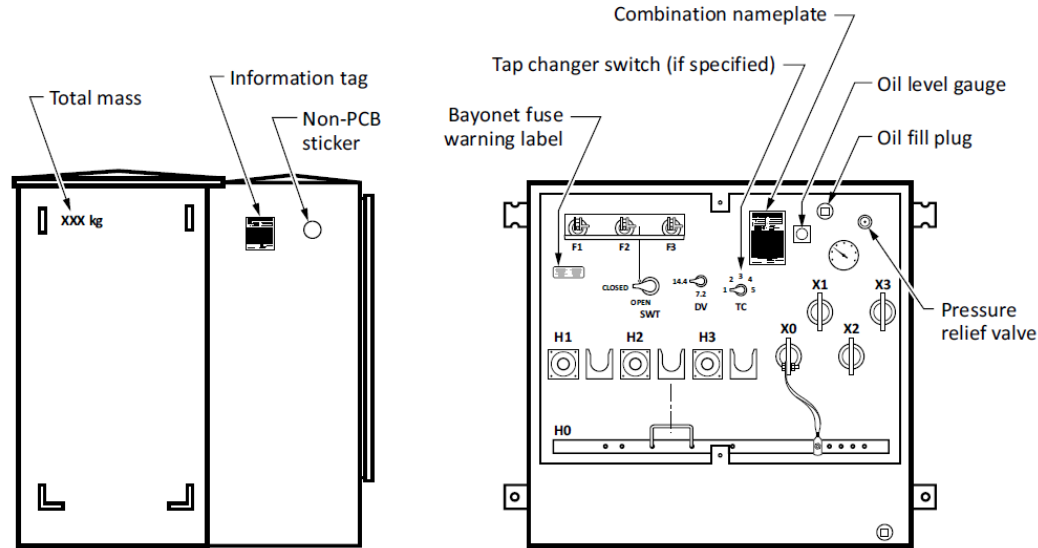
Transformer markings — Loop feed

(See Clauses [9.1.2](#), [9.4.2](#), [9.4.3.1](#), [9.4.4.1](#), [9.4.4.2](#), and [9.4.4.3](#).)



Substation Transformers---Warning Marking:Transformer markings — Radial feed

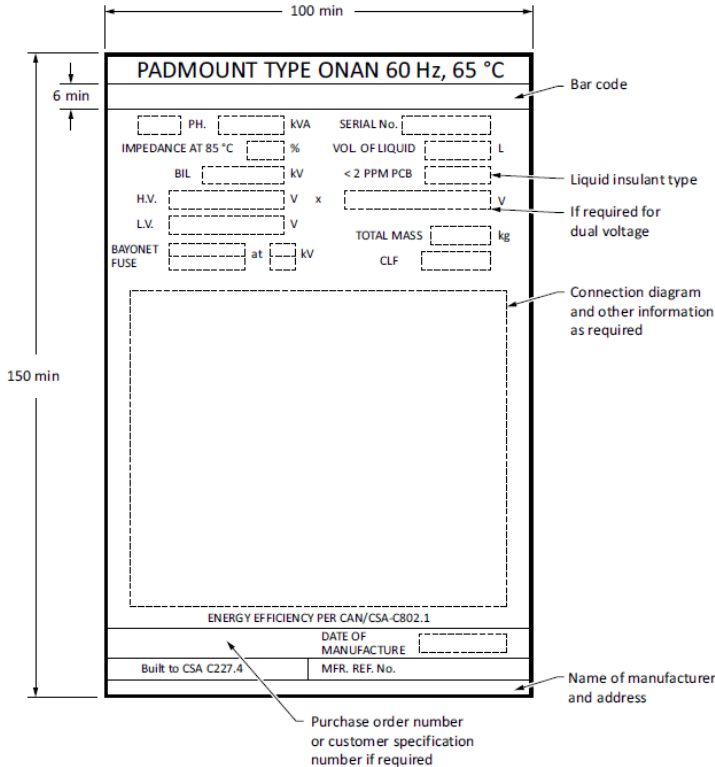
Figure 14 b)
Transformer markings — Radial feed
(See Clauses [9.1.2](#), [9.4.2](#), [9.4.3.1](#), [9.4.4.1](#), [9.4.4.2](#), and [9.4.4.3](#).)



Note: The locations of load-break, dual voltage, and tap changer switches may vary.

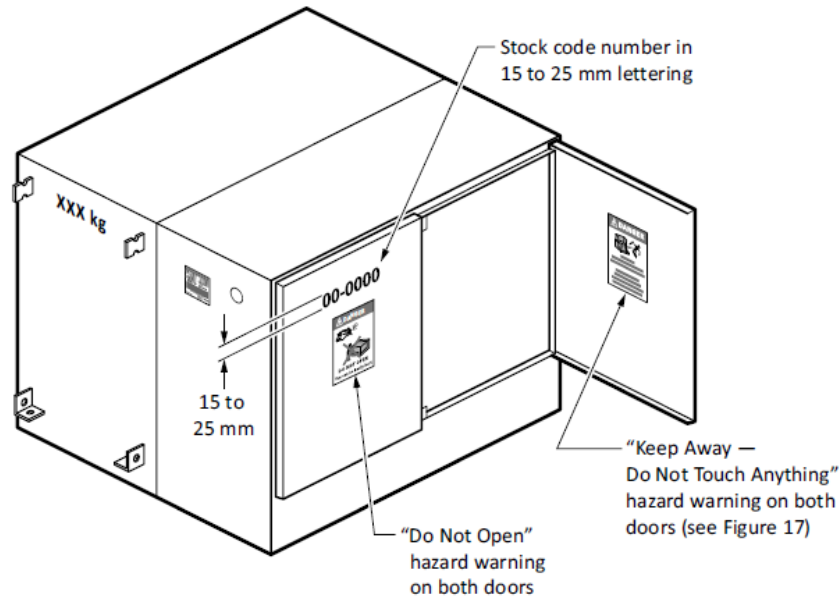
Three Phase Pad-mounted transformer ---- Nameplate(1)

Figure 13
Combination nameplate
(See Clauses 5.3.2 and 9.1.1.)



Substation Transformers---Warning Marking :Hazard warning sign placement

Figure 15
Hazard warning sign placement
(See Clauses [9.4.3.1](#), [9.4.3.4](#), [9.4.4.4](#), and [10](#).)

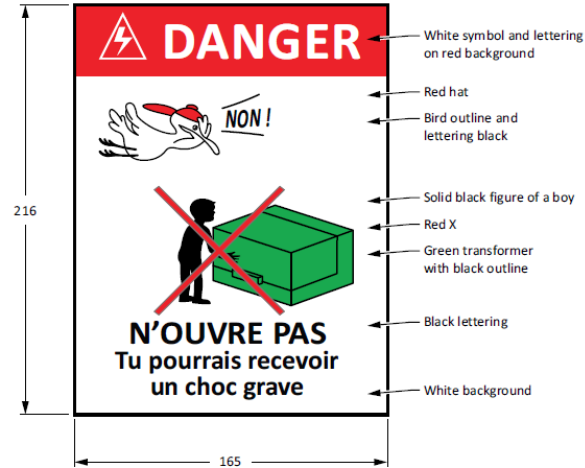


Substation Transformers---Warning Marking: DO NOT OPEN

Figure A.3
Outside warning label (English)
(See Clauses 9.4.3.4, A.1, and A.3.3.)



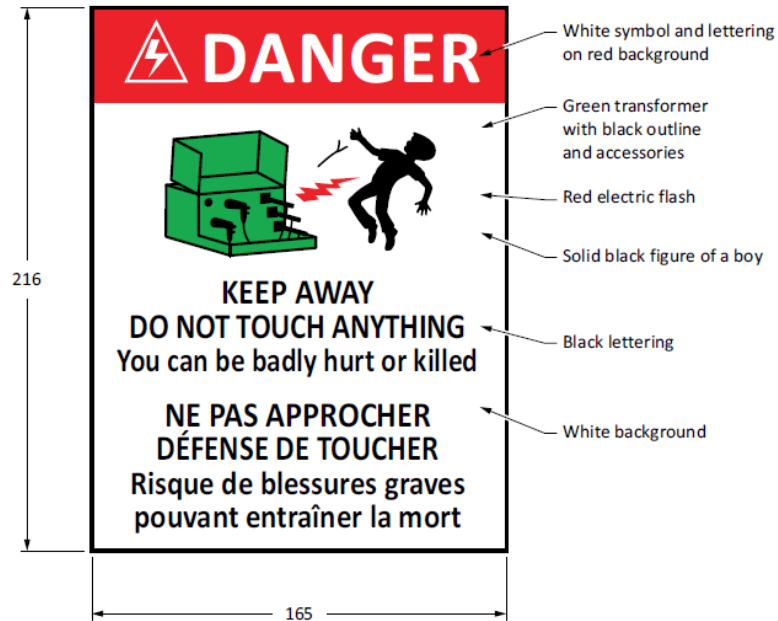
Figure A.4
Outside warning label (French)
(See Clauses 9.4.3.4, A.1, and A.3.3.)



Note: All dimensions are in millimetres.

Substation Transformers---Warning Marking

Figure 17
“Keep away” warning sign
(See Clauses [9.4.3.1](#) and [9.4.3.4](#) and Figure [15](#).)

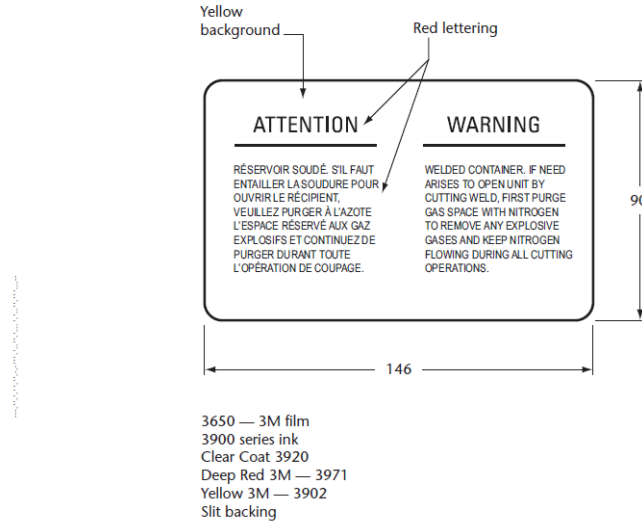


Note: All dimensions are in millimetres.

Substation Transformers---Warning Marking

Substation Transformers---Welded cover Marking

Welded cover

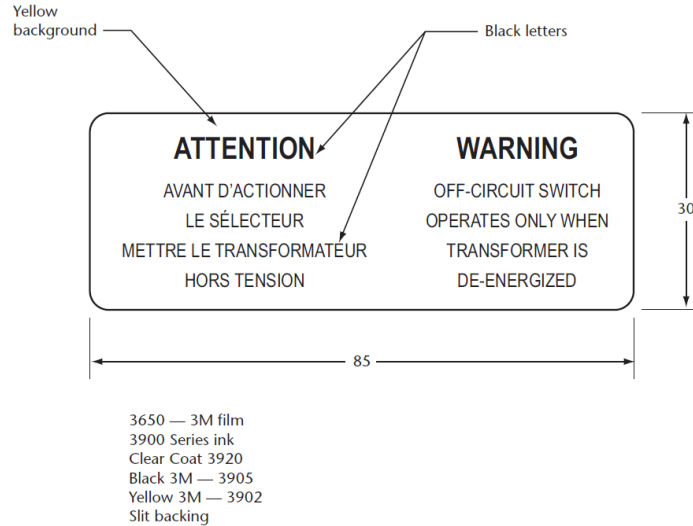


Note: All dimensions given are in millimetres.

Figure B.1
Welded container warning label
(See [Clauses 10.3.1](#), [B.1](#), and [B.3.1](#).)

Substation Transformers-----Off-circuit switches Marking

Off-circuit switches

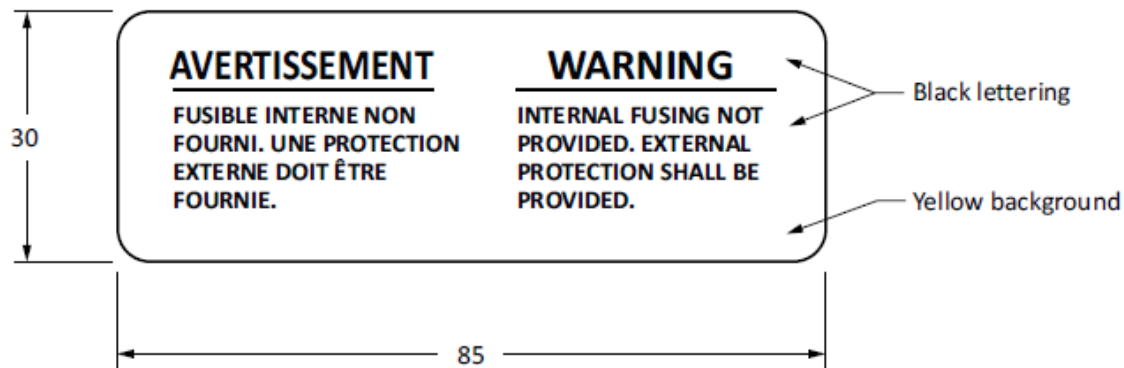


Substation Transformers-----Off-circuit switches Marking

9.4.3.5 Transformers without internal fusing

Transformers without internal fusing shall have a warning label on the faceplate of the transformer adjacent to the nameplate meeting the requirements of Figure A.5.

Figure A.5
No internal fuse warning
(See Clause [9.4.3.5](#).)

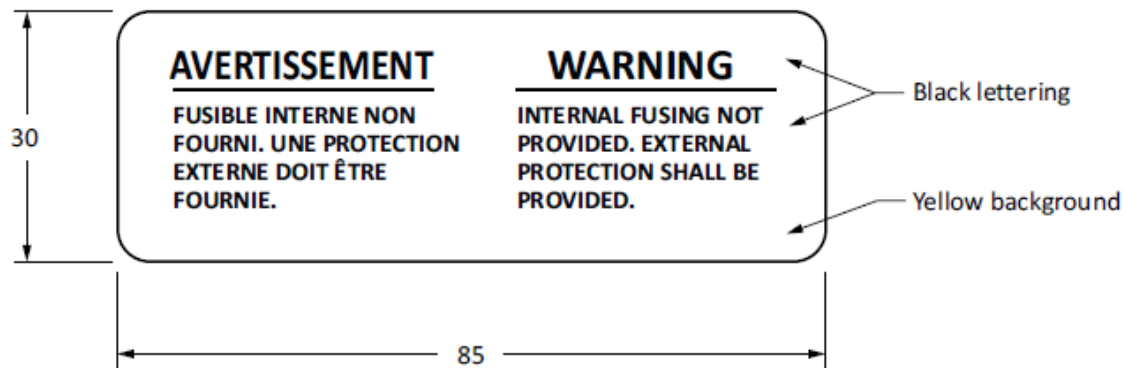


Substation Transformers-----Off-circuit switches Marking

9.4.3.5 Transformers without internal fusing

Transformers without internal fusing shall have a warning label on the faceplate of the transformer adjacent to the nameplate meeting the requirements of Figure A.5.

Figure A.5
No internal fuse warning
(See Clause [9.4.3.5.](#))

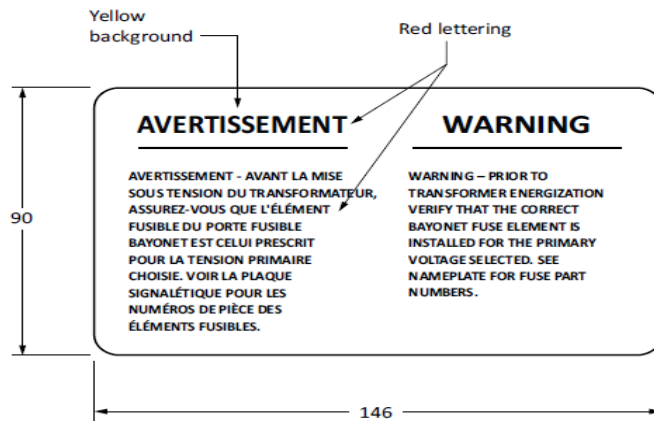


Substation Transformers----- Fusing for dual-voltage transformers

5.3.1.5 Fusing for dual-voltage transformers

For dual-voltage transformers, the CLF shall have a fuse voltage rating suitable for the higher voltage and a current rating adequate for the lower voltage. Bayonet fuses for both voltages and corresponding currents shall be provided, and the fuse for the higher voltage shall be installed unless otherwise specified by the purchaser. The manufacturer shall include a warning label at the dual-voltage selector switch to verify the correct bayonet fuses are installed for the voltage selected as shown in Figure A.6.

Figure A.6
Internal fuse warning for dual voltage transformers
(See Clause [5.3.1.5](#).)



HV Voltage bushing —IEEE 386

Connector

loadbreak connector

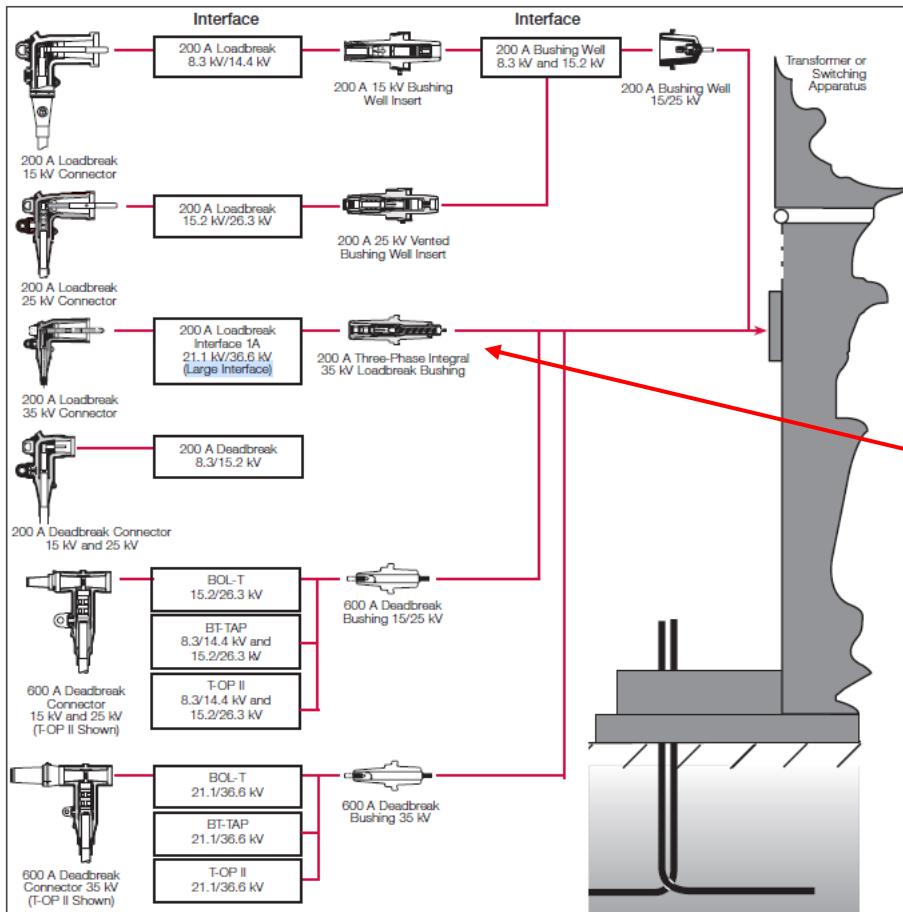
A separable insulated connector designed to close and **interrupt rated load current** or less on energized circuits under specified conditions. These connectors also have a fault close rating.

deadbreak connector:

A separable insulated connector designed to be separated and engaged **on de-energized circuits only**. This includes disconnectable buses.

Interface Description Per IEEE Std 386™ standard

Interface Description Per IEEE Std 386™ standard



loadbreak	deadbreak
200A	200A
-	600A
-	900A

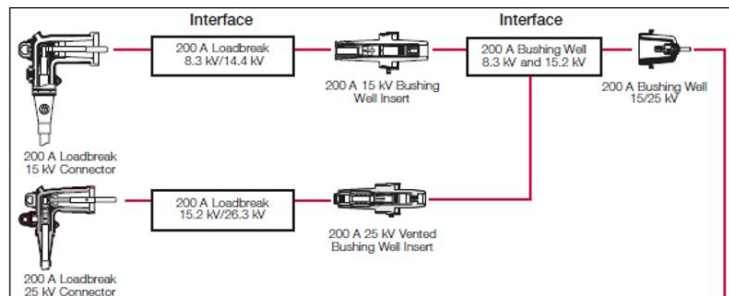
200 A integral loadbreak bushing specification information

- 200 A, 35 kV three-phase rated integral loadbreak bushing meeting the requirements of IEEE Std 386™ standard No. 1A (large 35 kV class interface).
- Voltage and current ratings in accordance with IEEE Std 386™ standard.

200 Amp Loadbreak Connector

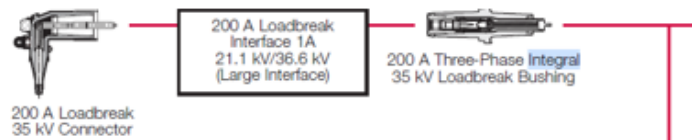
200 Amp Loadbreak Connector Detached 15 & 25 KV

- bushing well
- bushing insert
- bushing clamp
- ELBOW



35KV Integral loadbreak connector

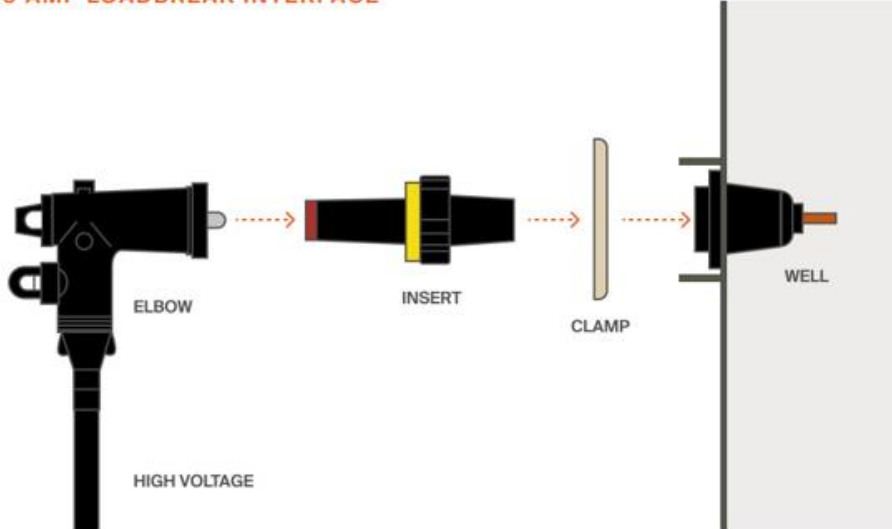
- Integral loadbreak bushing
- bushing clamp
- Load break connector



Loadbreak Connector

Primary/HV Padmount Transformer Bushing

200 AMP LOADBREAK INTERFACE



- bushing well
- bushing insert
- bushing clamp
- ELBOW

Voltage and current ratings and characteristics for loadbreak connectors

Table 1—Voltage ratings and characteristics for loadbreak connectors

Voltage class	Maximum voltage rating (kV rms) ^a	Withstand voltages		
		BIL and full wave (kV crest)	AC 60 Hz for 1 min (kV rms)	Partial discharge minimum extinction voltage (kV rms) ^b
15 kV	8.3 ^c	95	34	11
15 kV	8.3/14.4 ^d	95	34	11
25 kV	15.2 ^c	125	40	19
25 kV	15.2/26.3 ^d	125	40	19
28 kV	16.2/28.0 ^d	125	45	21.5
35 kV	21.1 ^c	150	50	26
35 kV	21.1/36.6 ^d	150	50	26

^aThe highest steady-state voltage across the open contacts that a loadbreak connector is rated to switch is the maximum phase-to-ground rms voltage for phase-to-ground rated devices or the maximum phase-to-phase rms voltage for phase-to-ground/phase-to-phase rated devices.

^bBased on a sensitivity of 5 pC (see 7.4).

^cPhase-to-ground.

^dPhase-to-ground/phase-to-phase.

Table 3—Current ratings and characteristics for loadbreak connectors

Continuous current rating (A rms)	Voltage class	Switching current rating		Fault-closure current rating ^b			Short-time current rating ^a		
		A rms, sym ^a	Number of operations	A rms, sym	Duration (seconds)	x/r	A rms, sym	Duration (seconds)	x/r
200	All	200	10	10 000	0.17	6	10 000	0.17	6
							3 500	3.00	6
600	15 kV	600	10	16 000	0.17	20	16 000	0.17	20
							10 000	3.00	20
600	25, 28 kV	600	5	10 000	0.17	20	10 000	0.17	20

Table 3—Current ratings and characteristics for loadbreak connectors (*continued*)

Continuous current rating (A rms)	Voltage class	Switching current rating		Fault-closure current rating ^b			Short-time current rating ^a		
		A rms, sym ^a	Number of operations	A rms, sym	Duration (seconds)	x/r	A rms, sym	Duration (seconds)	x/r
							10 000	3.00	20
600	35 kV	600	1	10 000	0.17	20	10 000	0.17	20
							10 000	3.00	20

^aConsult the manufacturer for ratings that exceed those listed in Table 3.

^bEquipment to which these assemblies are affixed may have lower safe limits of current performance.

Three Phase Pad-mounted transformer ----bushing well

Bushing wells are utilized for **200 amp loadbreak interfaces**. Bushing wells are typically designed for compatibility with **multiple voltage classes** such as **15kV, 25kV, and 35kV**. For this reason, a **single 35kV** rating is common since it allows **the same well** to be used **with inserts of differing ratings—such as 15kV, 25kV, and 35kV**. The well fits into a hole in the transformer front plate—just large enough for the shank to pass through, but small enough to allow the wider rim of its collar to rest against the exterior tank wall with a gasket seal between the two surfaces.

The middle of the well houses the conductor, a copper **$\frac{3}{8}$ "-16 threaded stud**, which provides **a termination point** for the winding leads of the transformer on one end, and a threaded insert on the other. The stud is designed to withstand a minimum torque of **17 ft. lbs** without breaking or stripping per IEEE Std 386. The **copper stud** can be either fixed or removable. The removable version allows for easy replacement if breakage occurs in the field.



Bushing Well



200 A HTN Tri-Clamp bushing well specification information

- Molded-in semi-conductive shield.
- 35 kV, 150 kV BIL.
- HTN material.
- Removable stud shall have provisions for easy removal of broken parts from both the bushing well and [insert](#).
- Voltage and current ratings in accordance with IEEE Std 386™ standard.

Transformer bushing clamp

The clamp fastens the well to the front plate of the transformer and secures the seal between the well gasket and tank wall. Typical configurations include **3-hole and 4-hole clamps**, which fit over fixed threaded studs in the transformer front plate.



Bushing Clamp

Catalog Section	Description	kV Class	Base Part Number	Notes
Temporary Grounding Sets				
	Single-Phase Three-Clamp Set Pad-mounted		133040	(1/0 Black Cable)
	Three-Phase Four-Clamp Set Pad-mounted		133040-1	(1/0 Black Cable)
			133040-2	(2/0 Black Cable)

Transformer bushing insert

The bushing insert **fits into the well** and provides the mating surface for cable fitted **loadbreak elbows**. The base of the insert contains a **$\frac{3}{8}$ " - 16 UNC copper threaded** section for screwing onto the well stud. At the front of the **threaded base lies a 5/16" hex broach which allows installation with a torque tool**. Maximum torque values for tightening inserts are usually provided by the bushing manufacturer. When purchased as a stand alone item, an additional torque tool may be ordered along with the insert to ensure a proper fit into the well. Failure of the bushing well stud may occur if higher- than-specified torque values are applied. As mentioned above, the minimum withstand torque for the copper stud the insert threads onto per IEEE Std 386 is **17 ft. lbs.**. The outside of the insert is insulated all the way around with a peroxide-cured **EPDM rubber material over a shield housing**, which surrounds the inner conductive portion of the bushing. An **arc snuffer** assembly at the tip of the bushing provides the ability to extinguish an **arc** when **breaking the load** at the elbow and bushing apparatus with **a hot stick**. Between the arc snuffer and hex broach lies an integral contact and piston assembly for transferring current to the bushing well stud; it also ensures proper contact during fault switching and fault-close operations.



Bushing Insert

Transformer bushing insert

- 200 Amp 15 kV Loadbreak Insert
- 200 Amp 25 kV Loadbreak Insert
- 200 Amp 35 kV Loadbreak Insert (Large Interface)
- 200 Amp 35 kV Loadbreak Insert (Small Interface)

Table 1—Voltage ratings and characteristics for loadbreak connectors

Voltage class	Maximum voltage rating (kV rms) ^a	Withstand voltages		
		BIL and full wave (kV crest)	AC 60 Hz for 1 min (kV rms)	Partial discharge minimum extinction voltage (kV rms) ^b
15 kV	8.3 ^c	95	34	11
15 kV	8.3/14.4 ^d	95	34	11
25 kV	15.2 ^c	125	40	19
25 kV	15.2/26.3 ^d	125	40	19
28 kV	16.2/28.0 ^d	125	45	21.5
35 kV	21.1 ^c	150	50	26
35 kV	21.1/36.6 ^d	150	50	26

^aThe highest steady-state voltage across the open contacts that a loadbreak connector is rated to switch is the maximum phase-to-ground rms voltage for phase-to-ground rated devices or the maximum phase-to-phase rms voltage for phase-to-ground/phase-to-phase rated devices.

^bBased on a sensitivity of 5 pC (see 7.4).

^cPhase-to-ground.

^dPhase-to-ground/phase-to-phase.

Transformer bushing insert

- **200 Amp 15 kV Loadbreak Insert**

The 200 amp 15kV Class insert is used for voltage applications requiring an insulation rating of 95kV BIL or lower. It is rated for a maximum continuous line-to-line voltage of 14.4 kV and a maximum continuous line-to-ground voltage of 8.3kV. The 15kV loadbreak insert is identified by a red tip surrounding the contact tube.



Bushing Insert

86

- **200 Amp 25 kV Loadbreak Insert**

The 25kV separable bushing insert has a 125kV BIL rating and is suitable for maximum continuous voltages of 26.3kV line-to-line and 15.2kV line-to-ground. A 28 kV rated version is also available for voltage applications requiring a higher voltage rating utilizing a standard 200 amp loadbreak interface. The 25 kV and 28 kV insert is easily identified by the blue tip around its contact tube.



Bushing Insert

- **200 Amp 25 kV Loadbreak Insert**

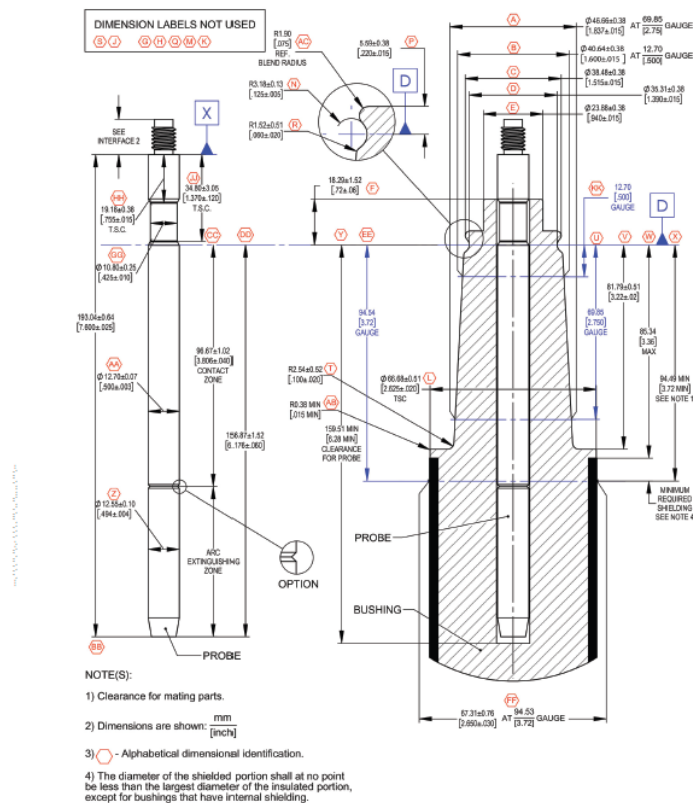


Figure 6—Interface 7A: a 200-A loadbreak interface, 25- and 28-kV class (previously Figure 7 in IEEE Std 386-2006)

Transformer bushing insert

200 Amp 35 kV Loadbreak Bushing (Large Interface)

The large interface 35kV loadbreak bushing (Figure 7–Interface 8 in IEEE Std 386-2016) utilizes a one piece integral bushing with no separable well and insert. It carries a 150 kV BIL rating, and it is suitable for continuous voltages up to 36.6kV line-to-line and 21.1kV line-to-ground. This type of bushing can also be identified by its purple tip near its locking groove (also referred to as the “purple cuff” or “large” version within the family of 35 kV loadbreak bushings).



Bushing Insert

200 Amp 35 kV Loadbreak insert (Large Interface)

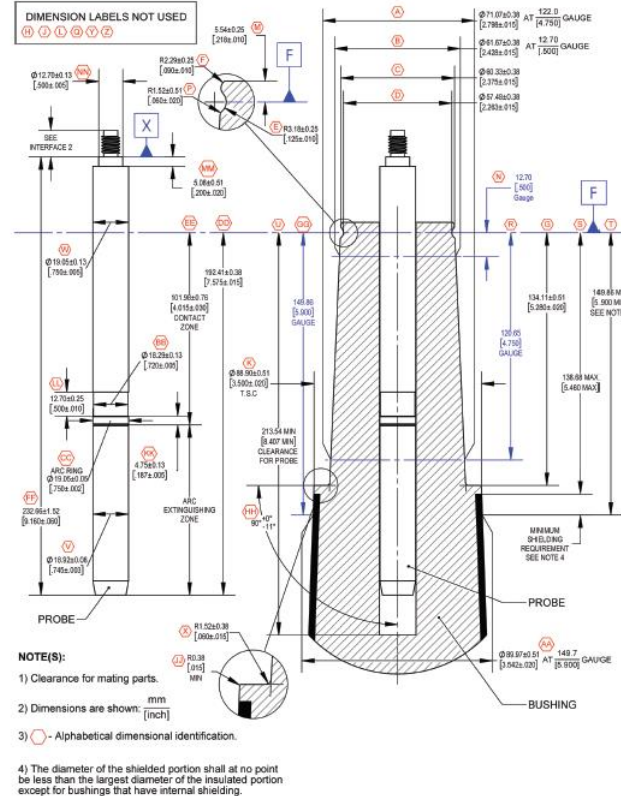


Figure 8—Interface 9: a 200-A loadbreak interface, single-phase, 35-kV class (large interface, previously Figure 9 in IEEE Std 386-2006)

200 Amp 35 kV Loadbreak Insert (Small Interface)

Unlike its counterpart, the small interface 35 kV bushing (Figure 9–Interface 7B in IEEE Std 386-2016) is made up of a separable well and insert. The small interface 35kV insert has the same 36.6kV max line-to-line and 21.1kV line-to-ground rating. The small interface insert is identified by its longer length and the yellow tip around its contact tube (also commonly called the “small” version within the family of 35 kV loadbreak bushings)

200 Amp 35 kV Loadbreak Insert (Small Interface)

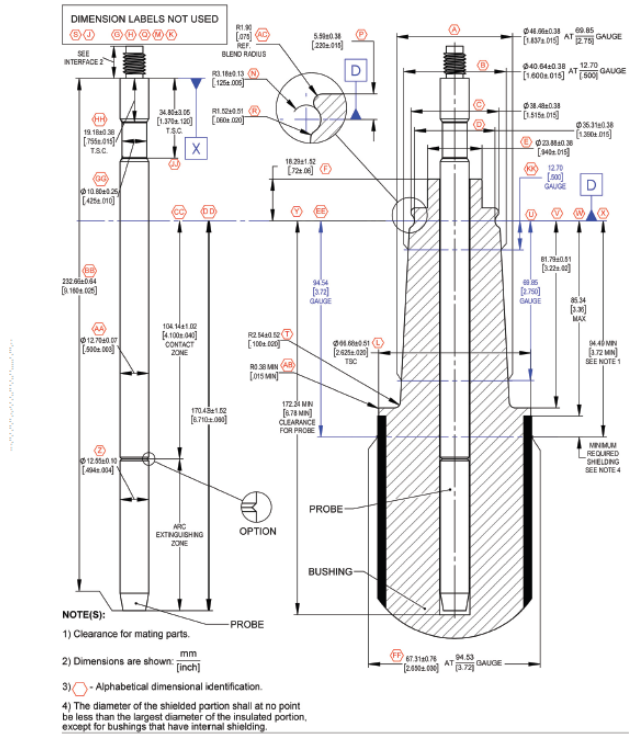


Figure 9—Interface 7B , a 200-A loadbreak interface, 35-kV class (small interface, previously Figure 7 in IEEE Std 386-2006)

Deadbreak connector

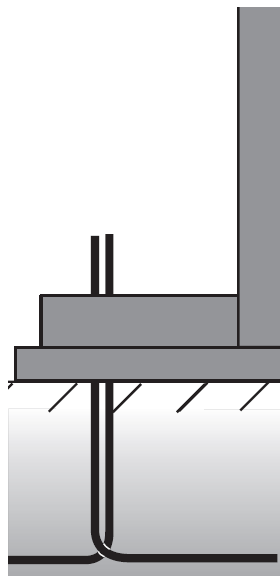
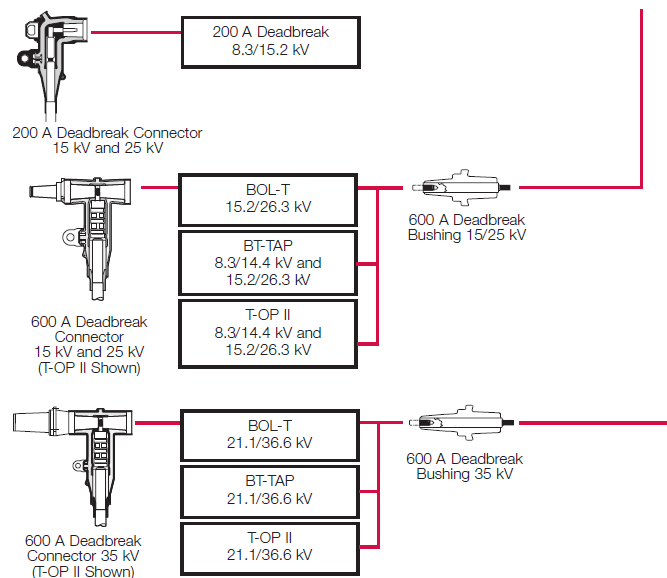


Table 2—Voltage ratings and characteristics of deadbreak connectors

Voltage class	Maximum voltage rating (kV rms) ^a	Withstand voltages		
		BIL and full wave (kV crest)	AC 60 Hz for 1 min (kV rms)	Partial discharge minimum extinction voltage (kV rms) ^b
15 kV	8.3	95	34	11
25 kV	15.2	125	40	19
28 kV	16.2	125	45	21.5
35 kV	21.1	150	50	26

^aPhase-to-ground.

^bBased on a sensitivity of 5 pC (see 7.4).

Table 4—Current ratings and characteristics for deadbreak connectors

Continuous current rating (A rms) ^a	Voltage class	Overload current ^{a,b,c} 4 h rating (A rms)	Short-time current rating		
			A rms, sym	Duration (seconds)	x/r
200	All	300	10 000	0.17	6
			3 500	3.00	6
600	All	900	25 000	0.17	20
			10 000	3.00	20
900	All	1200	40 000	0.17	20
			10 000	3.00	20

^aIn general, the overload capability of a connector exceeds its continuous current rating. Overload capability varies with environment, cable sizes, etc. The connector manufacturer's recommendations should be obtained for the particular combination involved.

^bConsult the manufacturer for ratings that exceed those listed in Table 4.

^cOne overload cycle during a 24 h period not to exceed 1500 h cumulative over the life of the connector.

Deadbreak connector

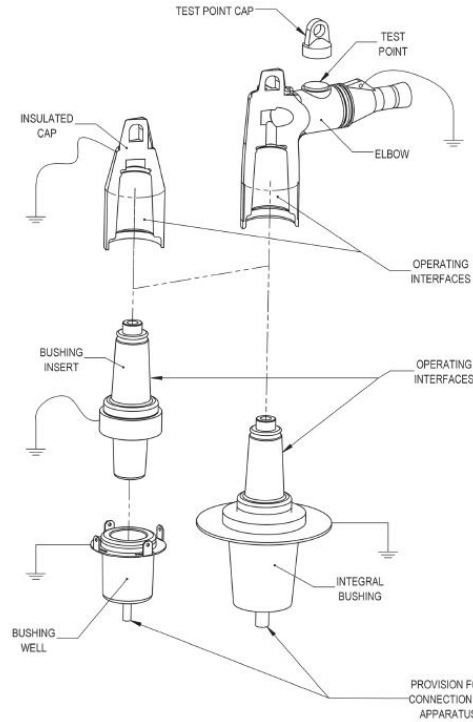


Figure 1—Interface 1: typical components of 200-A separable insulated connector system (previously Figure 1 in IEEE Std 386-2006)

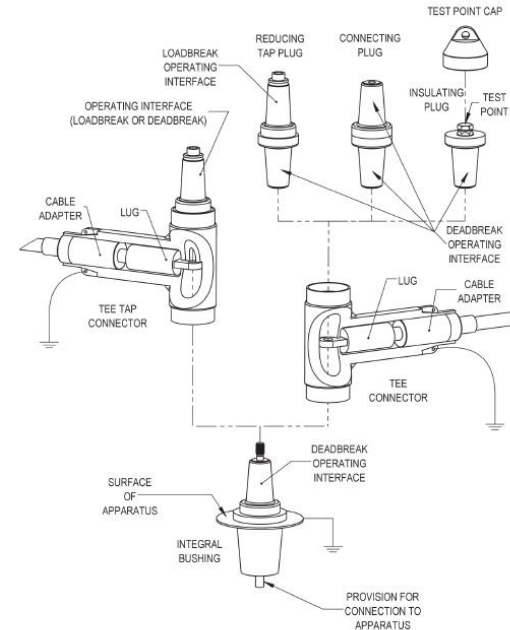


Figure 11—Typical components of a 600- or 900-A separable insulated connector system (previously Figure 2 in IEEE Std 386-2006)

200A Deadbreak bushing

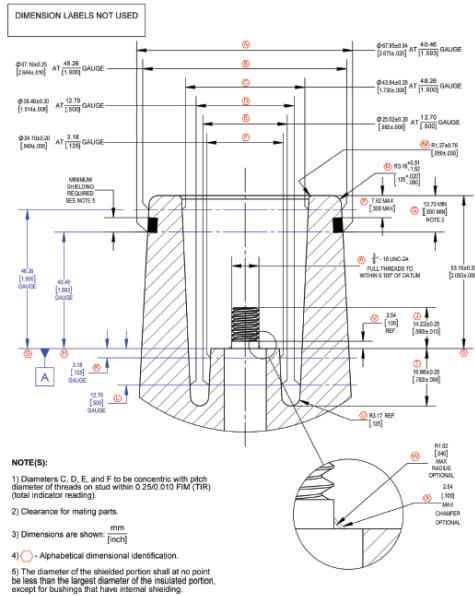


Figure 3—Interface 3: a 200-A deadbreak bushing well interface, 15-, 25-, 28-, and 35-kV class (previously Figure 3 in IEEE Std 386-2006)

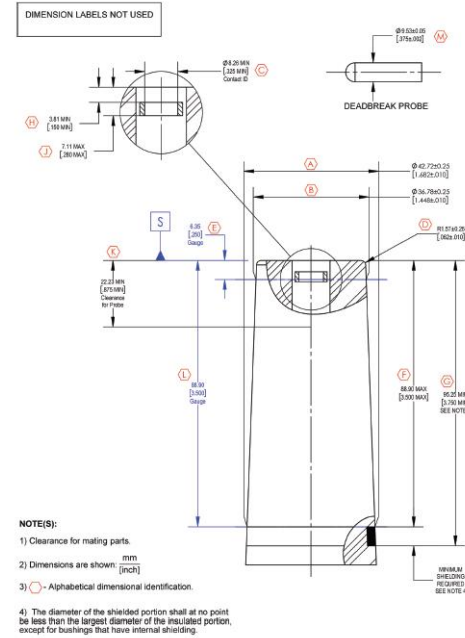


Figure 10—Interface 10: a 200-A deadbreak interface, 35-kV class (previously Figure 10 in IEEE Std 386-2006)

600 A Deadbreak bushing

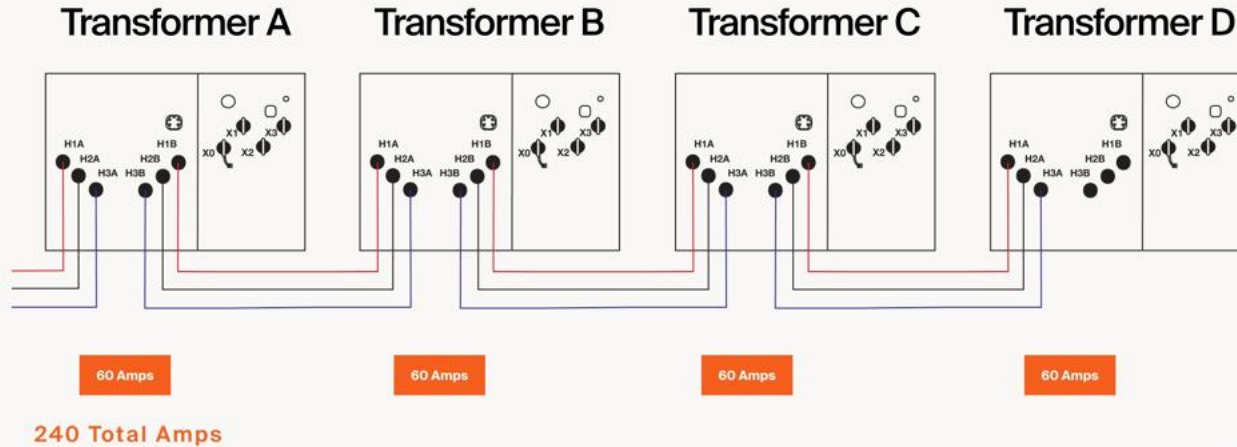
For primary transformer applications requiring a current rating above 200 amps, a 600 amp non-loadbreak (deadbreak) bushing is used. Deadbreak connections are also often required for HV connections with cable sizes 250 kcmil and above. Unlike the loadbreak interface, the deadbreak does not allow the load to be broken at the bushing. In order to disconnect HV cable connections at the transformer, current flow must be switched off upwind of the HV cables.

With deadbreak bushings, cable connectors are threaded onto the bushing with a metal stud. Another distinctive feature of deadbreak elbow connectors is that they allow for back to back cable splices at a single bushing (this is also referred to as piggybacking cables). All deadbreak bushings are built to conform with IEEE Std 386.

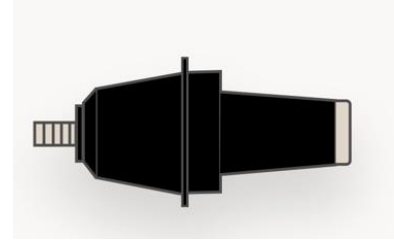
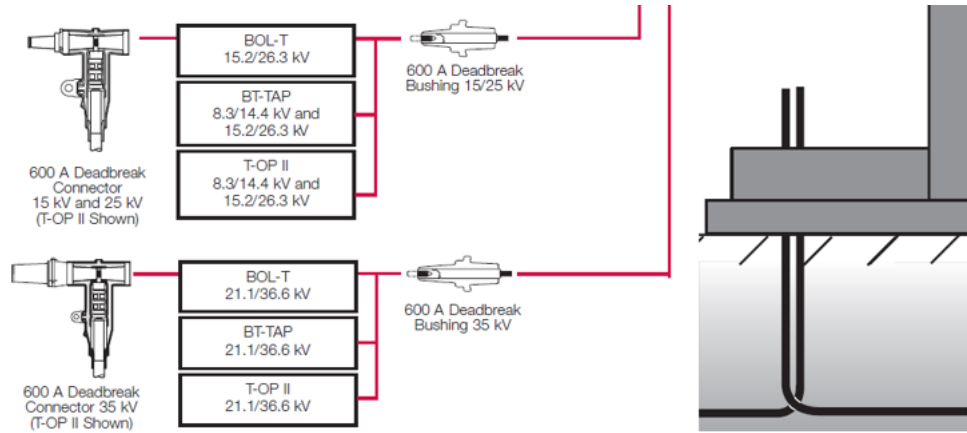
600 Amp bushings are required in loop feed transformer applications where there are multiple transformers in the loop. The bushings in the first transformer in the loop must support the amperage of all the downstream transformers in the loop. In the below diagram, "Transformer A" would require at least 600 amp deadbreak bushings since there is a total of 240 amps in the loop.

600 A Deadbreak bushing

600 Amp bushings for loop feed application

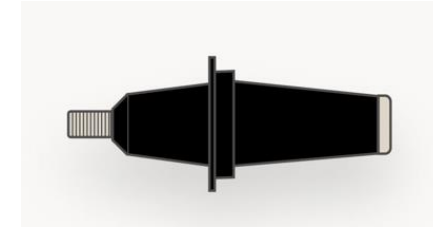


600 A Deadbreak bushing



600 Amp 15/25 kV

DEADBREAK BUSHING

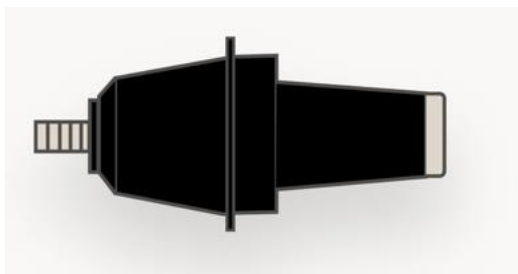


600 Amp 35 kV

DEADBREAK BUSHING

600 Amp 15/25 kV Deadbreak Bushing

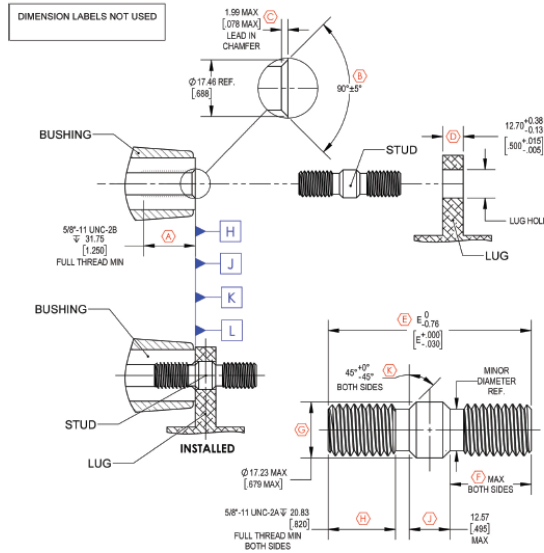
The 15/25 kV deadbreak bushing conforms to the requirements of IEEE Std 386, and it is compatible with **15kV Class (8.3kV Line-Ground)**, **25kV Class (15.2kV Line-Ground)**, and **28kV Class (16.2kV Line-Ground)** applications. It is an integral one-piece style bushing. As with all integral style bushings which are one piece, no drain wire is needed from the exterior of the bushing to ground.



600 Amp 15/25 kV

DEADBREAK BUSHING

600 Amp 15/25 kV Deadbreak Bushing



NOTE(S):

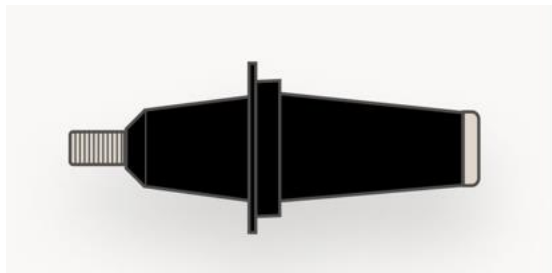
- 1) Clearance for mating parts.
- 2) Dimensions are shown: mm [inch]
- 3) () - Alphabetical dimensional identification.
- 4) Lug must allow stud to pass through hole.
- 5) Dimensions apply to all voltage classes except (E) and (F).

VOLTAGE CLASS:	(E)	(F)
15 / 25 / 28 kV Class	56.26 [2,215]	21.84 [860]
35 kV Class	62.61 [2,465]	25.02 [985]

Figure 12—Interface 19: a 600- and 900-A connection configuration, 15-, 25-, 28-, and 35-kV class

600 Amp 35 kV Deadbreak Bushing

The 35kV deadbreak version, like the 15/25kV, is an integral type bushing, rated 35kV Class (21.1kV Line-Ground). It is **longer than the 15/25kV** version, but a $\frac{5}{8}$ " stud is used for both.



600 Amp 35 kV

DEADBREAK BUSHING

600 Amp 35 kV Deadbreak Bushing

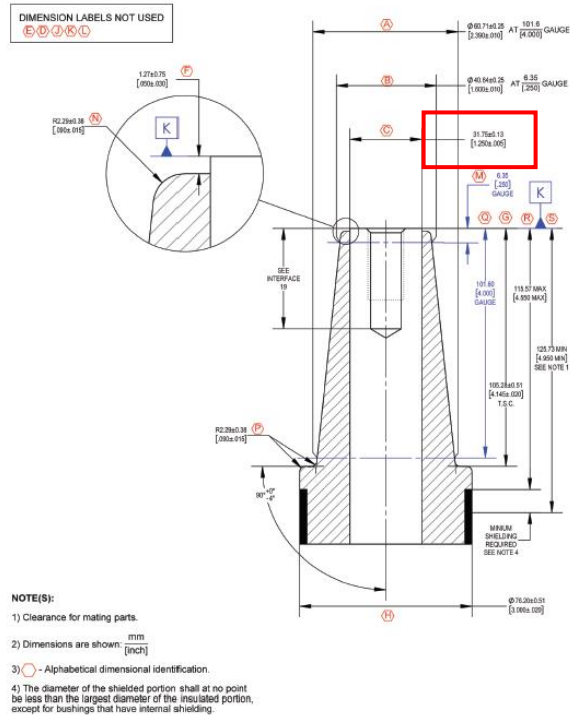


Figure 15—Interface 13: a 600- and 900-A deadbreak interface, 35-kV class (previously Figure 13 in IEEE Std 386-2006)

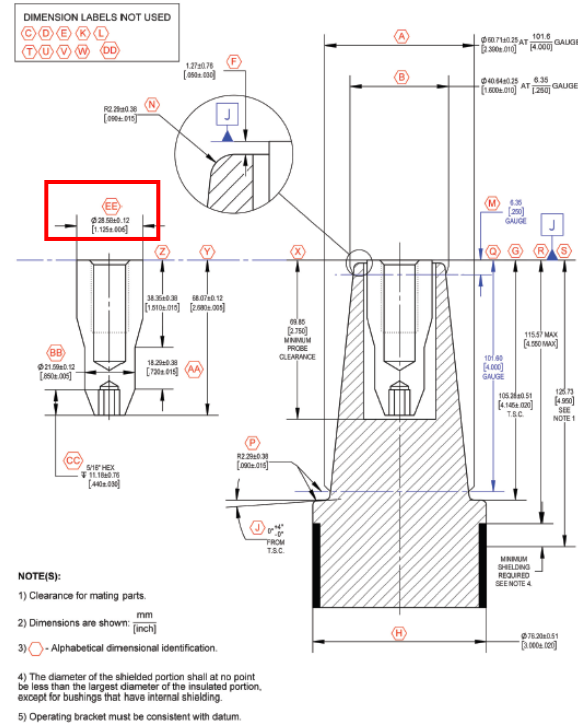
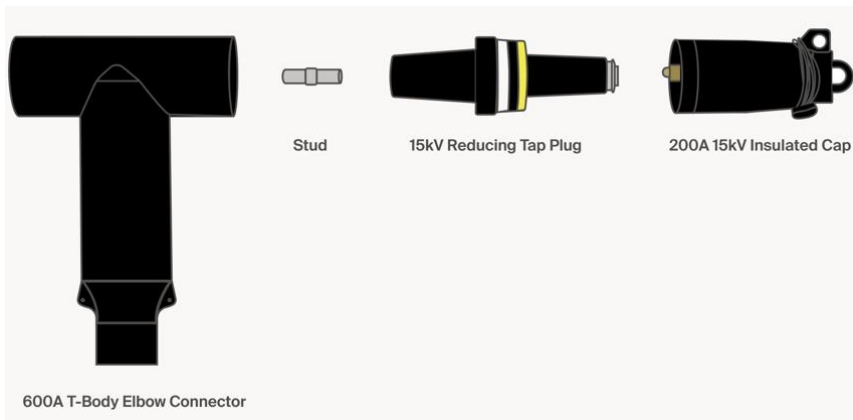


Figure 16—Interface 14: a 600- and 900-A deadbreak interface, 35-kV class (previously Figure 14 in IEEE Std 386-2006)

600A / 200 Amp Reducing Tap Plug

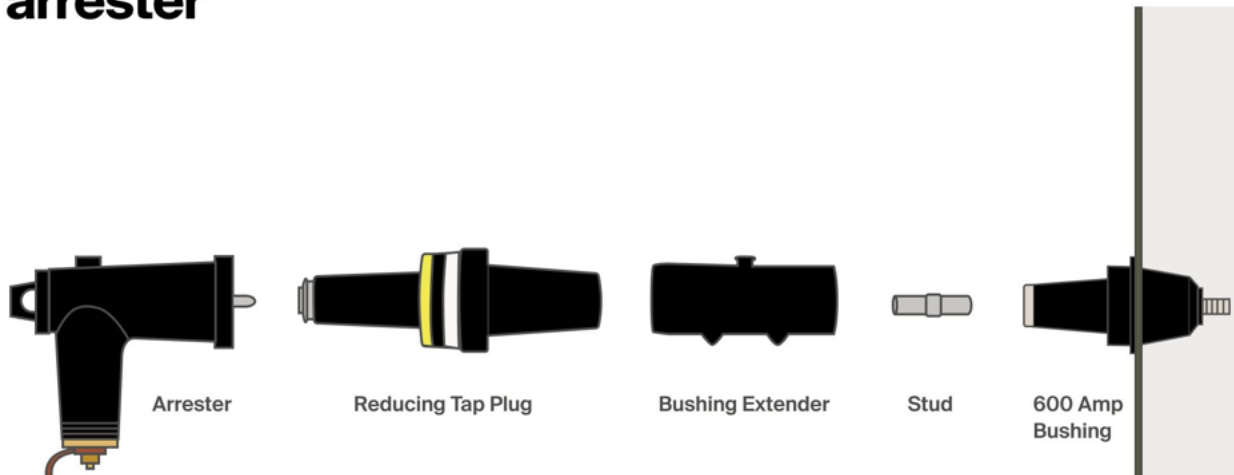
Some applications will **require 200 amp loadbreak** components to be fitted onto a 600 amp deadbreak interface. To accomplish this, a special **adapter** called **a reducing tap plug** is used. One end of the adapter is identical to the 600 amp bushing at the transformer, while the other end is configured to match the bushing interface of a 200 amp load break connection (available with 15kV, 25kV, and 35kV versions). Because the 600 amp end of the reducer is identical to the bushing at the transformer, a connection must be made with a deadbreak T-Body elbow or bushing extender.



600A / 200 Amp Reducing Tap Plug

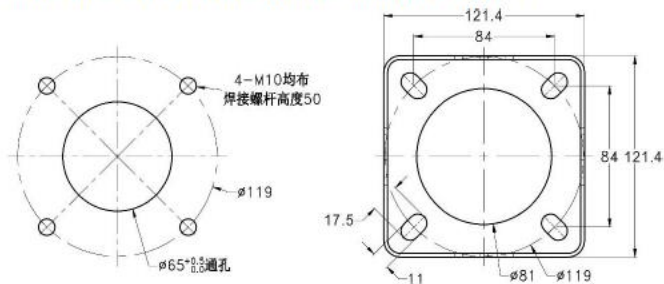
If a standalone 200 amp component is required with no cable connection at the transformer bushing, a bushing extender is used to connect the 600 amp end of the reducer to the bushing at the transformer faceplate. An example of this would be in the case of a six bushing transformer with three cables and three arresters connected at each bushing in the HV cabinet. For 600 amp bushing applications requiring surge arresters, a reducing tap plug is utilized to allow for the installation of standard 200 amp loadbreak arresters. For 600 amp applications where a reducing tap plug is required for arresters, provisions for additional cabinet space should be taken into consideration.

Reducing tap plug with bushing extender and arrester

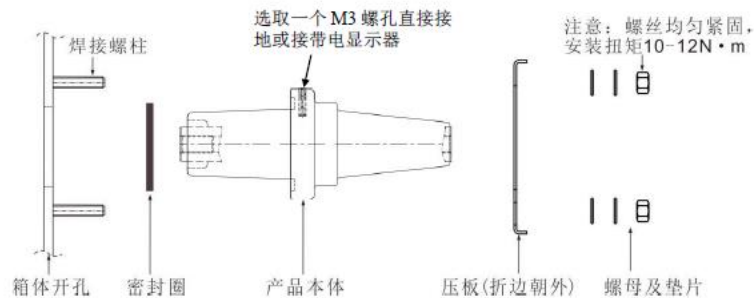


Voltage and current ratings and characteristics for deadbreak connectors

箱体开孔及压板尺寸（套管开孔处需做圆角处理并去除毛刺，以免划伤套管本体）：



装配顺序：



LV Voltage Padmount Spade Bushings

8.7.3 600 V and below terminals

8.7.3.1 General

The phase and neutral terminals shall be in accordance with Figure 4 or Figure 8 of this standard and arranged for vertical take-off. Terminal dimensions shall be as shown in Figure 19 of this standard or, if specified, Figure 20 of this standard.

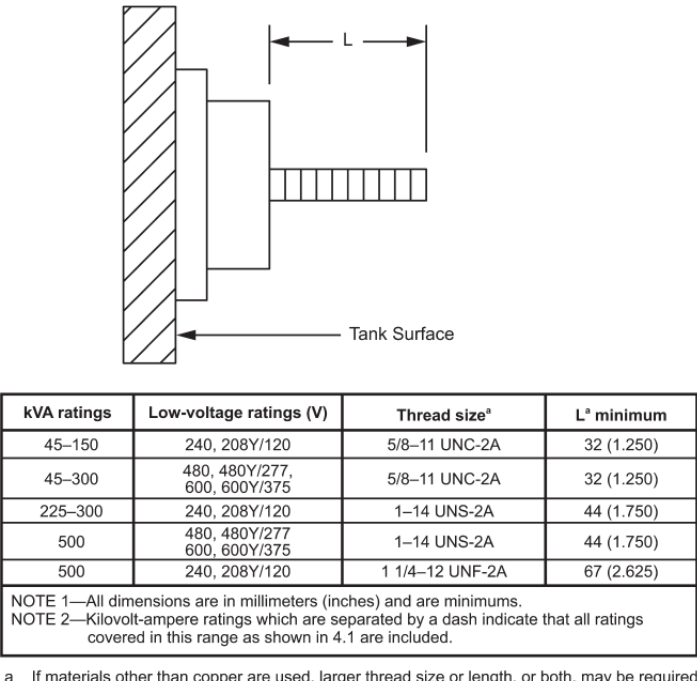
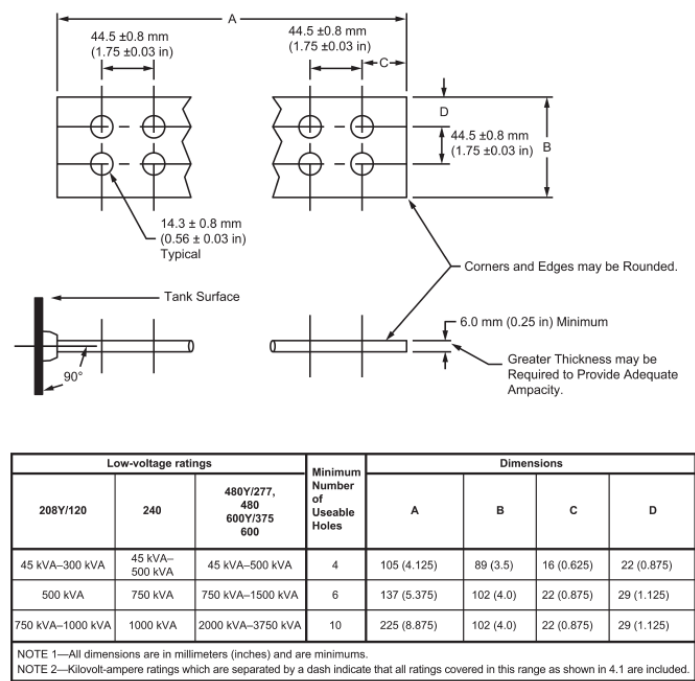
8.7.3.2 Support

Terminals equipped with 10 or more hole spades shall be furnished with additional support. A manufacturer designed support shall be attached at the end that is the farthest from the tank wall and mounted in such a way that it will not interfere with the use of any of the holes of the spade or any spade extenders/connectors that the user may want to install.

8.7.3.3 Bushing replacement

These bushings shall be externally replaceable. The inside terminal connections shall be externally removable through the connector's opening in the transformer tank, or accessible through a handhole to permit removal and replacement.

8.7.3 600 V and below terminals



kVA ratings	Low-voltage ratings (V)	Thread size ^a	L ^a minimum
45–150	240, 208Y/120	5/8–11 UNC-2A	32 (1.250)
45–300	480, 480Y/277, 600, 600Y/375	5/8–11 UNC-2A	32 (1.250)
225–300	240, 208Y/120	1–14 UNS-2A	44 (1.750)
500	480, 480Y/277, 600, 600Y/375	1–14 UNS-2A	44 (1.750)
500	240, 208Y/120	1 1/4–12 UNF-2A	67 (2.625)

NOTE 1—All dimensions are in millimeters (inches) and are minimums.
NOTE 2—Kilovolt-ampere ratings which are separated by a dash indicate that all ratings covered in this range as shown in 4.1 are included.

a If materials other than copper are used, larger thread size or length, or both, may be required.

6.1.3 Low-voltage bushings

6.1.3.1

Low-voltage terminals for all sizes of transformers shall be the spade type in accordance with Table 8 and Figure 11 and shall have a thickness that will provide adequate ampacity. Terminals shall be of plated corrosion-resistant metal, suitable for use with copper or aluminum connectors. Low-voltage terminals for stud-type bushings shall be of plated copper suitable for use with copper or aluminum connectors.

6.1.3.2

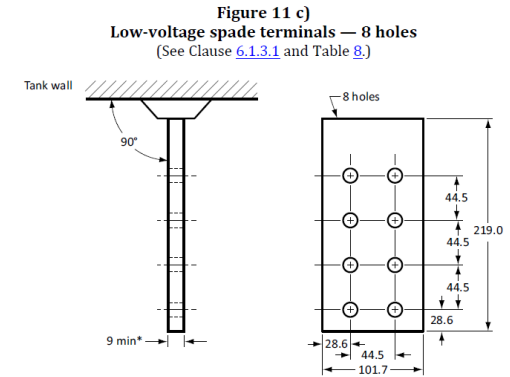
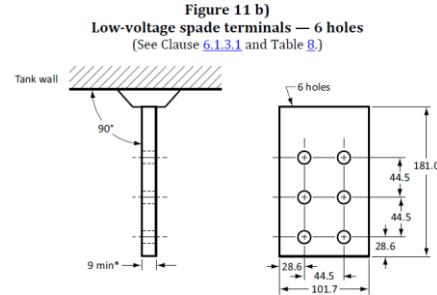
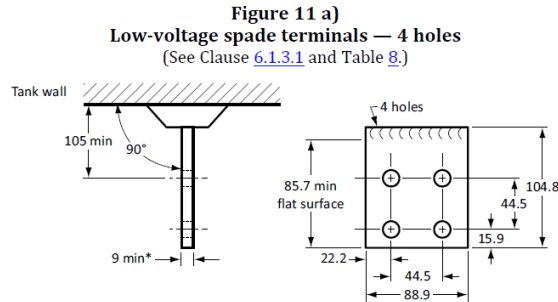
The low-voltage neutral shall be connected to an insulated terminal, designated X0, with a currentcarrying capacity equal to the other low-voltage terminals.

6.1.3.3

The low-voltage terminals shall be capable of withstanding a minimum vertical cantilever force of 750 Ncontinuously applied at the outermost hole of the terminal without causing a leak, damage, or permanent distortion.

Table 8
Low-voltage terminals
(See Clause [6.1.3.1](#).)

Transformer rating, kVA	Voltage rating of transformer, V		
	600Y/347	480Y/277	208Y/120
75–300	Figure 11 a)	Figure 11 a)	Figure 11 a)
500–1000	Figure 11 a)	Figure 11 a)	Figure 11 c)
1500–2000	Figure 11 b)	Figure 11 c)	Nonpreferred
2500–3000	Figure 11 c)	Figure 11 c)	Nonpreferred



* Greater thickness might be required for load capacity of transformer.

Notes for Figure 11 a), 11 b), and 11 c):

- 1) All dimensions are in millimetres. (Tolerances are not cumulative.)
- 2) All holes are $14.3 \text{ mm} \pm 0.8 \text{ mm}$ diameter.

There are many types and styles of secondary bushings for pad-mounted transformers. For low voltage secondaries 600 volts and below, the size of the bushing increases with the size of the transformer kVA. A 500 kVA transformer will have 4-hole spades, whereas a 2500 kVA will usually have 10 to 12 holes at each spade. As the rated secondary current of the transformer increases, more current carrying capacity is required for the conductors serving the load (meaning more cables at each spade).

For smaller bushings requiring fewer hole spades, a separate bushing with a screw on spade is commonly used. When required, secondary bushing supports are used to support any additional weight not supportable at the tank wall.

6-hole spade bushings↵



6-Hole Spade

SECONDARY/LV BUSHINGS

12-hole spade bushings↵



12-Hole Spade

SECONDARY/LV BUSHINGS

Universal NEMA Square Hole Pattern↵

Neutral terminal

8.7.4 Neutral terminal

8.7.4.1 Neutral terminal—600 V and below

The neutral terminal shall be either a **blade connected directly to the tank** or a **fully insulated terminal**. If a fully insulated terminal is used, a ground pad shall be provided on the outer surface of the tank. **One or more removable ground straps** suitably sized for the short-circuit rating of the transformer as defined in IEEE Std C57.12.00-2021 shall be provided and connected between the neutral terminal and the ground pad. For **wye-wye connected** units, the **high-voltage neutral shall be connected to the low-voltage neutral internally with provisions for opening** this connection for testing.

8.7.4.2 Neutral terminal—greater than 600 V

When provided and effectively grounded, the neutral bushing **may be two insulation classes below** that of the phase bushings or connectors. Otherwise, the neutral bushing shall be the insulation class of the phase bushings or connectors.

Table 7
Low-voltage ground strap ampacity
equivalence of ground strap
(See [Clause 6.4.2.](#))

Transformer size, kV•A	240/120 V low voltage, AWG
25	1 cu
50	1/0 cu
75	2/0 cu
100	4/0 cu
167	4/0 cu

Tap Changer --- optional

Pad-mounted Transformers-----De-energized tap changer

de-energized tap changer (DETC): A device for changing the tap of a winding, suitable for operation only when the transformer is de-energized. Syn: **off-circuit tap changer**

load tap changer (LTC): A device for changing the tap of a winding, suitable for operation while the transformer is energized or on load. Syn: **on-load tap changer (OLTC).**

off-circuit tap changer: See: **de-energized tap changer (DETC).**

on-load tap changer (OLTC): See: **Load tap changer (LTC).**

IEEE C57.12.34	C227.4 & C227.5
The switches should be clearly and permanently labeled as “Suitable for deenergized operation only.”	Figure A.2
Deenergized switches are normally located in and operated from the transformer compartment and not located in any auxiliary cabinet.	The means for operating these devices shall be installed in the cable entrance compartment
high-voltage taps shall be composed of two 2.5% above the high-voltage rating and two 2.5% below the high-voltage rating.	When required by the purchaser, transformers shall be equipped with four high voltage taps that are $\pm 2.5\%$ and $\pm 5\%$ of the rated voltage. All taps shall be full-capacity
Transformers with a 208Y/120 low-voltage rating may be specified with four 2.5% taps below the high-voltage rating	Transformers without taps are the standard
The operating handles should be secured by a bail or other means to guard against energized operation.	shall be externally operable lever-handle types, made of corrosion-resistant material.
The letter “A” or the Arabic numeral “1” shall be assigned to the voltage rating providing the maximum ratio of transformation---IEEE C57.12.36	The highest voltage tap shall be tap position 1 or A

Pad-mounted Transformers-----De energized tap changer



— 11 —

Product information—15/25/35kV 100A Tap-changer Switch

◆ General

The tap-changer switch is designed for use in distribution transformers to tap the winding terminals to get the different voltage value.

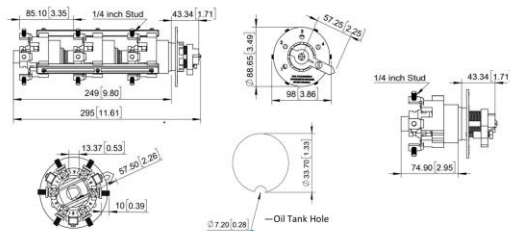
The switch body, rotor, and shaft are made of a high strength, glass-filled thermoplastic material. The operating shaft is sealed against leakage by two high temperature resistant O-Rings. The tap-changer switch is no excitation switch, operate it just in no load condition.

The switch has 6 contacts and five positions, can change the transformer voltage: -5%, -2.5%, 0, 2.5%, 5%

◆ Parameters

Rated Voltage (Phase-to-Phase)	15/25/35kV
Withstand Voltage (50/60Hz)	50kV/1min.
BIL and Full Wave Crest	150kV
Rated Current	100A

◆ Dimensions (All dimensions are in mm[inch])



◆ Order information

Description	Model
15/25/35kV 100A Single Phase Tap Changer	TP-S5
15/25/35kV 100A Three Phase Tap Changer	TP-T5

Note: Product related information is constantly updated. Please confirm relevant information with our company before placing an order.

Shanghai Gid Electric Co., Ltd.



www.gid-id.com



—Single Phase Tap-changer



—Three Phase Tap-changer

5.1.1 Rated characteristics for LTCs

The rated characteristics of an LTC are as follows:

- Rated through current
- Maximum rated through current
- Rated step voltage
- Maximum rated step voltage
- Rated frequency
- Rated insulation level

7.2 Design tests for DETCs

7.2.1 General

The following design tests shall be performed:

- Temperature rise of contacts
- Short-circuit current test
- Mechanical tests
- Dielectric tests

externally mounted tap changer: A tap changer mounted in a container outside the main transformer tank

internally mounted tap changer: A tap changer mounted inside the main transformer tank and immersed in the insulating liquid of the transformer and immersed in its own insulating liquid.



100 A



Volts L-L	DETC positions	Leads connected
1	1-2	1-2
2	2-3	2-3
3	3-4	3-4
4	4-5	4-5
5	5-6	5-6

De energized tap changer-----IEEE C57.131 **Temperature rise of contacts for LTCs**

Tests shall be performed to verify that the temperature rise above the medium surrounding each type of contact that carries current continuously in service does **not exceed** the values given in Table 6 when the contacts have reached a steady temperature when carrying **1.2 times the maximum rated through current**.

Table 6—Contact temperature rise limits for DETCs

Contact material	In air (K)	In liquid (K)
Plain copper	25	15
Silver-faced copper/alloys	40	15
Other materials	By agreement	15
NOTE—The above values are lower than for LTCs to guard against “pyrolitic” carbon formation, which can occur when contacts remain on one position for long periods. The temperature rise values of 15 K at 1.2 times the maximum rated through current approximate to a temperature rise of 11 K at 1.0 times the maximum rated through current.		

Deenergized tap changer-----IEEE C57.131 **Temperature rise**

Table 6—Contact temperature rise limits for DETCs

Contact material	In air (K)	In liquid (K)
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Other materials	By agreement	15

NOTE—The above values are lower than for LTCs to guard against “pyrolitic” carbon formation, which can occur when contacts remain on one position for long periods. The temperature rise values of 15 K at 1.2 times the maximum rated through current approximate to a temperature rise of 11 K at 1.0 times the maximum rated through current.

Deenergized tap changer-----IEEE C57.131 Short-circuit test for LTCs

All contacts of different design that carry current continuously shall be subjected to short-circuit currents, each **of at least 2 s ($\pm 10\%$) duration**. In the case of three-phase LTCs, it is sufficient to test the contacts of one phase only unless otherwise specified. Three applications shall be made with an **initial peak current of 2.5 ($\pm 5\%$) times the rms value of the shortcircuit test current**. The load impedance shall exhibit an X/R ratio of 12 or more in order to accomplish the desired degree of asymmetry. The contacts shall not be moved between these applications. When there are no facilities for point-on-wave switching, and it is not possible to obtain three short-circuit applications with initial peak current 2.5 times the rms value, the following test may be used. The rms value of the short-circuit test current may be increased so that the peak current is obtained for the three applications and the test duration reduced. When using this method, the product of the square of the increased rms current and the shorter test duration shall be not less than the product of the square of the rated short-circuit rms current and the 2 s duration. The value of the rms symmetrical short-circuit test current to be applied shall be equal to the maximum rated through current, multiplied by the factor shown in Figure 1.

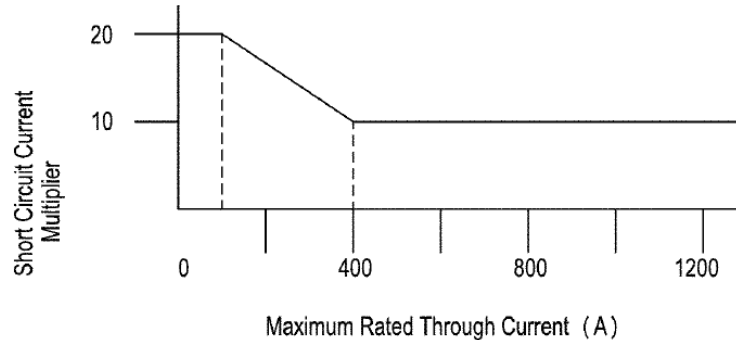


Figure 1—Short-circuit multiple of the maximum rated through current for LTCs

Deenergized tap changer-----IEEE C57.131 Short-circuit test for LTCs

All contacts of different design that carry current continuously shall be subjected to short-circuit currents, each **of at least 2 s ($\pm 10\%$) duration**. In the case of three-phase LTCs, it is sufficient to test the contacts of one phase only unless otherwise specified. Three applications shall be made with an **initial peak current of $2.5 (\pm 5\%)$ times the rms value of the shortcircuit test current**. The load impedance shall exhibit an X/R ratio of 12 or more in order to accomplish the desired degree of asymmetry. The contacts shall not be moved between these applications. When there are no facilities for point-on-wave switching, and it is not possible to obtain three short-circuit applications with initial peak current 2.5 times the rms value, the following test may be used. The rms value of the short-circuit test current may be increased so that the peak current is obtained for the three applications and the test duration reduced. When using this method, the product of the square of the increased rms current and the shorter test duration shall be not less than the product of the square of the rated short-circuit rms current and the 2 s duration. The value of the rms symmetrical short-circuit test current to be applied shall be equal to the maximum rated through current, multiplied by the factor shown in Figure 1.

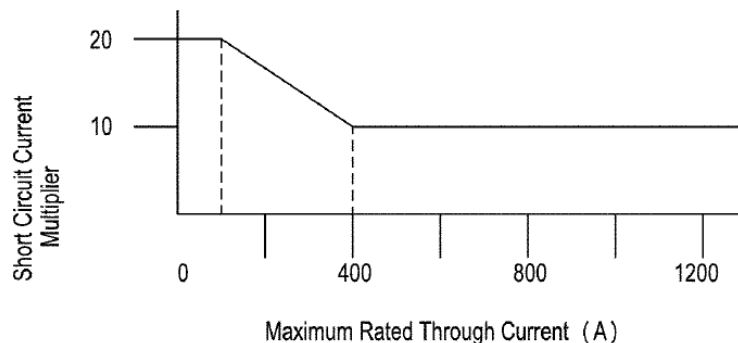


Figure 1—Short-circuit multiple of the maximum rated through current for LTCs

Deenergized tap changer-----IEEE C57.131 Mechanical tests for LTCs

If the load tap changer is of liquid-immersed design, it shall be **assembled and filled with clean liquid or immersed in a test tank filled with clean liquid**, and operated as for normal service conditions. The contacts shall not be energized and the full range of taps shall be utilized until a minimum of 500 000 tap change operations have been performed. At least **50 000 tap change operations** shall be carried out on the changeover selector.

For externally mounted LTCs, this test may **be performed at ambient temperature**. For internally mounted LTCs, half the number of operations shall be performed at a temperature of not less than 75 °C and half at a lower temperature (ambient or higher), for example during the heating or cooling period, with daily temperature cycles being permitted

Deenergized tap changer-----IEEE C57.131 Dielectric tests for LTCs

Dielectric tests shall be performed in the following sequence:

- a) Lightning impulse
- b) Switching impulse, when required
- c) Applied voltage
- d) Partial discharge, when required

5.2.6.3 Nature of tests

The insulation level of the LTC shall be demonstrated by dielectric tests performed on the following insulation distances:

- a) Test 1: live parts to ground
- b) Test 2: between phases, if applicable
- c) Test 3: between the first and last contacts of the tap selector or arcing tap switch and, where applicable, of the change-over selector
- d) Test 4: between any two contacts that connect across adjacent taps of the tapped winding, or any other contacts relevant to the LTC contact configuration.
- e) Test 5: between diverter switch contacts in their final open position

NOTE—The maximum voltage stress between contacts or terminals may depend on the position of the LTC. In such cases, the manufacturer shall state the tap position for the test.

5.2.6.4.1 Category 1

For test 1, the test voltages shall comply with appropriate values from Table 4. For tests 2, 3, 4, and 5, appropriate withstand values of lightning impulse voltage and, if applicable, of power frequency voltage with a duration of 60 s shall be determined and reported by the manufacturer of the LTC.

5.2.6.4.2 Category 2

For tests 1 and 2, test voltages shall comply with the appropriate values from Table 4. For tests 3, 4, and 5, appropriate withstand values of lightning impulse voltage and, if applicable, of power frequency voltage with a duration of 60 s shall be determined and reported by the manufacturer of the LTC. Test voltage levels for the partial discharge test, when required and as described in 5.2.6.9, shall comply with the appropriate values from Table 5.

Table 4—Insulation levels for LTCs

Nominal voltage level of LTC (kV)	Applied voltage test level (kV rms)	Basic lightning impulse level (kV crest)	Switching impulse level (kV crest)
15 and below	34	110	—
25	50	150	—
34.5	70	200	—
46	95	250	—
69	140	350	—
115	230	550	—
138	275	650	—
161	325	750	—
230	395	900	—
287	460	1050	—
345	520	1175	975

Liquid Level Indicator

Liquid-level indicating devices---Standard

Suitable means for indicating the correct liquid level at 25 °C shall be provided.

A.5 Liquid-level indicating devices

A liquid-level device is provided to indicate the level of insulating liquid in the transformer tank. The indicating devices provide either a “go” or “no go” indication, or they show a range from minimum to maximum.

A.5.1 Sight-type liquid-level indicators

This device is a plug equipped with a window, high-contrast floating ball, and reflector. It is located on the tank wall to view the desired liquid level and indicates a “go” or “no go” liquid-level indication.

A.5.2 Visual dial indicator

A.5.2.1 Display

This device is a needle indicator that provides continuous display of the liquid level from maximum to minimum. This can be a direct-drive or a gear-driven mechanism that provides amplified dial pointer movement relative to the corresponding float travel in the tank.

A.5.2.2 Auxiliary contacts

Lever-driven gauges can be equipped with integrated switches that allow for limit alarm control functions. Single-pole double-throw (SPDT) contacts may be connected to the movement to give a remote annunciation of a low or high liquid level. Depending on the model and manufacturer, the quantity and rating of these contacts can vary. Therefore, before the user applies these switches in control circuitry, the ratings must be obtained from the manufacturer.

magnetic-type liquid-level indicator shall be provided on all transformers.

All these devices shall be located in the cable entrance compartment.

Liquid level gauge

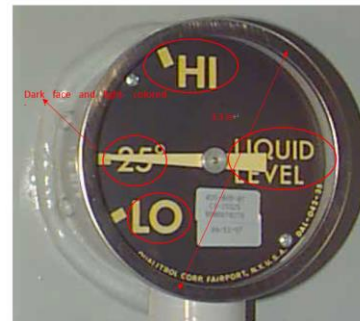


Figure 1. Liquid level gauge.



Liquid temperature indicating devices(Optional)

Liquid temperature indicating devices (optional)

Liquid temperature indicating devices

This device is furnished to reflect the top liquid temperature in the transformer tank in degrees Centigrade.

A.6.1 Temperature label

Temperature labels are temperature monitors consisting of one or more small heat-sensitive indicators sealed under transparent and heat-resistant windows. This application requires the use of reversible temperature labels so that they can change back and forth as needed to provide an indication of the present tank temperature. It can be inferred from this value the temperature of the liquid in the tank. Nonreversible temperature labels are also available. These provide indication of past operation at elevated temperatures. These labels are typically applied in the field when a normal liquid temperature measuring mechanism is not available.

A.6.2 Dial-type visual indicator

A dial-type visual indicator is a surface-mounted thermometer with temperature indication using a dial pointer and a maximum temperature pointer that may be reset. These dial-type visual indicators can be inserted directly into the tank for direct liquid contact or into a dry well, which allows them to be removed without lowering the liquid level.

A.6.3 Auxiliary contacts

Dial indication thermometers can be equipped with integrated switches that allow for limit alarm control functions. Adjustable switches may be connected to the movement for a high- or low-temperature alarm, or to energize a fan circuit. Depending on the model and manufacturer, the quantity and rating of these can vary. Therefore, before the user applies these switches in control circuitry, the ratings need to be obtained from the manufacturer.

A.6.4 Temperature transmitter

A temperature transmitter is an instrument that takes the input from a temperature sensor and converts the measurement to an instrument-level voltage or current output, or to a digital value. Another role that the temperature transmitter may play is to isolate the temperature signal by filtering any electromagnetic interference, as well as by amplifying and converting the temperature sensor's signal. This device enables the temperature measurement to be used directly to a display device at the transformer or interfaced to a supervisory control and data acquisition (SCADA) system or other distributed control system (DCS).

A.6.5 Temperature sensor

The temperature sensors can be a resistance temperature detector (RTD), thermistor, or thermocouple, which are immersed in the insulating liquid of the transformer. The location of this sensor needs to be discussed with the manufacturer as it could affect the insulation arrangement of the transformer

Product information—Transformer Oil Temperature Gauge

The transformer oil temperature gauge is suitable for the instrument installed on the side wall of the transformer for temperature measurement and indication. The temperature measurement range is 0~120°C.

- Case: 304 stainless steel, polish, non-vent.
- Pointer: aluminium, white painting, non-adjustable.
- Lens: polymethyl methacrylate.
- Accuracy: 3%
- Connection: Cr stainless steel M27 X 2.
- Tube & Socket & Movement: 304 stainless steel.
- Temperature gauge well: 304 stainless steel, polish, internal thread, M27 X 2, external thread, M27 X 21/2-14, diameter 12mm, length 173mm.

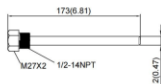
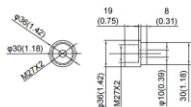
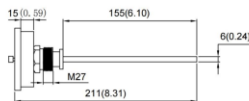
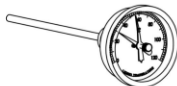


Figure 2. Liquid temperature gauge.

Description	Model
Transformer Oil Temperature Gauge	SID-YWJ-211-0

Shanghai Sid Electric Co., Ltd. www.sh-sid.com

Pressure-vacuum Gauge (Optional)

Pressure-vacuum Gauge (Optional)

The tank shall be of sufficient strength to withstand a range of gauge pressures between -35 kPa (-5 psig) to 49 kPa (7 psig) without permanent deformation, and 103 kPa (15 psig) without rupturing or affecting cabinet security as described in IEEE Std C57.12.28

Pressure-vacuum gauge



Figure 3. Pressure-vacuum gauge.

pressure-relief device (standard)

Accessories: Pressure- relief

pressure relief valve (PRV): An automatic one-way valve used to relieve excess pressure buildup.

pressure relief device (PRD): A mechanical device for the purpose of relieving liquid or gas pressure buildup.

pressure relief plug: A manually operated threaded tank fitting used to relieve excess pressure. The fitting shall allow slow release of pressure without completely removing the fitting from the tank.

pressure-vacuum bleeder valve: A valve designed to balance pressure and vacuum conditions in the sealed tank, by releasing or admitting air/gas at predetermined positive and negative settings.

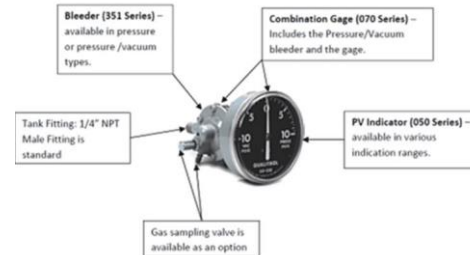
- Pull ring type PRV



- PRD



pressure-vacuum bleeder valve



Accessories: **pressure relief valve (PRV):**

To provide automatic relief of pressures that build up slowly in excess of normal operating pressures, **a replaceable PRV shall be provided on transformers 2500 kVA and below** that do not have a cover assembly designed to relieve excessive pressure. The PRV shall be designed and located to minimize liquid egress when relieving these excessive pressures. The PRV **shall be located above the 140 °C top liquid level**, by the manufacturer's calculation, and shall be located so as not to interfere with any other devices. The inlet port shall be 1/4 in or larger National Pipe Taper (NPT), sized for a specified minimum flow rate. Exposed parts shall be of weather- and corrosion-resistant materials. **Gaskets and O-rings shall withstand liquid vapor and a 105 °C temperature** continuously under operating conditions as described in IEEE Std C57.91, without seizing or deteriorating for the life of the transformer.

The PRV shall have a pull ring for manually reducing pressure to the atmosphere using a standard hookstick, and shall be capable of withstanding a static pull force of **112 N (25 lbf)** for 1 min without permanent deformation. The PRV shall withstand a static force of **445 N (100 lbf) for 1 min** applied normally to its longitudinal axis at the outermost extremity of its body.

When specified, the venting port, on the outward side of the PRV head set, shall be protected to prevent entry of dust, moisture, and insects before and after the PRV has actuated; or, a weather-cap-type indicator shall be provided, which will remain attached to the PRV and provide positive indication to an observer that the PRV has operated.

Venting and sealing characteristics shall be as follows:

- Cracking pressure: 69 kPa (gauge) $\pm$ 13 kPa (gauge) (10 psig $\pm$ 2 psig).**
- Resealing pressure: 42 kPa (gauge) (6 psig) minimum.**
- Zero leakage from resealing pressure to -56 kPa (gauge) (-8 psig).**
- Flow at 103 kPa (gauge) (15 psig) = 16.5 L/s (35 SCFM) minimum**, corrected for air pressure of 101 kPa (14.7 psi) (absolute) and air temperature of 21 °C.

- Flow rates available from 35 to 200 SCSM
- Designs accommodate a range of operating pressures (3.5 psi to 35 psi)

Model No.	Cracking Pressure		Resealing Pressure		Discharge Rate at 15 PSIG	Thread Size
	PSIG	KPA	PSIG	KPA		
Series 302						
302-010-01N	10 $\pm$ 2	69 $\pm$ 14	8 $\pm$ 2	55 $\pm$ 14	>50 SCFM	¼" NPT 18
302-010-02N	10 $\pm$ 2	69 $\pm$ 14	8 $\pm$ 2	55 $\pm$ 14	>50 SCFM	½" NPT 18
302-020-01I	10 $\pm$ 2	69 $\pm$ 14	8 $\pm$ 2	55 $\pm$ 14	>50 SCFM	¼" NPT 18
302-030-01I	10 $\pm$ 2	69 $\pm$ 14	8 $\pm$ 2	55 $\pm$ 14	>50 SCFM	¼" NPT 18



Accessories: **pressure relief device (PRD) & pressure-vacuum bleeder valve**

pressure relief device (PRD): A mechanical device for the purpose of relieving liquid or gas pressure buildup.

A PRD shall be provided on the cover for distribution substation transformers rated **above 2500 kVA** or having an insulation level above 200 kV BIL, and when specified for pad-mounted transformers. Alternative locations such as the tank sidewall may be specified. The minimum flow rate for cover-mounted PRDs should be of **142 m³/min (5000 SCFM)**, at an operating pressure of 69 kPa (gauge) (10 psig) when tested at **103 kPa (gauge) (15 psig)**. If a PRD is required or specified, the tank shall also **have a PRV to relieve the tank pressure manually, or a pressure-vacuum bleeder valve**. The use of the PRD on pad-mounted transformers shall not affect conformance with the enclosure security requirements of IEEE Std C57.12.28 or IEEE Std C57.12.29, as appropriate

5.1.7 Pressure-vacuum bleeder valve --- IEEE C57.12.36

A pressure-vacuum bleeder device, set to operate at the maximum operating pressures (positive and negative) indicated on the nameplate, shall be furnished on **transformers rated 2500 kVA and above**



Load-break switch(optional)

Dual-voltage (series multiple) switch(optional)

Accessories: Dual-voltage (series multiple) switch

This switch is used to change deenergized transformer windings between series and parallel to provide different common transformer voltage ratios, such as $12\ 470 \times 24\ 940\text{ V}$. Each voltage position should be clearly identifiable by a unique reference to the nameplate information.

Delta-wye switch (optional)

Accessories: Delta-wye switch

This switch is used to change deenergized transformer windings between delta and wye to provide different common transformer connections and ratios, such as 14 400 V / 24 940Y / 14 400 V. Each voltage position should be clearly identifiable by a unique reference to the nameplate information.

Valve

Accessories: **Drain valves, fill valves, and connections**

- fill valves
- Drain valves
- Sampling device
- lower fill valve
- Shut off valve
- Radiator Valve/ Radiator Butterfly Valves



Accessories: Drain valves, fill valves, and connections

A 1-in NPT upper plug (or cap) for filling and pressure testing shall be provided in the low-voltage compartment. A 1-in NPT drain plug (or cap) for transformers rated 75 kVA through 500 kVA and a 1-in NPT drain valve with built-in sampling device for transformers rated 750 kVA through 10 000 kVA shall be provided in the low-voltage compartment. Suitable means for indicating the correct liquid level at 25 °C shall be provided.



Under insulating liquid surge arrester(Optional)

Internal current transformer(Optional)

Winding hot-spot monitors(Optional)

Accessory boxes (Optional)

Pad-mounted Transformers-----

A 1-inch NPT upper plug (or cap) for filling and pressure testing shall be provided in the low-voltage compartment. A 1-inch NPT drain plug (or cap) for transformers rated 75 kVA through 500 kVA and a **1-inch NPT drain valve** with built-in sampling device for transformers rated 750 kVA through 10 000 kVA shall be provided in the **low-voltage compartment**. Suitable means for indicating the correct liquid level at 25 °C shall be provided.

A drain valve of **2.54 cm (1 in) with sampling device**, top filter-press connection (oil fill plug), and magnetic-type liquid-level indicator shall be provided on all transformers.

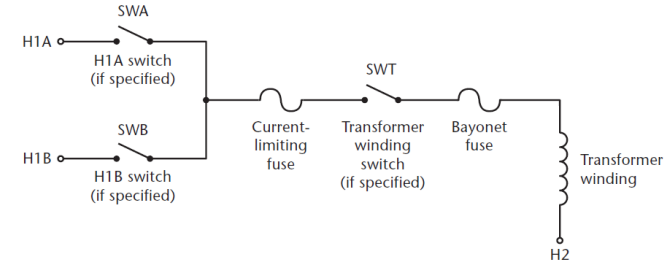


Overcurrent protection(optional)

Pad-mounted transformer-----Construction Fusing Systems

2.2.13.3.5 Combination of **Bayonet** and **Partial-Range Current-Limiting Fuses**

The most common method of protection for pad-mounted distribution transformers is the coordinated combination of a bayonet fuse (usually dual sensing) and a **partial-range current-limiting fuse (PRCL)**. The PRCL only responds to a **high fault current**, while the bayonet fuse is only capable of interrupting **low fault currents**. These fuses must be coordinated in such a way that any secondary fault will melt the bayonet fuse. Fault currents above the bolted secondary fault level are assumed to be due to **internal faults**. Thus **the PRCL, which is mounted inside the tank**, will operate only when the transformer has **failed** and **must be removed from service**.



All transformers shall be equipped with a **two-fuse system** in accordance with **Table 6**, consisting of a **replaceable dual-sensing bayonet fuse**, RTE-type or equivalent, and an **under-oil partial range backup current-limiting fuse (CLF)**,

A.3.5 Weak link fuse assembly (bayonet or internal cartridge)

The weak link fuse is an assembly designed to be located under insulating liquid to allow for the breaking of fault or load current. This is accomplished by utilizing a cartridge-type fuse. This fuse is designed to protect the distribution system from excessive **overloads and fault conditions in the transformer**.

Pad-mounted transformer-----Single Phase Construction *Bayonet fuse*

2.2.13.3.4 Bayonets

Pad-mounts and submersibles may use a primary link (expulsion fuse) that is **mounted internally** in the **transformer oil** but that can be **withdrawn for inspection** of the fuse element or to interrupt the primary feed. This device is called a bayonet and consists of a probe with a **cartridge** on the end that contains the **replaceable fuse** element. Fuses for bayonets may be **either fault sensing or dual sensing**.

A.3.5 Weak link fuse assembly (bayonet or internal cartridge)---IEEE C57.12.34

The weak link fuse is an assembly designed to be **located under insulating liquid** to **allow for the breaking of fault or load current**. This is accomplished by utilizing a cartridge-type fuse. This fuse is designed to protect the distribution system from excessive **overloads and fault conditions in the transformer**.

A.3.5.1 Mounting options

In the case of weak link fuses, use one of the following two installation methods:

- a) Bayonet type fuse canister with cartridge holder
- b) Internal cartridge in an insulated holder

Spill prevention



A.3.5.3.1 Current-sensing fuse

The current-sensing weak link fuse is a **single-element fuse that senses faults or overloads**. Current-sensing fuses **have higher melting temperature elements**, which tend to be insensitive to **high liquid temperatures**, and only open from excessive resistive heating caused by **a high current**.

A.3.5.3.2 Dual-sensing fuse

The dual-sensing weak link fuse comprises **a low melt temperature material**. These fuses provide protection against both **faults or overloads and high liquid temperature**. Because of its ability to sense and operate when there is higher liquid temperature, this link aids in preserving the life of a transformer as the life expectancy of a transformer is reduced when the transformer is exposed to long-term elevated temperatures

A.3.6 Isolation link

The **isolation link is located inside the tank**. It is not intended to **be externally accessible nor field replaceable**. It may be installed on an internal block or as an accessory part of the bayonet fuse assembly. The isolation link provides extra protection during refusing and switching operations and must be used in series with a bayonet-type fuse **if a partial-range, current-limiting fuse is not used**. Isolation links are not fuses and **do not have an interrupting rating** but will melt during most transformer failures preventing operating personnel from reenergizing the high-voltage circuit.

Pad-mounted transformer-----Single Phase Construction *Bayonet fuse*

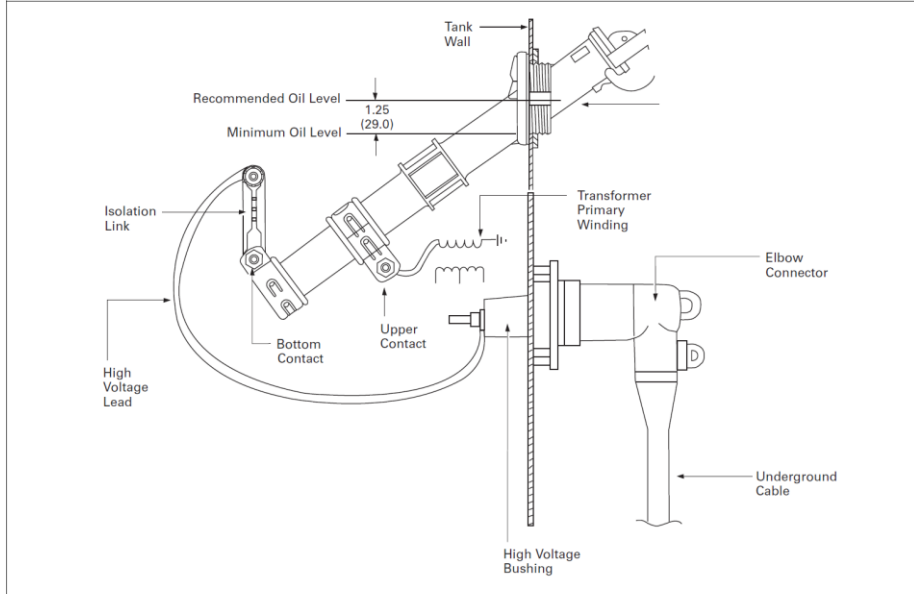


Figure 17.0-9. Bay-O-Net Assembly with Isolation Link

Note: Isolation link is not required if the Bay-O-Net fuse is used in series with a backup energy limiting fuse.

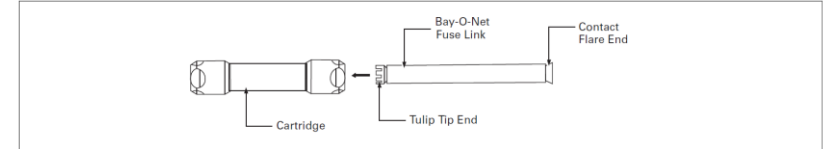


Figure 17.0-10. Insertion of Bay-O-Net Into Cartridge

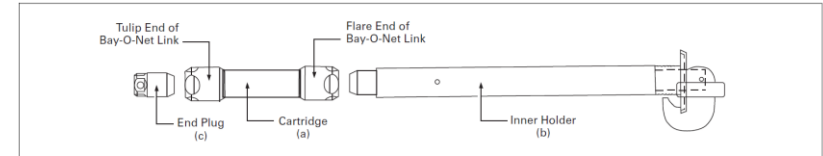


Figure 17.0-11. Assembly of Cartridge with Fuse Onto Inner Holder



Pad-mounted transformer----- *Partial-Range Current-Limiting Fuses*

A partial-range, current-limiting fuse is located **inside the tank**. This fuse is not intended to be **externally accessible nor field replaceable**. This fuse is defined by **IEEE Std C37.41 [B6].10** Because it is only a partialrange fuse, it is required that the fuse be paired with a **low-current series weak link** fuse to achieve full-range protection for the distribution transformer and isolation from the distribution system.

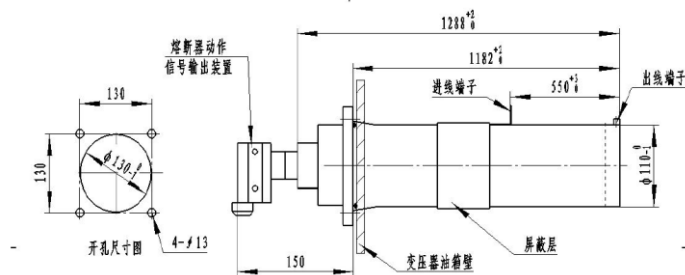
A.3.4 Full-range, current-limiting fuse

A.3.4.1 Dry well

A full-range, current-limiting fuse located in a dry-well canister should be of the nongassing type and explicitly designed for use in a dry-well canister. The canister can be provided as non-loadbreak. It is recommended that when using this type of fuse installation, a **disconnecting means** capable of breaking the full load current of the transformer be provided either in the fuse canister or **external to the canister utilizing an interlocking loadbreak switch** to prevent the fuse from being removed while the transformer is energized. For those applications where an interlocked loadbreak switch is not used in conjunction with the non-loadbreak fuseholder, a warning nameplate must be used as a **precaution against the energized operation of the fuseholder and a mechanism of securing the fuse canisters mechanically** to prevent inadvertent operation.

A.3.4.2 Submersible

The submersible full-range, current-limiting fuse is similar to a dry-well fuse canister fuse **except it is located under the insulating** liquid of the transformer. This makes it **useable with smaller clearances**, shorter creep paths, and simpler load break mechanisms. Submersible installations also eliminate damage from erosion and chemical changes due to weathering. This is normally accomplished by using a type of Bayonet fuse assembly that has been designed to accommodate a full-range current-limiting fuse.



Pad-mounted transformer ---- UL Safety mark NEMA 260

-- R/C (PGDQ2/8) or R/C (PGJI2/8)

- a. The outside warning label(s) (Figures 4-1 or 4-3) shall be mounted as close as is practical to the equipment door while providing maximum visibility.
- b. The inside danger label (Figures 4-2 or 4-4) shall be mounted as to be readily visible whenever the equipment door is open.



Figure 4-1
WARNING LABEL



Figure 4-2
DANGER LABEL

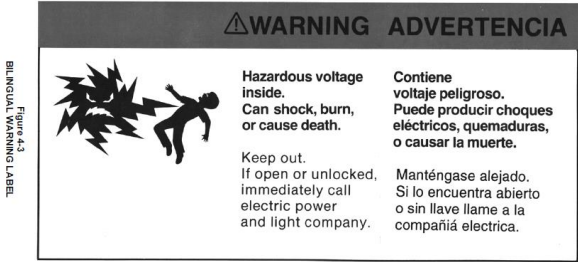


Figure 4-3
BILINGUAL WARNING LABEL

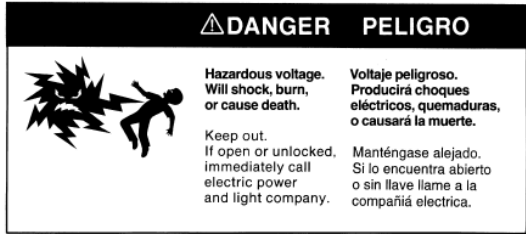


Figure 4-4
BILINGUAL DANGER LABEL



The color of the warning signal word field shall be orange, and the color of the danger signal word field shall be red. All label colors depicted in Figures 4-1 through 4-4 shall conform to the following Munsell Notations:

Black	N - 2.0 or lower
Orange	5.0 YR 6.0/15
Red	7.5 R 4.0/14
White	N - 9.0 or higher

The safety labels in Figures 4-1 and 4-2 shall be standard and shall be 4.5 inches, $\pm$ 0.25 inches (11.43 cm, $\pm$ 0.64 cm) wide and 8.0 inches, $\pm$ 0.50 inches (20.32 cm, $\pm$ 1.25 cm) long.3 The warning label (Figure 4-1) shall be affixed to the outside of the pad-mounted equipment. The danger label (Figure 4-2) shall be affixed to the inside of the pad-mounted equipment.

Thank you

